SHARP MEAN-FIELD ESTIMATES FOR DIFFUSIVE LOG/RIESZ GASES IN THE HILBERT–SCHMIDT REGIME

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Abstract. We study fixed-temperature logarithmic and Riesz gases after subtracting the leading mean-field contribution relative to a prescribed background law. For repulsive interactions in the Hilbert–Schmidt regime, we prove N-uniform bounds and quantitative convergence of the resulting modulated partition function to a normalization expressed by the Carleman–Fredholm determinant of the centered interaction operator. We show that the Hilbert–Schmidt threshold is sharp and obtain explicit lower bounds on the rate of divergence at and above it; these rates are expected to be nonoptimal. The proof combines positive-definite truncations and a low/high-frequency decomposition with exponential inequalities and Gaussian-chaos asymptotics for canonical degree-two U-statistics. As consequences, we establish entropic commutator estimates with the sharp $O(N^{-1})$ additive scale in the modulated-free-energy method, a static joint linear-statistics central limit theorem, and a dynamical central limit theorem for joint linear statistics at finitely many times. This extends the logarithmic partition-function estimates of the first two authors to the full Riesz Hilbert–Schmidt range and identifies the limiting determinant normalization. For the attractive logarithmic interaction at sufficiently small inverse temperature, we also prove analogous results. The principal results of this paper have been formalized in Lean 4, including all mathematical ingredients required by the formal proofs.

1. Introduction

For a symmetric two-body interaction potential W and one-body external potential V, consider the mean-field Hamiltonian

$$\mathcal{H}_N(X_N) \coloneqq \frac{1}{2N} \sum_{1\le i\neq j\le N} W(x_i,x_j) + \sum_{i=1}^N V(x_i), \qquad X_N \coloneqq (x_1,\dots,x_N). \tag{1.1}$$

The associated canonical Gibbs ensemble is

$$\mathrm{d}\mathbb{P}_{N,\beta}(X_N) = \frac{1}{Z_{N,\beta}} e^{-\beta \mathcal{H}_N(X_N)}\, \mathrm{d}X_N. \tag{1.2}$$

Here the normalizing constant $Z_{N,\beta}$ is the partition function. We work in the fixed-temperature mean-field scaling, in which β is independent of N; we refer to this as the diffusive temperature regime. Throughout the paper, unless explicitly indicated otherwise, the position variables belong to the whole space $\mathbb{R}^{\mathsf{d}}$; the flat-torus and kinetic variants are recorded in Remark 2.1.

The basic question of this paper is to identify the normalization which remains after the leading-order macroscopic mean-field contribution has been separated relative to a prescribed product background. Physically, the separation fixes the macroscopic density profile and asks what free-energy correction is left from the correlated microscopic fluctuations. A bounded residual normalization means that, after the mean-field cost has been removed, the singular interaction contributes only an

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$O(1)$ fluctuation factor. Divergence of this residual factor is the ultraviolet obstruction at the Hilbert–Schmidt threshold. Mathematically, this is the normalization needed to pass between product and modulated Gibbs references in the modulated-free-energy method [RS25]. Controlling this normalization also allows us to prove averaged versions of the commutator estimates, deterministic pointwise versions of which have been widely used in the mean-field and fluctuation theory of log/Coulomb/Riesz gases; see Serfaty's lecture notes [Ser26] and the third author's recent survey [Ros25].

For repulsive logarithmic and Riesz interactions in the Hilbert–Schmidt regime, we prove sharp fixed-temperature partition-function estimates, including a quantitative convergence rate to the limiting determinant normalization and the sharpness of the square-integrability threshold. We then use these bounds as exponential-moment inputs for entropic commutator estimates governing variations of the modulated energy. Finally, we derive sharp mean-field estimates for interacting diffusions, a static central limit theorem for joint linear statistics under the modulated Gibbs law, and a dynamical central limit theorem for joint linear statistics at finitely many times. For the attractive logarithmic interaction at sufficiently small inverse temperature, we also prove analogous results. These results are used in the authors' companion works [DGR26d, DG26] to obtain sharp two-sided bounds for the size of chaos for diffusive log/Riesz gas measures in several statistical divergences up to the optimal Hilbert–Schmidt threshold and to prove sharp quantitative CLTs for fluctuations around the mean-field limit.

The precise statements, their relation to previous work, and the proof strategy are given in Section 2.

2. Main results and proof strategy

2.1. Modulated Gibbs normalization for log/Riesz gases.

We first isolate the modulated normalization relative to a prescribed background law, then identify its Hilbert–Schmidt determinant limit and specialize the construction to logarithmic and Riesz interactions.

Splitting relative to a background law.

Fix a probability measure μ on $\mathbb{R}^{\mathsf{d}}$. When μ is absolutely continuous with strictly positive density, write $\mathrm{d}\mu = Z_\mu^{-1} e^{-U}\,\mathrm{d}x$, with $U = -\log\mu$ understood up to an additive constant. For integers $N \geq 1$, write $[N] \coloneqq \{1,\dots,N\}$. Set

$$h_\mu(x) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} W(x,y)\,\mathrm{d}\mu(y), \qquad I_\mu \coloneqq \frac{1}{2}\iint_{(\mathbb{R}^{\mathsf{d}})^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y). \tag{2.1}$$

We use the notation

$$\tilde{f}(x,y) \coloneqq f(x,y) - \int_{\mathbb{R}^{\mathsf{d}}} f(x,z)\,\mathrm{d}\mu(z) - \int_{\mathbb{R}^{\mathsf{d}}} f(z,y)\,\mathrm{d}\mu(z) + \int_{(\mathbb{R}^{\mathsf{d}})^2} f(z,w)\,\mathrm{d}\mu(z)\,\mathrm{d}\mu(w) \tag{2.2}$$

for the Hoeffding centering of a two-body kernel. The modulated energy relative to μ is characterized by the identity[1]

$$\frac{1}{2N}\sum_{1\leq i\neq j\leq N} W(x_i,x_j) = N\mathsf{F}_N(X_N,\mu) + \sum_{i=1}^N h_\mu(x_i) - NI_\mu. \tag{2.3}$$

Equivalently,

$$\mathsf{F}_N(X_N,\mu) = \frac{1}{2N^2}\sum_{1\leq i\neq j\leq N} \tilde{W}(x_i,x_j) - \frac{1}{N^2}\sum_{i=1}^N h_\mu(x_i) + \frac{1}{N}I_\mu. \tag{2.4}$$

[1]This identity is often called the splitting formula (relative to thermal equilibrium) in the log/Coulomb/Riesz literature [Ser26, Chapter 5, Section 5.1]. It is a special case of the Hoeffding decomposition of a U-statistic.

Consequently,

$$Z_{N,\beta} = Z_\mu^N e^{\beta N I_\mu} \mathbb{E}_{\mu^{\otimes N}} \left[\exp\left\{ -\beta N \mathsf{F}_N(X_N,\mu) - \sum_{i=1}^N (\beta V + \beta h_\mu - U)(x_i) \right\} \right]. \tag{2.5}$$

If the one-body term is absorbed into the reference law, equivalently if $U = \beta(V + h_\mu)$ up to an additive constant, the expectation in (2.5) reduces to the modulated partition function

$$\mathsf{K}_{N,\beta}(\mu) \coloneqq \mathbb{E}_{\mu^{\otimes N}}[\exp\{-\beta N \mathsf{F}_N(X_N,\mu)\}]. \tag{2.6}$$

The condition $U = \beta(V + h_\mu)$ is precisely the Euler–Lagrange relation satisfied by the thermal equilibrium measure introduced below. Conversely, for $\beta > 0$, (2.6) is an ordinary Gibbs partition function with one-body potential $\beta^{-1}U - h_\mu$, up to the deterministic factor in (2.5). Thus, the adjective "modulated" refers to the centered normal form obtained after subtracting the mean-field contribution.

Associated to a configuration $X_N = (x_1, \dots, x_N)$, denote the empirical measure by

$$\mu_N \coloneqq \frac{1}{N} \sum_{i=1}^N \delta_{x_i}. \tag{2.7}$$

The modulated Gibbs measure is

$$\mathrm{d}\mathbb{Q}_{N,\beta}(\mu)(X_N) \coloneqq \frac{1}{\mathsf{K}_{N,\beta}(\mu)} e^{-\beta N \mathsf{F}_N(X_N,\mu)} \, \mathrm{d}\mu^{\otimes N}(X_N). \tag{2.8}$$

When $\mu = \mu_\beta$ is the thermal equilibrium measure (see, e.g., [Ser26, Section 2.5, especially Proposition 2.19]), characterized by the Euler–Lagrange relation for the macroscopic/mean-field free energy

$$\mathrm{d}\mu_\beta = Z_{\mu_\beta}^{-1} e^{-\beta(V + h_{\mu_\beta})} \, \mathrm{d}x, \tag{2.9}$$

the condition $U = \beta(V + h_\mu)$ above is exactly satisfied. Hence, the modulated Gibbs measure coincides with the original canonical ensemble: $\mathbb{Q}_{N,\beta}(\mu_\beta) = \mathbb{P}_{N,\beta}$. If $\beta > 0$, the associated modulated free energy

$$\mathsf{E}_N(f_N,\mu) \coloneqq \frac{1}{\beta N} \mathsf{H}(f_N|\mu^{\otimes N}) + \mathbb{E}_{X_N \sim f_N}[\mathsf{F}_N(X_N,\mu)], \qquad f_N \in \mathcal{P}((\mathbb{R}^{\mathsf{d}})^N), \tag{2.10}$$

satisfies

$$\mathsf{E}_N(f_N,\mu) = \frac{1}{\beta N} \left(\mathsf{H}(f_N|\mathbb{Q}_{N,\beta}(\mu)) - \log \mathsf{K}_{N,\beta}(\mu) \right). \tag{2.11}$$

Jensen's inequality and (2.4) give the order-one lower bound

$$\mathsf{K}_{N,\beta}(\mu) \geq \exp\Big(-\beta N \mathbb{E}_{\mu^{\otimes N}}[\mathsf{F}_N(X_N,\mu)] \Big) = \exp\left(\beta I_\mu\right), \tag{2.12}$$

provided that μ has finite energy. The three questions of interest to us are boundedness of (2.6) uniformly in N, identification of its limit, and sharpness of the assumptions under which such a bound is possible.

Hilbert–Schmidt centered kernels and limiting normalization.

The limiting normalization is governed by the quadratic fluctuation operator associated with the Hoeffding-centered interaction. If $\tilde{W} \in L^2(\mu^{\otimes 2})$, then $\tilde{W}$ defines a Hilbert–Schmidt operator on $L^2(\mu)$, which we denote by $T_\mu : L^2(\mu) \to L^2(\mu)$:

$$T_\mu f(x) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \tilde{W}(x,y) f(y) \, \mathrm{d}\mu(y). \tag{2.13}$$

We write $(\lambda_j)_{j\geq 1}$ for the nonzero eigenvalues of T_μ, repeated with algebraic multiplicity. For $\beta \geq 0$, define

$$\det_2(\mathrm{Id} + \beta T_\mu) \coloneqq \prod_{j\geq 1}(1+\beta\lambda_j)e^{-\beta\lambda_j}. \tag{2.14}$$

This is the Carleman–Fredholm, or regularized, determinant; see, for example, Simon [Sim05, Chapter 9]. Unlike the ordinary Fredholm determinant $\det(\mathrm{Id} + \beta T_\mu) = \prod_j(1+\beta\lambda_j)$, which requires T_μ to be trace class, $\det_2$ is well-defined for Hilbert–Schmidt operators. The subscript 2 records the second-order regularization: the linear trace contribution is subtracted. In the repulsive positive-definite cases below, T_μ is nonnegative on the centered subspace, so the real logarithm is unambiguous and

$$\log\det_2(\mathrm{Id} + \beta T_\mu) = \sum_{j\geq 1}(\log(1+\beta\lambda_j) - \beta\lambda_j). \tag{2.15}$$

In these positive-definite cases, we set

$$\mathsf{K}_{\infty,\beta}(\mu) \coloneqq e^{\beta I_\mu}\det_2(\mathrm{Id} + \beta T_\mu)^{-1/2}. \tag{2.16}$$

At the product level, the centered two-body term is the canonical statistic

$$W_N \coloneqq \frac{1}{N}\sum_{1\leq i\neq j\leq N}\tilde{W}(\xi_i,\xi_j), \qquad \xi_1,\dots,\xi_N \overset{\text{iid}}{\sim} \mu. \tag{2.17}$$

This is the fluctuation variable whose Laplace transform is exactly the centered two-body factor in the modulated partition function. The origin of the determinant is transparent from the Gaussian-chaos limit. Writing $(Z_j)_{j\geq 1}$ for independent standard Gaussians, the series $\sum_j \lambda_j(Z_j^2 - 1)$ converges in L^2 and

$$\mathbb{E}\left[\exp\left\{-\frac{\beta}{2}\sum_j \lambda_j(Z_j^2-1)\right\}\right] = \prod_j e^{\beta\lambda_j/2}(1+\beta\lambda_j)^{-1/2} = \det_2(\mathrm{Id}+\beta T_\mu)^{-1/2}. \tag{2.18}$$

For a finite-rank kernel, the corresponding limit of W_N follows immediately from the multivariate central limit theorem; the general Hilbert–Schmidt case is obtained by spectral approximation. The main theorem below proves this normalization for the singular interactions to which we now specialize.

Logarithmic and Riesz interactions.

For the log/Riesz specialization, we take $W(x,y) = \mathsf{g}(x-y)$, where

$$\mathsf{g}(x) = \begin{cases} -\log|x|, & \mathsf{s} = 0, \\ \frac{1}{\mathsf{s}}|x|^{-\mathsf{s}}, & 0 < \mathsf{s} < \mathsf{d}. \end{cases} \tag{2.19}$$

The objects h_μ, I_μ, F_N, $\mathsf{K}_{N,\beta}$, $\mathbb{Q}_{N,\beta}$, and T_μ introduced above are henceforth understood for this interaction. With an abuse of notation, we use the same symbol for an absolutely continuous measure and for its Lebesgue density; thus statements such as $\mu \in L^p$, $\|\mu\|_{L^p}$, and pointwise bounds on μ refer to this density.

Up to a normalizing constant, these interactions are characterized as fundamental solutions of the fractional Laplacian, $(-\Delta)^{\frac{\mathsf{d}-\mathsf{s}}{2}}\mathsf{g} = c_{\mathsf{d},\mathsf{s}}\delta_0$. We restrict to the *potential* regime $\mathsf{s} < \mathsf{d}$, in which g is locally integrable, thereby excluding the *hypersingular* case $\mathsf{s} \geq \mathsf{d}$, studied for instance in [HSST21, HLSS18]. The case $\mathsf{s} = \mathsf{d} - 2$ corresponds to the classical *Coulomb* potential from electrostatics/gravitation. Based on this threshold, it is convenient to call the interaction *sub-Coulomb* if $\mathsf{s} < \mathsf{d} - 2$ and *super-Coulomb* if $\mathsf{s} > \mathsf{d} - 2$. More generally, interactions of the form (2.19) play an important role in physics [DRAW02], approximation theory [BHS19], and machine learning [AHS23, HWAH24, HBGS23, HHA+24]. In the singular regime considered below, these same kernels are also central in the analysis of Coulomb/Riesz gases and singular mean-field limits.

For this interaction, the modulated energy F_N originated in the study of the statistical mechanics of Coulomb/Riesz gases [SS15, RS16, PS17, LS17, Ser23] and later was used in the derivation of mean-field dynamics [Due16, Ser20, NRS22] and following works; see [Ser26, Chapter 4] for a comprehensive discussion of the modulated energy. For a nonsingular positive-definite kernel, the quadratic form $\iint_{(\mathbb{R}^{\mathsf{d}})^2} \mathsf{g}(x-y)\ \mathrm{d}(\nu-\mu)(x)\ \mathrm{d}(\nu-\mu)(y)$ is the squared maximum mean discrepancy (MMD) between ν and μ [GBR+12, MFSS17]. Thus, F_N may be viewed, up to the conventional factor $1/2$, as the diagonal-renormalized log/Riesz analogue of this MMD energy with $\nu = \mu_N$.

The positive definiteness of the repulsive kernels implies that T_μ is nonnegative on the centered subspace. For $0 < \mathsf{s} < \mathsf{d}/2$ and $\mu \in L^\infty$, the local singularity satisfies $\mathsf{g} \in L^2_{\mathrm{loc}}$ and hence $\tilde{\mathsf{g}} \in L^2(\mu^{\otimes 2})$; the logarithmic singularity is locally square-integrable under the logarithmic integrability assumptions stated below. Thus, the range $\mathsf{s} < \mathsf{d}/2$ is precisely the Hilbert–Schmidt range for the Riesz kernels.

Remark 2.1 (Torus and kinetic variants)**.** All of the results below also hold on the flat torus $\mathbb{T}^{\mathsf{d}}$ where (2.19) is replaced by the periodic (spectral) logarithmic/Riesz potential. We do not state the torus case separately because the proofs are unchanged after these notational substitutions.

The static spatial results also have an immediate kinetic variant. If the starting Hamiltonian $\mathcal{H}_N$ is augmented by the standard quadratic kinetic energy, then the velocity variables tensorize as an independent Maxwellian factor. Consequently, the spatial modulated partition-function and entropic commutator estimates carry over verbatim for spatial observables. The diffusive mean-field convergence application below is specific to the overdamped dynamics and is not part of this kinetic extension. For a kinetic mean-field convergence result that exploits precisely this Maxwellian factorization, see the work of Duerinckx and Jabin [DJ26] discussed in Section 2.5.

2.2. **Partition-function estimates.** Our first set of results controls the normalization of the modulated Gibbs measure. The main theorem gives the fixed-temperature uniform bound and the regularized-determinant limit. The Hilbert–Schmidt threshold is sharp, as quantified after the theorem. The convergence rate in (2.23) is likewise not expected to be optimal; see Remark 2.4.

Theorem 2.2 (Repulsive logarithmic/Riesz case)**.** *Fix $\beta \geq 0$ and assume $0 \leq \mathsf{s} < \mathsf{d}/2$ and $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$; if $\mathsf{s} = 0$, assume in addition*

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} |\log|x-y||\ \mathrm{d}\mu^{\otimes 2}(x,y) < \infty. \tag{2.20}$$

(i) (Uniform bound.) *There is a finite constant $C_{\mathrm{ub}}(\beta,\mu)$ such that*

$$\sup_{N\geq 1} \mathsf{K}_{N,\beta}(\mu) \leq C_{\mathrm{ub}}(\beta,\mu). \tag{2.21}$$

(ii) (Quantitative convergence.) *If $\mathsf{s} = 0$, assume moreover that*

$$\int_{\mathbb{R}^{\mathsf{d}}} e^{-4\beta h_\mu}\ \mathrm{d}\mu < \infty, \qquad \tilde{\mathsf{g}} \in L^2(\mu^{\otimes 2}). \tag{2.22}$$

Then there are constants $C_{\mathrm{rate}}(\beta,\mu) < \infty$ and, in the logarithmic case, $c_{\log}(\beta,\mu) > 0$ such that, for all $N \geq 1$,

$$\left|\mathsf{K}_{N,\beta}(\mu) - \mathsf{K}_{\infty,\beta}(\mu)\right| \leq C_{\mathrm{rate}}(\beta,\mu) \begin{cases} (\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}, & 0 < \mathsf{s} < \mathsf{d}/2, \\ (e+N)^{-c_{\log}(\beta,\mu)}, & \mathsf{s} = 0. \end{cases} \tag{2.23}$$

The threshold sharpness is quantitative. For every $\beta > 0$ and $\mathsf{d}/2 \leq \mathsf{s} < \mathsf{d}$, Proposition 5.10 gives constants $c, C > 0$ such that, for all sufficiently large N,

$$\mathsf{K}_{N,\beta}(\mu) \geq C^{-1}(\log N)^c, \qquad \mathsf{s} = \mathsf{d}/2, \tag{2.24}$$

$$\log \mathsf{K}_{N,\beta}(\mu) \geq c(\log N)^{2-\mathsf{d}/\mathsf{s}} - C, \qquad \mathsf{d}/2 < \mathsf{s} < \mathsf{d}. \tag{2.25}$$

These explicit lower rates are not expected to be optimal; see Remark 5.11.

For $\beta > 0$, define the macroscopic free energy

$$\mathcal{F}_{\beta,V}(\mu) \coloneqq \frac{1}{\beta}\int_{\mathbb{R}^{\mathsf{d}}} (\beta V + \log \mu)\, \mathrm{d}\mu + I_\mu. \tag{2.26}$$

Here, the combined integrand is required to be integrable and I_μ finite. If μ_β satisfies the thermal Euler–Lagrange relation and has finite absolute pair energy, then

$$\int_{\mathbb{R}^{\mathsf{d}}} (\beta V + \log \mu_\beta)\, \mathrm{d}\mu_\beta = -\log Z_{\mu_\beta} - 2\beta I_{\mu_\beta}, \tag{2.27}$$

so $-\beta\mathcal{F}_{\beta,V}(\mu_\beta) = \log Z_{\mu_\beta} + \beta I_{\mu_\beta}$. Substitution into the exact reduction (2.5), followed by Theorem 2.2(ii), gives the following equilibrium consequence.

Corollary 2.3 (Equilibrium free-energy correction)**.** *Fix $\beta > 0$, and let μ_β be an absolutely continuous thermal equilibrium measure satisfying*

$$\mathrm{d}\mu_\beta = Z_{\mu_\beta}^{-1} e^{-\beta(V+h_{\mu_\beta})}\, \mathrm{d}x \tag{2.28}$$

and the hypotheses of Theorem 2.2(ii). Then the canonical partition function satisfies the exact identity

$$Z_{N,\beta} = e^{-\beta N \mathcal{F}_{\beta,V}(\mu_\beta)} \mathsf{K}_{N,\beta}(\mu_\beta), \tag{2.29}$$

and consequently

$$\log Z_{N,\beta} = -\beta N \mathcal{F}_{\beta,V}(\mu_\beta) + \beta I_{\mu_\beta} - \frac{1}{2}\log\det{}_2(\mathrm{Id} + \beta T_{\mu_\beta}) + O_{\beta,\mu_\beta}(r_{N,\mathsf{s}}), \tag{2.30}$$

where

$$r_{N,\mathsf{s}} \coloneqq \begin{cases} (\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}, & 0 < \mathsf{s} < \mathsf{d}/2, \\ (e+N)^{-c_{\log}(\beta,\mu_\beta)}, & \mathsf{s} = 0. \end{cases} \tag{2.31}$$

The entropy rewriting (2.11) also gives the exact variational identity

$$\inf_{f_N \in \mathcal{P}((\mathbb{R}^{\mathsf{d}})^N)} \mathsf{E}_N(f_N, \mu) = -\frac{1}{\beta N}\log \mathsf{K}_{N,\beta}(\mu), \tag{2.32}$$

with the infimum attained by $\mathbb{Q}_{N,\beta}(\mu)$. Thus, the determinant limit identifies the coefficient of the order-N^{-1} modulated-free-energy floor appearing in the diffusive application.

Remark 2.4 (On the convergence rate)**.** The emphasis in Theorem 2.2(ii) is on the convergence itself, valid up to the sharp threshold $\mathsf{s} < \mathsf{d}/2$; the rate in (2.23) is the one the method yields without further effort, and we have not sought to optimize it. We expect the optimal rate to be of order $1/N$ in the logarithmic case —attainable by a tailored version of Stein's method— and to be s-dependent in the range $0 < \mathsf{s} < \mathsf{d}/2$. This is the subject of ongoing work by the authors.

Remark 2.5 (On the exponential integrability of h_μ)**.** Since $\mu \in L^\infty$, the potential h_μ is always bounded above, so the first condition in (2.22) constrains only its lower tail, that is, the decay of μ.

For later use in the repulsive logarithmic/Riesz case, fix an admissible uniform partition-function constant

$$C_{\mathrm{pf}}(\beta,\mu) \coloneqq \max\{e, C_{\mathrm{ub}}(\beta,\mu)\}. \tag{2.33}$$

The dependence of admissible choices of C_{ub}, C_{pf}, the attractive constants β_0 and $C_{\mathrm{ub}}^{\mathrm{att}}$, and the later commutator constants is summarized in Appendix A.

The same fixed-temperature estimate also interpolates with the sharp pointwise lower bound for the modulated energy as the inverse temperature grows with N. The precise statement, including the specialization to power-law temperature scales and the microscopic-endpoint comparison, is recorded in Corollary 5.7 and proved in Section 5.4.

Next, we turn to the attractive logarithmic case, corresponding to replacing g with $\mathsf{g}_{\mathrm{att}} \coloneqq \log|x|$. We complete the partition-function discussion with a high-temperature, or small-inverse-temperature, uniform bound and a quantitative determinant limit, proved in Section 6; the superscript att denotes the corresponding quantities defined above with g replaced by $\mathsf{g}_{\mathrm{att}}$. The restriction to the logarithmic kernel is essential. For the attractive Riesz sign, i.e. the kernel $-\mathsf{g}$ with $0 < \mathsf{s} < \mathsf{d}$, the modulated partition function is infinite for every $\beta > 0$ and every $N \geq 2$ whenever μ has positive density on some open set, since the near-diagonal factor $e^{c|x-y|^{-\mathsf{s}}}$, $c > 0$, is not locally integrable. The attractive logarithmic weight is instead only polynomial, of order $|x-y|^{-c\beta}$ near the diagonal, and this is precisely the integrability that the smallness $\beta < \beta_0$ exploits.

Theorem 2.6 (Attractive logarithmic case, small β)**.** *Let $\mu \in \mathcal{P}(\mathbb{R}^{\mathsf{d}})$ be such that $\mathsf{g}_{\mathrm{att}} \in L^1(\mu^{\otimes 2})$, and assume that there is a constant $C_{\mathrm{mom}} < \infty$ such that*

$$\sup_{y\in\mathbb{R}^{\mathsf{d}}} \int_{\mathbb{R}^{\mathsf{d}}} |\tilde{\mathsf{g}}_{\mathrm{att}}(x,y)|^p \, \mathrm{d}\mu(x) \leq C_{\mathrm{mom}}^p \, p! \qquad \text{for every } p \geq 1. \tag{2.34}$$

Then there is a universal constant $C_* < \infty$, the constant of the moment bound (6.5) *in the proof, such that, setting*

$$\beta_0 \coloneqq \frac{1}{C_* C_{\mathrm{mom}}}, \tag{2.35}$$

for every fixed $0 \leq \beta < \beta_0$ with $e^{2\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$, there is a finite constant $C_{\mathrm{ub}}^{\mathrm{att}}(\beta,\mu)$ satisfying

$$\sup_{N\geq 1} \mathsf{K}_{N,\beta}^{\mathrm{att}}(\mu) \leq C_{\mathrm{ub}}^{\mathrm{att}}(\beta,\mu). \tag{2.36}$$

Moreover, assume $\mu \in L^\infty(\mathbb{R}^{\mathsf{d}})$, let T_μ^{att} be the Hilbert–Schmidt operator with kernel $\tilde{\mathsf{g}}_{\mathrm{att}}$, and set $\mathsf{K}_{\infty,\beta}^{\mathrm{att}}(\mu) \coloneqq e^{\beta I_\mu^{\mathrm{att}}} \det_2\big(\mathrm{Id} + \beta T_\mu^{\mathrm{att}}\big)^{-1/2}$. There is a threshold

$$0 < \beta_{\mathrm{q}} = \beta_{\mathrm{q}}(\mathsf{d}, \|\mu\|_{L^\infty}, C_{\mathrm{mom}}) \leq \frac{\beta_0}{2} \tag{2.37}$$

such that, for every fixed $0 < \beta < \beta_{\mathrm{q}}$ with $e^{4\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$, there are constants $C_{\mathrm{rate}}^{\mathrm{att}}(\beta,\mu) < \infty$ and $c_{\mathrm{att}}(\beta,\mu) > 0$ satisfying

$$\Big|\mathsf{K}_{N,\beta}^{\mathrm{att}}(\mu) - \mathsf{K}_{\infty,\beta}^{\mathrm{att}}(\mu)\Big| \leq C_{\mathrm{rate}}^{\mathrm{att}}(\beta,\mu)(e+N)^{-c_{\mathrm{att}}(\beta,\mu)} \tag{2.38}$$

for every $N \geq 1$. The determinant defining $\mathsf{K}_{\infty,\beta}^{\mathrm{att}}(\mu)$ is positive.

For a periodic uniform background, the one-body term vanishes and the additional exponential-integrability hypothesis in the quantitative assertion is automatic.

Remark 2.7 (On the perturbative threshold and periodic uniform backgrounds)**.** The sufficient threshold β_0 furnished by the preceding factorial-moment argument is not claimed to be sharp and should not be identified with the global-minimization, stability, or collision thresholds of the attractive logarithmic gas. For the periodic logarithmic interaction with uniform background in dimensions $\mathsf{d} \geq 2$, the authors prove in forthcoming work [DGR26a] that the macroscopic log-HLS inequality yields the Carleman–Fredholm determinant limit throughout the strict global-minimization regime. Combined with the sharp inequalities of Lin and Lucia in dimension two [LL07] and of Lei and the third author in dimensions $\mathsf{d} \geq 3$ [LR26], this gives the limit for $0 \leq \beta < 2\mathsf{d}$ in dimensions $2 \leq \mathsf{d} \leq 10$ and for $0 \leq \beta < \beta_{\mathrm{s}}(\mathsf{d})$ in dimensions $\mathsf{d} \geq 12$. In dimension eleven, the limit holds for $0 \leq \beta < \beta_{\mathrm{gm}}(11) < \beta_{\mathrm{s}}(11)$, although the exact value of $\beta_{\mathrm{gm}}(11)$ remains unknown. The one-dimensional case follows from the exact circle formula in Section 6.3, where the detailed phase diagram and the remaining conjectural asymptotics are recorded.

The moment hypothesis (2.34) keeps the supremum over the second variable outside the x-integration. It is therefore weaker than

$$\int_{\mathbb{R}^{\mathsf{d}}} \left(\sup_{y \in \mathbb{R}^{\mathsf{d}}} |\tilde{\mathsf{g}}_{\mathrm{att}}(x,y)| \right)^p \, \mathrm{d}\mu(x) \leq C^p p!, \tag{2.39}$$

which takes the pointwise supremum before integrating and is the norm considered by Jabin–Wang [JW18]; that stronger norm excludes the Keller–Segel kernel.

The exponential-integrability condition mirrors Remark 2.5, with the opposite sign. Since the attractive potential grows at infinity, it constrains the upper tail of h_μ^{att}, and hence the decay of μ. Indeed, $\mathsf{g}_{\mathrm{att}} \in L^1(\mu^{\otimes 2})$ implies $\int_{\mathbb{R}^{\mathsf{d}}} \log(1+|y|) \, \mathrm{d}\mu(y) < \infty$, while

$$h_\mu^{\mathrm{att}}(x) \leq \log(1+|x|) + \int_{\mathbb{R}^{\mathsf{d}}} \log(1+|y|) \, \mathrm{d}\mu(y). \tag{2.40}$$

Consequently, $e^{2\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$ whenever $\int_{\mathbb{R}^{\mathsf{d}}} |x|^{2\beta} \, \mathrm{d}\mu(x) < \infty$. The local singularity, which may drive h_μ^{att} to $-\infty$, enters with the favorable sign.

The proof of the uniform bound and determinant limit in Theorem 2.2 begins with the canonical statistic (2.17). Positive-definite truncation and a low/high-frequency decomposition reduce the singular kernel to data controlled by the master layer-cake estimate Proposition 3.5; the high-frequency L^2, mixed, and operator norms are small precisely when $\mathsf{s} < \mathsf{d}/2$. The logarithmic case uses the same mechanism after the infrared part has been paired with the one-body term through the cancellation in Lemma 4.3. The resulting uniform exponential moments are proved in Proposition 5.1.

The determinant limit then follows by Hilbert–Schmidt approximation and the bounded-feature Laplace estimate Proposition 5.4, while Lemma 5.6 shows that the centered one-body average contributes a lower-order error. Beyond the Hilbert–Schmidt threshold, Lemma 5.9 replaces the centered kernel by its conditional expectation on a dyadic spatial partition whose scale is coupled to N. Conditional Jensen reduces the original partition function to the resulting finite-rank coarse-grained kernel, whose Hilbert–Schmidt norm diverges; the quantitative finite-particle estimate in Proposition 5.4 then gives the explicit lower rates in Proposition 5.10.

The attractive logarithmic theorem follows a genuinely different perturbative route, since positivity of the interaction is no longer available. After the one-body factor is separated, centering forces the surviving moments of the pair statistic to involve repeated indices; decoupling by Proposition 3.3 and the factorial-moment hypothesis (2.34) make the resulting exponential series summable for $\beta < \beta_0$. A feature-space expansion for the positive quadratic exponential, combined with logarithmic truncation, gives the quantitative determinant convergence in (2.38).

2.3. **Entropic commutators and diffusive mean-field estimates.** Our second set of results concerns so-called commutator estimates for the variations of the modulated energy along transport fields. Let us briefly explain the context and motivation for these estimates.

The modulated energy approach to mean-field convergence consists of establishing a Gronwall relation for $\mathsf{F}_N(X_N^t, \mu^t)$, where X_N^t is a solution of the microscopic system and μ^t is a solution of the limiting mean-field equation. An essential point in this approach is to control the derivative of F_N along a transport $v : \mathbb{R}^{\mathsf{d}} \to \mathbb{R}^{\mathsf{d}}$. In the modulated-free-energy method, this same derivative is the transport commutator left after the relative Fisher-information term has been put in favorable sign.

For applications to the mean-field limit, v is the velocity field of the limiting mean-field dynamics. For applications to central limit theorems (CLTs) for the fluctuations (the next-order description), v is the gradient of a test function evolved along the adjoint linearized mean-field flow, and these inequalities are at the core of the "transport" approach to fluctuations of canonical Gibbs ensembles [LS18, Ser23, PS25]. The same quantity also appears in the loop, or Dyson–Schwinger, equations in the random matrix theory literature [BG13, BG24, BGK15]. These identities are obtained by

differentiating the Gibbs integral under infinitesimal changes of variables and thus admit a transport interpretation.

The higher-order commutators below are the corresponding higher variations of the modulated energy. As in the first-order case, deterministic commutator estimates seek to control these quantities by $C_v(\mathsf{F}_N(X_N,\mu)+C_\mu N^{-\alpha})$ [RS26, HCRS25]. Second-order estimates are important for the aforementioned transport method for studying the fluctuations of Coulomb/Riesz gases. Estimates beyond second order are also useful, allowing one to obtain finer estimates on the fluctuations of Riesz gases to treat a broader class of interactions [PS25] and also to compute the asymptotics of arbitrary-order cumulants [Ros].

We now use the partition-function control to prove the averaged, entropic counterpart of these deterministic commutator bounds when the point configuration is distributed according to the modulated Gibbs measure. The main point is to prove a uniform unnormalized exponential-moment bound under $\mu^{\otimes N}$, then convert it by Donsker–Varadhan into an entropic estimate for N-particle laws.

For a Lipschitz transport $v:\mathbb{R}^{\mathsf{d}}\to\mathbb{R}^{\mathsf{d}}$ and an integer $n\geq 1$, define

$$\mathsf{A}_n(v,X_N,\mu)\coloneqq\int_{(\mathbb{R}^{\mathsf{d}})^2\setminus\triangle}\nabla^{\otimes n}\mathsf{g}(x-y):(v(x)-v(y))^{\otimes n}\,\mathrm{d}(\mu_N-\mu)^{\otimes 2}(x,y). \tag{2.41}$$

Since $\mathsf{A}_n(-v,X_N,\mu)=(-1)^n\mathsf{A}_n(v,X_N,\mu)$, a sign can be absorbed into v only when n is odd. We therefore keep the explicit sign parameter $\sigma\in\{\pm1\}$ in the all-order estimate.

Theorem 2.8. *Fix an integer $n\geq 1$. Assume $0\leq\mathsf{s}<\mathsf{d}/2$, $\mu\in L^\infty\cap\mathcal{P}(\mathbb{R}^{\mathsf{d}})$, and $v:\mathbb{R}^{\mathsf{d}}\to\mathbb{R}^{\mathsf{d}}$ is globally Lipschitz; if $\mathsf{s}=0$, assume in addition (2.20). Then there exist a threshold $\beta_{\mathrm{loc}}=\beta_{\mathrm{loc}}(\mu,v,n)>0$ and, for every $\beta>0$, constants $\varepsilon_{\beta,n}>0$ and $C_{\mathrm{em}}(\beta,\mu,v,n)<\infty$, made explicit in Remark A.1, such that, for every $\sigma\in\{\pm1\}$,*

$$\sup_{N\geq1}\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu)}\Big]\leq C_{\mathrm{em}}(\beta,\mu,v,n). \tag{2.42}$$

Consequently, for every $f_N\in\mathcal{P}((\mathbb{R}^{\mathsf{d}})^N)$ with finite relative entropy with respect to $\mathbb{Q}_{N,\beta}(\mu)$, the observable $N\mathsf{A}_n(v,\cdot,\mu)$ is f_N-integrable and

$$\Big|\mathbb{E}_{X_N\sim f_N}\Big[N\mathsf{A}_n(v,X_N,\mu)\Big]\Big|\leq C_{\beta,n,v}N\mathsf{E}_N(f_N,\mu)+C_{\beta,\mu,v,n}, \tag{2.43}$$

where E_N is the modulated free energy (2.10) and

$$C_{\beta,n,v}\coloneqq\frac{\beta}{\varepsilon_{\beta,n}}\qquad\text{and}\qquad C_{\beta,\mu,v,n}\coloneqq\frac{1}{\varepsilon_{\beta,n}}(\log C_{\mathrm{pf}}(\beta,\mu)+\log C_{\mathrm{em}}(\beta,\mu,v,n)). \tag{2.44}$$

Although this dependence is suppressed in the notation, β_{loc}, $\varepsilon_{\beta,n}$, and hence $C_{\beta,n,v}$ depend on the background μ as well as on β, v, and n. We write $C_{\beta,n,v}(\mu)$ when this dependence must be displayed; the detailed dependence of the exponential-moment and entropic constants is recorded in Appendix A.

Diffusive mean-field application. We now turn to the application to mean-field convergence and propagation of chaos for interacting systems of diffusive gradient flows. Consider the overdamped N-particle diffusion

$$\mathrm{d}x_i^t=-\frac{1}{N}\sum_{\substack{1\leq j\leq N\\ j\neq i}}\nabla\mathsf{g}(x_i^t-x_j^t)\,\mathrm{d}t+\sqrt{\frac{2}{\beta}}\,\mathrm{d}B_i^t,\qquad i\in[N], \tag{2.45}$$

where $B_1,\dots,B_N$ are independent Brownian motions.

Establishing the mean-field limit means proving convergence, as $N\to\infty$, of the empirical measure $\mu_N^t\coloneqq N^{-1}\sum_{i=1}^N\delta_{x_i^t}$ associated to a solution of (2.45). Assuming the initial empirical measures

converge to a sufficiently regular density μ^0, a formal Itô calculation leads one to expect that, for $t > 0$, μ_N^t converges to the solution of the McKean–Vlasov equation

$$\partial_t \mu^t = \frac{1}{\beta}\Delta\mu^t + \operatorname{div}\Big(\mu^t \nabla h_{\mu^t}\Big), \qquad h_{\mu^t} = \mathsf{g} * \mu^t, \tag{2.46}$$

which is completely deterministic, the noise having been averaged out to become diffusion.

Convergence of the empirical measure is qualitatively equivalent to propagation of molecular chaos for exchangeable particle laws; we refer the reader to [Szn91, HM14] for classical treatments and [CD22a, CD22b] for modern reviews. This latter notion means that, if the initial law f_N^0 is μ^0-chaotic, in the sense that the k-point marginals $f_{N;k}^0 \rightharpoonup (\mu^0)^{\otimes k}$ as $N \to \infty$ for every fixed k, then the evolved law f_N^t is μ^t-chaotic.

For positive-temperature gradient dynamics, the modulated energy combines naturally with relative entropy in the form of the modulated free energy[2] [BJW20, BJW19, BJW23, RS25, CdCRS25b]. More generally, the modulated-free-energy method combines two complementary controls: relative entropy controls the marginals, while the modulated energy controls the empirical measure. Differentiating $\mathsf{E}_N(f_N^t, \mu^t)$ along the N-particle flow and along (2.46) produces a relative Fisher-information term with favorable sign and a transport commutator. For the singular logarithmic and Riesz interactions considered here, the principal analytic input is the control of this commutator furnished by Theorem 2.8; the corresponding deterministic first-variation estimates are developed in [HCRS25, Ros25]. The Fisher-information term is discarded below, while the commutator is exactly the $n = 1$ case of Theorem 2.8. For the repulsive logarithmic interaction, the first two authors showed that the commutator contribution in the modulated-free-energy differential inequality is only $O(N^{-1})$ after time integration, eliminating the extra factor of $\log N$ present in cruder deterministic estimates [DG25]. The present theorem proves the corresponding conclusion in the locally square-integrable repulsive log/Riesz regime using the partition-function and entropic-commutator estimates developed in this paper.

Theorem 2.9 (Diffusive mean-field convergence)**.** *Fix $T_* \in (0, \infty]$ and $\beta > 0$. Let $(\mu^t)_{0 \le t < T_*}$ solve* (2.46)*, and assume that this solution admits a globally*[3] *Lipschitz velocity field u^t satisfying*

$$\partial_t \mu^t + \operatorname{div}(\mu^t u^t) = 0. \tag{2.47}$$

Let $(f_N^t)_{0 \le t < T_*}$ be a curve of N-particle probability laws and write

$$\mathsf{E}_N^t \coloneqq \mathsf{E}_N(f_N^t, \mu^t). \tag{2.48}$$

Assume that $t \mapsto \mathsf{E}_N^t$ is absolutely continuous and that the pair (f_N^t, μ^t) satisfies the modulated-free-energy dissipation inequality

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{E}_N^t \le \frac{1}{N}\Big|\mathbb{E}_{X_N \sim f_N^t}\Big[N\mathsf{A}_1(u^t, X_N, \mu^t)\Big]\Big|, \qquad \text{a.e. } t \in (0, T_*). \tag{2.49}$$

Assume also that, for a.e. $t \in (0, T_*)$, the hypotheses of Theorem 2.8 are satisfied with $n = 1$, $\mu = \mu^t$, and $v = u^t$, at the fixed inverse temperature β.

Set

$$\mathcal{A}^t \coloneqq \exp\Big(\int_0^t C_{\beta,1,u^\tau}(\mu^\tau)\, \mathrm{d}\tau\Big), \tag{2.50}$$

$$\mathcal{B}^t \coloneqq \int_0^t \exp\Big(\int_\tau^t C_{\beta,1,u^r}(\mu^r)\, \mathrm{d}r\Big) C_{\beta,\mu^\tau,u^\tau,1}\, \mathrm{d}\tau, \tag{2.51}$$

[2] In the sub-Coulomb Riesz regime, the mean-field limit can also be treated by the modulated energy alone, as shown in [RS23]. We use the modulated-free-energy formulation here because it is the setting in which the partition-function and entropic-commutator estimates enter.

[3] For sufficiently regular solutions, this velocity is $u^t = -\nabla h_{\mu^t} - \beta^{-1}\nabla \log \mu^t$.

where the constants are those of Theorem 2.8. If these quantities are finite for every $t < T_*$, then

$$\mathsf{E}_N(f_N^t, \mu^t) \le \mathcal{A}^t \mathsf{E}_N(f_N^0, \mu^0) + \frac{\mathcal{B}^t}{N}, \qquad 0 \le t < T_*. \tag{2.52}$$

The $O(1/N)$ scale in the estimate (2.52) is the iid-chaotic scale. The diagonal-exclusion correction in the modulated energy is already of order N^{-1}, and one should not expect a generally smaller additive floor in this topology. This does not contradict the sharpness statement in the work of Chodron de Courcel, Serfaty, and the third author [CdCRS25b]. Their $N^{\mathsf{s}/\mathsf{d}-1}$ scale is sharp when the temperature is allowed to vanish with N, or when the initial randomization already produces a modulated free energy of order $N^{\mathsf{s}/\mathsf{d}-1}$, which is the case in the usual log/Riesz gas temperature scaling or the corresponding low-temperature regime; see, e.g., [Ser26, Section 3.2, especially (3.2.4)]. In the genuinely diffusive regime considered here, with fixed positive noise and iid-scale initial modulated free energy, the additive error is of the same order as the initial error.

Remark 2.10 (Uniform-in-time propagation and generation)**.** Evidently, the same estimate is uniform in time if the coefficients generated by the full velocity $u^t = -\nabla h_{\mu^t} - \beta^{-1}\nabla \log \mu^t$ are integrable strongly enough that $\sup_{t\ge 0} \mathcal{A}^t < \infty$ and $\sup_{t\ge 0} \mathcal{B}^t < \infty$. This is the abstract form of the usual uniform-in-time propagation-of-chaos mechanism in the modulated-free-energy method. If, in addition, the nonpositive Fisher-information term in the dissipation identity is retained and a uniform modulated logarithmic Sobolev inequality for $\mathbb{Q}_{N,\beta}(\mu^t)$ is available, then the same argument gives generation of chaos down to the $O(1/N)$ diffusive floor; this is the mechanism emphasized in [RS25]. For the attractive logarithmic gas, forthcoming work [DGR26b] develops the modulated-free-energy approach in the periodic and quadratically confined planar settings, including uniform-in-time estimates near equilibrium.

The proof of Theorem 2.8 uses the same exponential-moment mechanism as the partition-function theorem with the transport variation inserted. The local numerator estimate Lemma 7.2 absorbs the short-range commutator into the modulated energy, and the Hölder bootstrap in Section 7.4 removes the small-inverse-temperature restriction by invoking the partition-function bound at a larger inverse temperature. The Donsker–Varadhan formula then gives the entropic inequality. Inserting its first-order case into the modulated-free-energy dissipation inequality and applying Gronwall's lemma proves Theorem 2.9. The corresponding interpolation between the fixed-temperature entropic estimate and the deterministic pointwise estimate is stated and proved in Corollary 7.3.

2.4. **Static and dynamical fluctuations.** We first record the static fluctuation consequence of the partition-function convergence. For a real-valued test function $\phi \in L^2(\mu)$, set

$$S_N(\phi) \coloneqq \sqrt{N}\Big(\int_{\mathbb{R}^{\mathsf{d}}} \phi \, \mathrm{d}\mu_N - \int_{\mathbb{R}^{\mathsf{d}}} \phi \, \mathrm{d}\mu\Big), \qquad \bar\phi \coloneqq \phi - \int_{\mathbb{R}^{\mathsf{d}}} \phi \, \mathrm{d}\mu, \tag{2.53}$$

and define

$$\Sigma_{\beta,\mu}(\phi,\psi) \coloneqq \Big\langle \bar\phi, (\mathrm{Id} + \beta T_\mu)^{-1}\bar\psi \Big\rangle_{L^2(\mu)}. \tag{2.54}$$

Since T_μ is bounded, nonnegative, and self-adjoint on $L^2_0(\mu)$, the covariance form $\Sigma_{\beta,\mu}$ is well defined and continuous on $L^2(\mu) \times L^2(\mu)$.

Proposition 2.11 (Static linear-statistics CLT)**.** *Assume the hypotheses of Theorem 2.2, including* (2.22) *when* $\mathsf{s} = 0$*. Fix $\beta \ge 0$ and real-valued functions $\phi_1, \dots, \phi_k \in L^2(\mu)$.*

Then, for every $\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k$,

$$\lim_{N\to\infty} \mathbb{E}_{X_N \sim \mathbb{Q}_{N,\beta}(\mu)}\left[\exp\left(\mathrm{i}\sum_{i=1}^k \theta_i S_N(\phi_i)\right)\right] = \exp\left(-\frac{1}{2}\sum_{i,j=1}^k \theta_i\theta_j \Sigma_{\beta,\mu}(\phi_i,\phi_j)\right). \tag{2.55}$$

Consequently, $(S_N(\phi_1), \dots, S_N(\phi_k))$ *converges in law under* $\mathbb{Q}_{N,\beta}(\mu)$ *to a centered Gaussian vector with covariance matrix* $(\Sigma_{\beta,\mu}(\phi_i, \phi_j))_{i,j=1}^k$.

If, in addition, $\phi_1, \dots, \phi_k \in L^\infty(\mu)$*, then, for every* $\theta \in \mathbb{R}^k$*,*

$$\lim_{N\to\infty} \mathbb{E}_{X_N \sim \mathbb{Q}_{N,\beta}(\mu)} \left[\exp\left(\sum_{i=1}^k \theta_i S_N(\phi_i) \right) \right] = \exp\left(\frac{1}{2} \sum_{i,j=1}^k \theta_i \theta_j \Sigma_{\beta,\mu}(\phi_i, \phi_j) \right). \tag{2.56}$$

Remark 2.12 (Equilibrium and dynamical interpretation)**.** When $\mu = \mu_\beta$ is the thermal equilibrium law, $\mathbb{Q}_{N,\beta}(\mu_\beta) = \mathbb{P}_{N,\beta}$, so Proposition 2.11 is a central limit theorem for the original canonical ensemble. The $L^2(\mu)$ test class is the natural and essentially optimal class for a general $\sqrt{N}$-scale Gaussian theorem with the covariance above. Indeed, since T_μ is nonnegative,

$$\frac{1}{1 + \beta \|T_\mu\|_{\mathrm{op}}} \|\bar{\phi}\|^2_{L^2(\mu)} \le \Sigma_{\beta,\mu}(\phi, \phi) \le \|\bar{\phi}\|^2_{L^2(\mu)}. \tag{2.57}$$

Thus, the covariance norm is equivalent to the centered $L^2(\mu)$ norm. The boundedness assumption in the final assertion is used only to justify convergence of the two-sided moment generating functions. The proposition also identifies the initial Gibbs characteristic function entering the dynamical central limit theorem below.

We now turn to the fluctuation process associated with (2.45)–(2.46). For a bounded test ϕ, write

$$S_N^t(\phi) := \sqrt{N} \Big(\int_{\mathbb{R}^{\mathsf{d}}} \phi \, \mathrm{d}\mu_N^t - \int_{\mathbb{R}^{\mathsf{d}}} \phi \, \mathrm{d}\mu^t \Big). \tag{2.58}$$

For $\nu \in \mathcal{P}(\mathbb{R}^{\mathsf{d}})$, define the adjoint linearized operator on C^2 test functions at points x where the integral below is absolutely convergent,

$$(\mathcal{L}_{\beta,\nu}\phi)(x) := \frac{1}{\beta}\Delta\phi(x) - \int_{\mathbb{R}^{\mathsf{d}} \setminus \{x\}} \nabla \mathsf{g}(x-y) \cdot (\nabla\phi(x) - \nabla\phi(y)) \, \mathrm{d}\nu(y). \tag{2.59}$$

Theorem 2.13 (Dynamical linear-statistics CLT)**.** *Let* $0 \le \mathsf{s} < \mathsf{d}/2$*,* $\beta > 0$*, and* $T > 0$*. Let* $(\mu^t)_{0\le t\le T}$ *be a mean-field trajectory and* $(f_N^t)_{0\le t\le T}$ *the corresponding particle laws. Assume the following.*

(i) The particle system and mean-field equation are well posed in a class for which the time-dependent Itô formulas used below are valid; the Brownian motions are independent of the initial configuration; $f_N^0 = \mathbb{Q}_{N,\beta}(\mu^0)$*; and the estimate* (2.52) *holds. Moreover,*

$$\sup_{0\le t\le T} \Big(\|\mu^t\|_{L^\infty} + \mathcal{A}^t + \mathcal{B}^t \Big) < \infty. \tag{2.60}$$

If $\mathsf{s} = 0$*, assume in addition that*

$$\inf_{0\le t\le T} I_{\mu^t} > -\infty. \tag{2.61}$$

(ii) There are constants $0 < c_{\beta,T} \le C_{\beta,T} < \infty$ *and* $C_{2\beta,T} < \infty$ *such that*

$$\inf_{N\ge 1} \inf_{0\le t\le T} \mathsf{K}_{N,\beta}(\mu^t) \ge c_{\beta,T}, \qquad \sup_{N\ge 1} \sup_{0\le t\le T} \mathsf{K}_{N,\beta}(\mu^t) \le C_{\beta,T}, \tag{2.62}$$

$$\sup_{N\ge 1} \sup_{0\le t\le T} \mathsf{K}_{N,2\beta}(\mu^t) \le C_{2\beta,T}, \tag{2.63}$$

and μ^0 *satisfies the hypotheses of Proposition 2.11.*

(iii) There is a real linear test space $\mathscr{D}_T \subset C_b^2(\mathbb{R}^{\mathsf{d}})$ *such that, for every* $t \in [0,T]$ *and* $\phi \in \mathscr{D}_T$*, the terminal-value problem*

$$-\partial_r f^r = \mathcal{L}_{\beta,\mu^r} f^r, \qquad 0 \le r \le t, \qquad f^t = \phi, \tag{2.64}$$

has a unique real-valued classical solution, denoted by $f^r = \mathsf{U}^{r,t}\phi$*, for which* f*,* ∇f*,* $\nabla^{\otimes 2} f$*, and* $\partial_r f$ *are jointly continuous and*

$$\sup_{0\le r\le t}\Big(\|f^r\|_{L^\infty} + \|\nabla f^r\|_{L^\infty} + \|\nabla^{\otimes 2} f^r\|_{L^\infty} + \|\partial_r f^r\|_{L^\infty}\Big) < \infty. \tag{2.65}$$

The bound in (2.65) *is uniform over each fixed finite family of terminal pairs* (t,ϕ)*.*

Fix $k \ge 1$*, times* $t_1,\dots,t_k \in [0,T]$*, and tests* $\phi_1,\dots,\phi_k \in \mathscr{D}_T$*. Then*

$$\Big(S_N^{t_1}(\phi_1),\dots,S_N^{t_k}(\phi_k)\Big) \Longrightarrow (G_1,\dots,G_k), \tag{2.66}$$

where the vector on the right is centered Gaussian with covariance

$$\begin{aligned}\mathbb{E}[G_a G_b] &= \Sigma_{\beta,\mu^0}\Big(\mathsf{U}^{0,t_a}\phi_a, \mathsf{U}^{0,t_b}\phi_b\Big)\\ &\quad + \frac{2}{\beta}\int_0^{t_a\wedge t_b}\int_{\mathbb{R}^{\mathsf{d}}} \nabla \mathsf{U}^{r,t_a}\phi_a \cdot \nabla \mathsf{U}^{r,t_b}\phi_b \,\mathrm{d}\mu^r \,\mathrm{d}r.\end{aligned} \tag{2.67}$$

The uniform background controls in hypothesis (i), the partition-function bounds in hypothesis (ii), and the Hessian bound in hypothesis (iii) imply the family-uniform exponential commutator estimate (8.27) used below; see Remark A.1 and Lemma 8.1.

Remark 2.14 (Backward adjoint regularity)**.** The backward-adjoint regularity required in hypothesis (iii) of Theorem 2.13 is proved in the forthcoming work of Huang, Serfaty, and the third author [HRS].

Remark 2.15 (Toward a path-space central limit theorem)**.** Theorem 2.13 identifies the finite-dimensional distributions of any possible path-space limit. Consequently, if the fluctuation fields are tight in a separable continuous-path space in which the admissible tests form a measure-determining class, and if the Gaussian process characterized by (2.67) has trajectories in that space, then every subsequential limit has the same law. Thus, the remaining step toward a path-space central limit theorem is the corresponding tightness and trajectory-regularity estimate.

For the static theorem, the linear tilt is first combined with the joint Gaussian/degenerate-chaos limit and the determinant normalization to prove the bounded-test Laplace-transform formula. A truncation argument, using a uniform $L^2(\mu^{\otimes N})$ bound on the density of $\mathbb{Q}_{N,\beta}(\mu)$ relative to $\mu^{\otimes N}$, then extends the weak central limit theorem to arbitrary $L^2(\mu)$ tests; see Section 8.1. For the dynamical theorem, the backward adjoint equation cancels the linearized drift and the finite-N identity Proposition 8.2 expresses each terminal statistic as an initial statistic, a martingale, and a transport-commutator remainder. The static central limit theorem identifies the initial term, while Lemma 8.1, using the family-uniform form of the entropic commutator estimate, makes the remainder negligible. A characteristic-function argument, together with convergence of the martingales' predictable quadratic covariations to their deterministic mean-field limits, gives the time-integrated part of the covariance in (2.67).

2.5. **Relation to previous work.** There is a large literature on full Gibbs partition functions for logarithmic, Coulomb, and Riesz gases. At the macroscopic level, first-order free-energy asymptotics and empirical-measure large deviations identify the mean-field variational problem and its equilibrium measures; at the next orders, microscopic screening, renormalized energies, empirical-field large deviations, local laws, rigidity, and Coulomb/Riesz free-energy expansions enter the description. For a review of this body of results, see Serfaty's lecture notes [Ser26, Chapters 3, 5, and 8–13]. A further distinction is the temperature scale. In many of the microscopic equilibrium results, the relevant inverse-temperature scale is the one at which nearest-neighbor Riesz costs enter the Gibbs weight; in the present normalization this means $\beta = \beta_N$ of order $N^{1-\mathsf{s}/\mathsf{d}}$, and $\beta_N \sim N$ in the logarithmic case. The present paper instead works in the fixed-temperature diffusive regime $\beta_N = O(1)$, which is a high-temperature regime from that perspective. Those results are substantially more refined at the

microscopic level than the present work. They concern the full confined ensemble and its equilibrium structure. On the other hand, these results do not imply those of the present paper because of the differences in temperature regimes. In the present paper, the singularity of the interaction is not controlled through the screening and renormalized-energy mechanisms that describe microscopic equilibrium structure. The residual normalization is instead a high-temperature fluctuation problem for the centered two-body statistic, and its sharp behavior is governed by the Hilbert–Schmidt threshold and the Carleman–Fredholm determinant.

Central limit theorems for equilibrium linear statistics of logarithmic, Coulomb, and Riesz gases have been obtained in several complementary temperature and spatial regimes. At the diffusive, or high-temperature, scale considered here, one-dimensional logarithmic ensembles were treated for circular and real beta ensembles in [HL21, NT18, DGM24, MM25], generally for smooth or polynomial test functions and, in some cases, with quantitative normal-approximation estimates. At stronger, non-diffusive interaction scales, quantitative or mesoscopic central limit theorems for one-dimensional beta ensembles were proved in [BLS18, LLW19, BL18, Lam21, BMP22], while two-dimensional Coulomb fluctuations at the conventional interaction scale were treated at macroscopic and mesoscopic scales in [LS18, BBNY19]. Serfaty's fluctuation and free-energy theory includes broad, possibly particle-number-dependent temperature regimes, including the diffusive regime in dimension two; the corresponding higher-dimensional central limit theorem is conditional on a no-phase-transition assumption and a sufficiently accurate free-energy expansion [Ser23]. Recent work of Peilen and Serfaty treats super-Coulombic Riesz gases in arbitrary dimension, in a restricted range of inverse powers and inverse temperatures, and obtains a central limit theorem at small mesoscopic scales [PS25]. Boursier's fixed-temperature circular Riesz theorem is a particularly relevant low-regularity result, covering interval counts and certain algebraically singular test functions [Bou21]. By contrast, Proposition 2.11 applies at the diffusive scale, for arbitrary prescribed bounded backgrounds satisfying the partition-function hypotheses, to any fixed finite family of real-valued square-integrable test functions throughout the logarithmic/Riesz Hilbert–Schmidt range. For bounded tests, it also gives convergence of the two-sided moment generating functions for every real tilt. It does not address particle-number-dependent mesoscopic tests, convergence of the full spatial fluctuation field, or a quantitative normal-approximation rate.

A separate literature in one dimension gives all-order expansions of log-gas and beta-ensemble partition functions by loop equations and topological recursion; see, for example, [BG13, BG24, BGK15, BLS18, DG24]. We emphasize that these results use the special structure of the one-dimensional logarithmic potential, which does not carry over to higher dimensions or to Riesz interactions.

The dynamical motivation comes from the modulated-energy and modulated-free-energy approach to singular mean-field limits. The method was developed for Coulomb/Riesz and related singular interactions in [Ser20, Due16, NRS22, RS23, JW18, BJW20, BJW23], and the modulated Gibbs viewpoint was used in [RS25] to connect propagation and generation of chaos with uniform modulated logarithmic Sobolev inequalities. In this framework, (2.11) shows that $\log \mathsf{K}_{N,\beta}(\mu)$ is the exact defect in changing the reference entropy from $\mu^{\otimes N}$ to the modulated Gibbs measure. The present paper supplies the corresponding order-one partition-function estimate in the locally square-integrable repulsive log/Riesz regime.

Fluctuation-process central limit theorems for regular mean-field particle systems are classical; see, for instance, [ST85, FM97], and [Due21] for a modern, quantitative perspective. For singular diffusions, Wang, Zhao, and Zhu prove convergence of the fluctuation process to a generalized Ornstein–Uhlenbeck process, with the two-dimensional Biot–Savart law as a principal application, while Cecchin and Nikolaev obtain quantitative weak convergence in negative Sobolev spaces and treat both viscous-vortex and repulsive-Coulomb models [WZZ23, CN25]. Duerinckx and Jabin establish correlation estimates and a fixed-time central limit theorem for Langevin particle systems with square-integrable singular interactions; their analysis is presented for underdamped dynamics,

while the corresponding correlation estimates extend, with minor adaptations, to overdamped systems [DJ25, Remark 1.2]. Their hierarchical argument does not, however, identify time correlations along trajectories. In the special one-dimensional logarithmic setting, global fluctuation results for Dyson dynamics were obtained by Bender [Ben08] and Unterberger [Unt18], while Nakano, Trinh, and Trinh [NTT23] prove process-level central limit theorems for moment observables of high-temperature beta-Dyson and beta-Laguerre systems. Theorem 2.13 differs in starting from a correlated modulated Gibbs law and in treating the full Hilbert–Schmidt logarithmic/Riesz class under the stated mean-field and backward-adjoint assumptions. It identifies joint linear statistics at finitely many times, and hence their time correlations, but does not provide a quantitative convergence rate or path-space tightness. Once the static fluctuation theorem and the entropic commutator estimate are available, the final probabilistic stage of the proof is comparatively direct and elementary: the backward adjoint equation cancels the linearized drift, entropy control removes the commutator remainder, and a finite-dimensional martingale characteristic-function argument identifies the covariance.

Several works are closer at the level of partition functions. Ben Arous and Brunaud treated the regular case of bounded continuous mean-field interactions by Laplace-method arguments, obtaining nondegenerate Gaussian fluctuations and qualitative limits involving the Carleman–Fredholm determinant for the modulated partition function [BAB90]. The present theorem can be viewed as the singular log/Riesz analogue with the sharp Hilbert–Schmidt threshold made explicit. An earlier logarithmic equilibrium predecessor is the work of Grotto and Romito [GR20], who treat the two-dimensional case on the flat torus with uniform background. They prove convergence of the corresponding partition functions and Gaussian fluctuations. Although their paper is written in the two-dimensional setting, the argument may be generalized, with some modifications, to higher-dimensional logarithmic kernels. Their work does not, however, provide the commutator estimates or the modulated-free-energy mean-field application developed here. The first two authors recently proved the corresponding arbitrary-background logarithmic result: for repulsive logarithmic kernels, on both the torus and the whole space, they obtain an N-uniform modulated partition-function bound for arbitrary reference laws with bounded density [DG25]. The proof is Nelson/φ_2^4-inspired [Nel66]: a Donsker–Varadhan reduction is combined with heat regularization, layer-cake estimates, Besov control, and high-moment correlation inequalities. The present argument uses a different U-statistic route, with positive-definite truncations and GLZ tail bounds, and in addition identifies the Carleman–Fredholm determinant limit and the sharp log/Riesz Hilbert–Schmidt threshold.

During the preparation of the present manuscript, Duerinckx and Jabin obtained closely related high-temperature estimates [DJ26]. Their main analysis is presented on the torus but is extendable to the whole space as explained in Appendix A of their work. In the homogeneous torus setting, they prove the uniform square-integrable bound, the qualitative determinant limit describing the normalization of equilibrium quadratic fluctuations, and qualitative divergence outside the square-integrable class. Their main result concerns small fluctuations of the Newtonian dynamics around Gibbs equilibrium and proves convergence to the linearized Vlasov equation, reducing weighted L^p estimates on the kinetic Gibbs equilibrium to spatial partition-function bounds through the Maxwellian factorization described in Remark 2.1. The proof is math-physics inspired, relying on a cluster/Mayer expansion [Bry86]. Duerinckx and Jabin note that their main result extends to arbitrary inverse temperature $\beta < \infty$ as soon as the corresponding spatial partition functions remain uniformly bounded; see [DJ26, Remark 1.2]. The present paper supplies the needed partition-function estimates at every fixed $\beta < \infty$ for the repulsive logarithmic/Riesz interactions in the Hilbert–Schmidt regime. More generally, it addresses the modulated partition function around an arbitrary bounded reference density μ in the whole-space setting, with the torus variant described in Remark 2.1, and proves quantitative convergence to the Carleman–Fredholm determinant associated with T_μ. We also prove sharpness of the Hilbert–Schmidt threshold, with explicit nonoptimal

divergence rates at and above it. The proof uses canonical U-statistics and decoupling inequalities rather than a cluster expansion.

The overlapping preprint of Wang and Zhao [WZ26] develops all-temperature partition-function bounds using a Mayer–Penrose expansion. Their preliminary bounds were communicated to the first author before their preprint appeared. Following discussions in which they learned of our work, they extended their results to include a Carleman–Fredholm determinant limit, commutator estimates, and quantitative fixed-time fluctuations. The present paper uses a different argument based on canonical U-statistics.

On the attractive side, Grotto recently considered the two-dimensional periodic logarithmic interaction with uniform background, in the equivalent negative-temperature point-vortex formulation. For sufficiently small attractive inverse temperature, he proves qualitative convergence of the partition function to the corresponding Carleman–Fredholm determinant and convergence of the fluctuation field to the associated Gaussian energy–enstrophy ensemble [Gro26]. In contrast, the quantitative conclusion of Theorem 2.6 gives a power rate of determinant convergence for arbitrary bounded backgrounds satisfying the stated factorial-moment and exponential-integrability hypotheses. His proof is perturbative and based on a Mayer/cluster expansion. The work [Gro26] does not address the entropic commutator estimates developed here or their applications to quantitative dynamical mean-field limits and dynamical central limit theorems for joint linear statistics at finitely many times.

Finally, we note that the proof belongs to the probabilistic theory of degenerate U-statistics. After Hoeffding centering, (2.17) is a canonical degree-two statistic. Finite-rank kernels reduce to the classical Gaussian-chaos limit for degenerate quadratic forms [Ser80, dJ90]. The singular infinite-dimensional problem is controlled by decoupling and exponential inequalities for canonical U-statistics, in particular the estimates of de la Peña and Montgomery-Smith and those of Giné, Latała, and Zinn [dlPMS95, GLZ00]. This explains both the appearance of the determinant (2.18) and the sharpness of the $L^2(\mu^{\otimes 2})$ threshold.

2.6. **Proof strategy and logical structure.** This subsection records the logical dependence among the main results. More detailed roadmaps of the individual arguments are given at the beginning of the corresponding proof sections.

The main Hilbert–Schmidt branch of the argument is organized around a common exponential-moment input. The master layer-cake estimate of Section 3, together with the deterministic truncation package in Section 4, yields uniform product-space exponential moments for the centered interaction. Hilbert–Schmidt approximation and the bounded-feature Laplace estimate then give the regularized-determinant limit. Applying the same mechanism to a transport variation gives the exponential commutator estimate; Donsker–Varadhan converts it into Theorem 2.8, and the first-order case closes the Gronwall inequality in Theorem 2.9.

The fluctuation results form the second downstream branch. The determinant limit first identifies the static tilted Laplace transforms for bounded tests and hence the covariance in Proposition 2.11. A uniform $L^2(\mu^{\otimes N})$ bound on the Gibbs density, followed by truncation, extends the weak central limit theorem to arbitrary $L^2(\mu)$ tests. For the dynamical theorem, the backward adjoint equation cancels the linearized drift and expresses each terminal linear statistic as the sum of an initial statistic, a martingale, and a transport-commutator remainder. The static central limit theorem identifies the initial term, while entropy transfer and the exponential commutator estimate make the remainder negligible. A characteristic-function argument, together with convergence of the martingales' predictable quadratic covariations to their deterministic mean-field limits, produces the time-integrated part of the limiting covariance. The hypotheses in Theorem 2.13 provide the time-dependent Itô identities and backward adjoint evolution, while the uniform density and logarithmic-energy controls, together with the adjoint Hessian bound, allow Theorem 2.8 to be applied uniformly along the mean-field trajectory.

Two arguments are logically separate from this chain. The attractive logarithmic theorem follows a perturbative route, since positivity of the interaction is no longer available. After the one-body factor is separated, centering forces the surviving moments of the pair statistic to involve repeated indices; decoupling and the factorial-moment hypothesis make the resulting exponential series summable at sufficiently small inverse temperature. A feature-space expansion for the positive quadratic exponential, combined with logarithmic truncation, gives the quantitative determinant convergence in (2.38). Beyond the Hilbert–Schmidt threshold, the centered kernel is replaced by its conditional expectation on a dyadic spatial partition whose scale is coupled to N. Conditional Jensen reduces the original partition function to the resulting finite-rank coarse-grained kernel, whose Hilbert–Schmidt norm diverges; the bounded-feature Laplace estimate in Proposition 5.4 then yields the quantitative divergence rates. In the logarithmic Hilbert–Schmidt argument, the further non-generic ingredient is the infrared cancellation in Lemma 4.3.

2.7. **Organization of the paper.** The remainder of the paper is organized as follows. Section 3 develops the master layer-cake principle for degenerate U-statistics, and Section 4 collects the deterministic truncation estimates. Section 5 proves the repulsive logarithmic/Riesz partition-function estimates, their temperature interpolation, and the quantitative sharpness of the Hilbert–Schmidt threshold. Section 6 proves the attractive logarithmic uniform bound and quantitative determinant limit, and then treats the periodic uniform-background phase diagram. Section 7 proves the entropic commutator estimate and the diffusive mean-field application, while Section 8 proves the static and dynamical fluctuation theorems. Section 9 records consequences and open problems, and Appendix A gives the detailed dependence of constants.

2.8. **Lean autoformalization.** Formal counterparts of the principal partition-function estimates (Lean source), entropic-commutator estimates (Lean source), static central limit theorem (Lean source), conditional mean-field theorem (Lean source), and dynamical central limit theorems (Riesz; logarithmic) have been developed in Lean 4. The source is available in [DGR26c]. The formal proofs also include the versions or specializations of external results used as proof ingredients. The accepted proof packages were checked by Lean's kernel and independently by Nanoda, using only Lean's standard axioms of propositional extensionality, classical choice, and quotient soundness. These checks apply to the formal statements and proofs, with their stated hypotheses.

The autoformalization was carried out using Anthropic's Claude and OpenAI's ChatGPT/Codex under the authors' close supervision. The AI tools produced the formal code; the authors directed the work and reviewed the mathematical content. The authors take responsibility for the statements and arguments presented here. The revisions following autoformalization of version 1, contained in the present version, make explicit domains, integrability conditions, and auxiliary hypotheses needed in the proofs.

2.9. **Statement on AI use.** The authors used generative AI tools (OpenAI's ChatGPT and Codex, and Anthropic's Claude) during the development and preparation of this paper to identify potentially relevant literature, explore and test mathematical arguments, improve the exposition, check internal consistency and cross-references, and carry out the autoformalization described in Section 2.8. The authors treated all AI-generated output as provisional material requiring independent verification and did not rely on it as an authority, checking literature suggestions against the relevant sources and independently working through mathematical suggestions before deciding whether to incorporate them into the paper. The authors made all final decisions concerning the manuscript and take full responsibility for its contents.

3. A master layer-cake estimate for degenerate U-statistics

This section develops the probabilistic engine of the paper. The decoupling and GLZ tail bounds for degenerate U-statistics, recalled in Section 3.1, are packaged in Section 3.2 into a master layer-cake estimate for exponential moments, which is the result applied in Sections 5 and 7.

3.1. Decoupling and GLZ tail bounds.

We recall from Giné, Latała, and Zinn [GLZ00, Theorem 3.3] the following tail bound for U-statistics.

Proposition 3.1. *There exists a universal constant $L > 0$ with the following property. Let $\{h_{ij}\}_{1\le i,j\le n}$ be bounded, completely degenerate kernels, let $\{\xi_i^{(1)}, \xi_j^{(2)}\}_{1\le i,j\le n}$ be independent random variables in a measurable space $(S, \mathscr{S})$, and define*

$$A := \max_{1\le i,j\le n} \|h_{ij}\|_{L^\infty}, \qquad C^2 := \sum_{1\le i,j\le n} \mathbb{E}\big[h_{ij}^2(\xi_i^{(1)}, \xi_j^{(2)})\big], \tag{3.1}$$

$$B^2 := \max_{1\le i,j\le n} \max\Big\{ \Big\| \sum_{i'=1}^n \mathbb{E}\big[h_{i'j}^2(\xi_{i'}^{(1)}, \cdot)\big] \Big\|_{L^\infty}, \ \Big\| \sum_{j'=1}^n \mathbb{E}\big[h_{ij'}^2(\cdot, \xi_{j'}^{(2)})\big] \Big\|_{L^\infty} \Big\}, \tag{3.2}$$

$$D := \sup \ \mathbb{E}\Big[\sum_{1\le i,j\le n} h_{ij}(\xi_i^{(1)}, \xi_j^{(2)})\, f_i(\xi_i^{(1)})\, g_j(\xi_j^{(2)}) \Big], \tag{3.3}$$

where the supremum in D runs over all $f_i, g_j : S \to \mathbb{R}$ with $\mathbb{E}[\sum_i f_i^2(\xi_i^{(1)})] \le 1$ and $\mathbb{E}[\sum_j g_j^2(\xi_j^{(2)})] \le 1$. Then, for every $t > 0$,

$$\mathbb{P}\Big(\Big| \sum_{1\le i,j\le n} h_{ij}(\xi_i^{(1)}, \xi_j^{(2)}) \Big| \ge t \Big) \le L \exp\Big(-\frac{1}{L} \min\Big\{ \Big(\frac{t}{C}\Big)^2, \ \frac{t}{D}, \ \Big(\frac{t}{B}\Big)^{2/3}, \ \Big(\frac{t}{A}\Big)^{1/2} \Big\} \Big). \tag{3.4}$$

Remark 3.2. In all applications below, the variables are iid with law μ and a single symmetric kernel is repeated over the pairs of indices: $h_{ij} = h$ for $i \ne j$, and $h_{ii} = 0$. Exploiting this repetition, the four quantities of Proposition 3.1 reduce to N-weighted norms of h:

$$A \le \|h\|_{L^\infty}, \qquad B^2 \le (N-1)\|h\|_{L^\infty L^2(\mu)}^2, \qquad C^2 = N(N-1)\|h\|_{L^2(\mu^{\otimes 2})}^2, \qquad D \le N\|\mathcal{T}_h\|_{\mathrm{op}}, \tag{3.5}$$

where $\|h\|_{L^\infty L^2(\mu)} := \sup_x \big(\int_S h(x,y)^2 \, \mathrm{d}\mu(y)\big)^{1/2}$ and $\mathcal{T}_h f := \int_S h(\cdot, y) f(y) \, \mathrm{d}\mu(y)$, with operator norm $\|\mathcal{T}_h\|_{\mathrm{op}}$ on $L^2(\mu)$. These are the norms entering the master layer-cake estimate below (Proposition 3.5), which consumes Proposition 3.1 in the decoupled form (3.9).

We recall from de la Peña and Montgomery-Smith [dlPMS95, Theorem 1] the following decoupling inequality.

Proposition 3.3. *For every $k \ge 2$ there exists a constant $C_k > 0$, depending only on k, with the following property. Let $\{\xi_i\}_i$ be a sequence of independent random variables in a measurable space $(S, \mathscr{S})$, let $\{\xi_i^{(j)}\}_i$, $1 \le j \le k$, be independent copies of $\{\xi_i\}$, and let $f_{i_1\cdots i_k} : S^k \to E$ be measurable functions with values in a Banach space $(E, \|\cdot\|)$. Then, for every $n \ge k$ and every $t > 0$,*

$$\mathbb{P}\Big(\Big\| \sum_{1\le i_1\ne\cdots\ne i_k\le n} f_{i_1\cdots i_k}(\xi_{i_1}^{(1)}, \dots, \xi_{i_k}^{(1)}) \Big\| \ge t \Big) \le C_k \, \mathbb{P}\Big(C_k \Big\| \sum_{1\le i_1\ne\cdots\ne i_k\le n} f_{i_1\cdots i_k}(\xi_{i_1}^{(1)}, \dots, \xi_{i_k}^{(k)}) \Big\| \ge t \Big), \tag{3.6}$$

the summations ranging over all tuples $(i_1, \dots, i_k) \in [n]^k$ with distinct entries.

Remark 3.4 (Moment decoupling with an explicit constant)**.** For $k = 2$ and canonical kernels, the companion moment comparison holds with the explicit constant 4: if $f_{ij} : S^2 \to E$, $1 \le i \ne j \le n$, are such that $f_{ij}(\xi_i, \xi_j)$ and $f_{ij}(\xi_i^{(1)}, \xi_j^{(2)})$ are Bochner integrable and

$$\mathbb{E}[f_{ij}(\xi_i, y)] = 0 \quad \text{for a.e. } y, \qquad \mathbb{E}[f_{ij}(x, \xi_j)] = 0 \quad \text{for a.e. } x, \tag{3.7}$$

then, for every nondecreasing convex $\Phi : [0, \infty) \to [0, \infty)$,

$$\mathbb{E}\,\Phi\bigg(\Big\| \sum_{1\le i\ne j\le n} f_{ij}(\xi_i, \xi_j)\Big\|\bigg) \le \mathbb{E}\,\Phi\bigg(4\Big\| \sum_{1\le i\ne j\le n} f_{ij}(\xi_i^{(1)}, \xi_j^{(2)})\Big\|\bigg). \tag{3.8}$$

This form, with $\Phi(x) = x^{2k}$, is used in the moment computation of Section 6.

3.2. **The master layer-cake estimate.** The partition-function bounds of Section 5 and the local numerator estimate of Section 7 both rest on the same mechanism: a layer-cake decomposition in which, at height u, the singular and infrared parts of the interaction are removed by deterministic estimates, and the tail of the remaining centered U-statistic is controlled by the following decoupled form of the GLZ bound. Applying Proposition 3.3 with $k = 2$ and $E = \mathbb{R}$ and then Proposition 3.1, we obtain that if $\xi_1, \dots, \xi_n$ are iid, then for all $t > 0$,

$$\begin{aligned}\mathbb{P}\bigg(\Big| \sum_{1\le i_1\ne i_2\le n} h(\xi_{i_1}, \xi_{i_2})\Big| \ge t\bigg) &\le C_2 L \exp\bigg(-\frac{1}{L}\min\bigg\{\Big(\frac{t}{C_2 C}\Big)^2,\ \frac{t}{C_2 D},\ \Big(\frac{t}{C_2 B}\Big)^{2/3},\ \Big(\frac{t}{C_2 A}\Big)^{1/2}\bigg\}\bigg)\\ &\le L' \exp\bigg(-\frac{1}{L'}\min\bigg\{\Big(\frac{t}{N\|h\|_{L^2(\mu^{\otimes 2})}}\Big)^2,\ \frac{t}{N\|\mathcal{T}_h\|_{\mathrm{op}}},\\ &\qquad \Big(\frac{t}{N^{1/2}\|h\|_{L^\infty L^2(\mu)}}\Big)^{2/3},\ \Big(\frac{t}{\|h\|_{L^\infty}}\Big)^{1/2}\bigg\}\bigg),\end{aligned} \tag{3.9}$$

for a universal constant $L' > 0$, where the final line is by Remark 3.2 and a relabeling of constants. Hereafter, we recycle notation and write L in lieu of L'.

We state and prove the layer-cake mechanism once; in the statement, $\tilde{q}$ denotes the μ-centering (2.2) of a kernel q, and all norms of kernels are computed on $S = \operatorname{supp}\mu$.

Proposition 3.5 (Master layer-cake estimate)**.** *Let $\mu \in \mathcal{P}(\mathbb{R}^{\mathsf{d}})$, and let $L \ge 1$ be the universal constant in* (3.9)*. Let $g : (\mathbb{R}^{\mathsf{d}})^2 \to \mathbb{R}$ be a symmetric, measurable two-body kernel with $g \in L^1(\mu^{\otimes 2})$ and $w : \mathbb{R}^{\mathsf{d}} \to \mathbb{R}$ a measurable one-body function, and consider the statistics*

$$\Xi_N(X_N) \coloneqq -\frac{1}{2N}\sum_{1\le i\ne j\le N} \tilde{g}(x_i, x_j) + \frac{1}{N}\sum_{i=1}^N w(x_i), \qquad N \ge 1. \tag{3.10}$$

Let $U_0 \ge 1$ be a threshold, and suppose that for every $N \ge 1$ and every $u \ge U_0$ there exist a bounded symmetric kernel $q_{N,u}$ and a number $\mathsf{m}_{N,u} > 0$ such that the following hold.

(M1) (Event reduction) *For $\mu^{\otimes N}$-a.e. configuration X_N,*

$$\Xi_N(X_N) \ge u \quad \Longrightarrow \quad -\frac{1}{2N}\sum_{1\le i\ne j\le N} \tilde{q}_{N,u}(x_i, x_j) \ge u. \tag{3.11}$$

(M2) (Deterministic diagonal bound) *For every configuration X_N,*

$$-\frac{1}{2N}\sum_{1\le i\ne j\le N} \tilde{q}_{N,u}(x_i, x_j) \le \mathsf{m}_{N,u}. \tag{3.12}$$

(M3) (Norm bounds)

$$\|\tilde{q}_{N,u}\|^2_{L^\infty L^2(\mu)} \le \frac{1}{L}, \qquad \|\tilde{q}_{N,u}\|^2_{L^2(\mu^{\otimes 2})} \le \frac{1}{L}, \qquad \|\mathcal{T}_{\tilde{q}_{N,u}}\|_{\mathrm{op}} \le \frac{1}{L}. \tag{3.13}$$

Suppose, finally, that there exists $N_0 \ge 1$ such that

$$2L^2 \sup_{u \ge U_0} \min\{u,\ \mathsf{m}_{N,u}\}\,(1 + \|\tilde{q}_{N,u}\|_{L^\infty}) \le N \qquad \text{for every } N \ge N_0, \tag{3.14}$$

the supremum being finite also for $N < N_0$. Then, writing S_N for the supremum in (3.14)*,*

$$\sup_{N \ge 1} \mathbb{E}_{\mu^{\otimes N}}\big[e^{\Xi_N}\big] \le \max\Big\{\max_{1 \le N < N_0} e^{U_0 + \mathsf{S}_N},\ e^{U_0} + L\Big\}, \tag{3.15}$$

the inner maximum being absent when $N_0 = 1$.

Remark 3.6 (Uniformity in families)**.** The bound (3.15) depends on the data only through L, U_0, and the suprema S_N in (3.14), monotonically. Consequently, if $\{\Xi_N^{(\lambda)}\}_{\lambda \in \Lambda}$ satisfy (M1)–(M3) for each $\lambda \in \Lambda$ with a common threshold U_0, and if there are λ-independent majorants $\mathsf{m}^{(\lambda)}_{N,u} \le \mathsf{m}_{N,u}$ and $\|\tilde{q}^{(\lambda)}_{N,u}\|_{L^\infty} \le \mathsf{a}_{N,u}$ with, for some $N_0 \ge 1$,

$$2L^2 \sup_{u \ge U_0} \min\{u,\ \mathsf{m}_{N,u}\}\,(1 + \mathsf{a}_{N,u}) \le N \qquad \text{for every } N \ge N_0, \tag{3.16}$$

and if, for each $N < N_0$, the corresponding supremum $\mathsf{S}^{(\lambda)}_N$ is bounded uniformly in $\lambda \in \Lambda$, then $\sup_{\lambda \in \Lambda} \sup_{N \ge 1} \mathbb{E}_{\mu^{\otimes N}}\big[e^{\Xi_N^{(\lambda)}}\big] < \infty$. In Section 5, λ is the near-field truncation scale $\eta \in [0, \eta_0]$ in the Riesz case and the infrared cutoff $T \ge 2$ in the logarithmic case.

Proof of Proposition 3.5. Fix N_0 as in the statement. We will use repeatedly the following observation: for every $u \ge U_0$ with $u > \mathsf{m}_{N,u}$, the event $\{\Xi_N \ge u\}$ is $\mu^{\otimes N}$-null, by (M1) and (M2).

The case $N < N_0$ (L^∞ bound). By the observation, Ξ_N can reach a height $u \ge U_0$ only if $u \le \mathsf{m}_{N,u}$, and then $u = \min\{u, \mathsf{m}_{N,u}\} \le \mathsf{S}_N$ by the definition of S_N. Hence $\Xi_N \le \max\{U_0, \mathsf{S}_N\} \le U_0 + \mathsf{S}_N$ $\mu^{\otimes N}$-a.e., and

$$\mathbb{E}_{\mu^{\otimes N}}\big[e^{\Xi_N}\big] \le e^{U_0 + \mathsf{S}_N}, \qquad N < N_0. \tag{3.17}$$

The case $N \ge N_0$ (layer-cake bound). We claim that

$$\mu^{\otimes N}(\Xi_N \ge u) \le L\, e^{-2u} \qquad \text{for every } u \ge U_0. \tag{3.18}$$

By the observation, we may assume $u \le \mathsf{m}_{N,u}$, in which case (3.14) gives the lower bound

$$2L^2\, u\,(1 + \|\tilde{q}_{N,u}\|_{L^\infty}) \le 2L^2\, \mathsf{S}_N \le N. \tag{3.19}$$

By (M1) and the decoupled GLZ tail estimate (3.9), applied to the bounded canonical kernel $\tilde{q}_{N,u}$ at level $t = 2Nu$,

$$\begin{aligned}\mu^{\otimes N}(\Xi_N \ge u) &\le \mu^{\otimes N}\bigg(\Big|\sum_{1 \le i \ne j \le N} \tilde{q}_{N,u}(x_i, x_j)\Big| \ge 2Nu\bigg) \\ &\le L \exp\left(-\frac{1}{L}\min\left\{\frac{4u^2}{\|\tilde{q}\|^2_{L^2(\mu^{\otimes 2})}},\ \frac{2u}{\|\mathcal{T}_{\tilde{q}}\|_{\mathrm{op}}},\ \frac{N^{1/3}(2u)^{2/3}}{\|\tilde{q}\|^{2/3}_{L^\infty L^2(\mu)}},\ \left(\frac{2Nu}{\|\tilde{q}\|_{L^\infty}}\right)^{1/2}\right\}\right),\end{aligned} \tag{3.20}$$

where, within this proof, we abbreviate $\tilde{q} = \tilde{q}_{N,u}$. We check that each of the four entries is at least $2Lu$, so that

$$2Lu \le \min\left\{\frac{4u^2}{\|\tilde{q}\|^2_{L^2(\mu^{\otimes 2})}},\ \frac{2u}{\|\mathcal{T}_{\tilde{q}}\|_{\mathrm{op}}},\ \frac{N^{1/3}(2u)^{2/3}}{\|\tilde{q}\|^{2/3}_{L^\infty L^2(\mu)}},\ \left(\frac{2Nu}{\|\tilde{q}\|_{L^\infty}}\right)^{1/2}\right\}. \tag{3.21}$$

The bounds for the first two entries follow from (3.13) and $1 \le U_0 \le u$. The third entry satisfies $2Lu \le N^{1/3}(2u)^{2/3}\|\tilde{q}\|_{L^\infty L^2(\mu)}^{-2/3}$ if and only if $2L^3\|\tilde{q}\|_{L^\infty L^2(\mu)}^2 u \le N$, which follows by (3.13) and (3.19). The fourth entry satisfies $2Lu \le (2Nu/\|\tilde{q}\|_{L^\infty})^{1/2}$ if and only if $2L^2\|\tilde{q}\|_{L^\infty} u \le N$, which again holds by (3.19). This proves (3.21), hence the claim (3.18), and the layer-cake formula gives

$$\mathbb{E}_{\mu^{\otimes N}}[e^{\Xi_N}] \le e^{U_0} + \int_{U_0}^{\infty} e^u\, \mu^{\otimes N}(\Xi_N \ge u)\, \mathrm{d}u \le e^{U_0} + L\int_{U_0}^{\infty} e^{-u}\, \mathrm{d}u \le e^{U_0} + L, \qquad N \ge N_0. \tag{3.22}$$

Taking the maximum of the two bounds gives (3.15). □

4. Truncation estimates for the log/Riesz kernels

This section collects the deterministic truncation estimates for the interaction kernels, used in Sections 5 and 7: the positive-definite physical-space truncations imported from [HCRS25] (Section 4.1), and the Fourier truncations and deterministic diagonal bounds built on them (Section 4.2).

4.1. **Positive-definite truncations.**

The following positive-definite truncation package is extracted from the third author's work with Hess-Childs and Serfaty [HCRS25, Subsection 2.1, Lemma 2.2, Proposition 2.9]. The lemma is valid for the full range $0 < \mathsf{s} < \mathsf{d}$; the stricter condition $\mathsf{s} < \mathsf{d}/2$ imposed in this paper enters only when the singular remainder or the centered kernel is estimated in L^2.

In the sequel, every free truncation parameter η is taken in $0 < \eta \le \eta_0$, where $\eta_0 > 0$ denotes a fixed small truncation scale, unless explicitly stated otherwise.

Lemma 4.1 (Potential truncation)**.** *Let $\phi \in \mathcal{S}(\mathbb{R}^{\mathsf{d}})$ be nonnegative, radial, decreasing, and satisfy $\widehat{\phi} \ge 0$, and set $\phi_t(x) \coloneqq t^{-\mathsf{d}}\phi(x/t)$. For all potential truncations, we fix $\phi(x) = (2\pi)^{-\mathsf{d}/2}e^{-|x|^2/2}$.*

For $0 < \mathsf{s} < \mathsf{d}$, set

$$\mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}^{-1} = \mathsf{s}\int_0^\infty r^{\mathsf{s}}\phi(r)\,\frac{\mathrm{d}r}{r}, \tag{4.1}$$

$$\mathsf{g}(x) = \mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}\int_0^\infty t^{\mathsf{d}-\mathsf{s}}\phi_t(x)\,\frac{\mathrm{d}t}{t}, \qquad x \ne 0. \tag{4.2}$$

For $\eta > 0$, define the scale-η truncated potential and potential difference

$$\mathsf{g}_\eta(x) \coloneqq \mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}\int_\eta^\infty t^{\mathsf{d}-\mathsf{s}}\phi_t(x)\,\frac{\mathrm{d}t}{t}, \qquad \mathsf{f}_\eta(x) \coloneqq \mathsf{g}(x) - \mathsf{g}_\eta(x) = \mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}\int_0^\eta t^{\mathsf{d}-\mathsf{s}}\phi_t(x)\,\frac{\mathrm{d}t}{t}. \tag{4.3}$$

Then, by radial monotonicity,

$$\|\mathsf{g}_\eta\|_{L^\infty} = \mathsf{g}_\eta(0) = C_{\phi,\mathsf{d},\mathsf{s}}\eta^{-\mathsf{s}}, \qquad \mathsf{g}_\eta \ge 0, \qquad \widehat{\mathsf{g}_\eta} \ge 0, \qquad \widehat{\mathsf{f}_\eta} \ge 0, \tag{4.4}$$

with the Fourier inequalities understood in the sense of tempered distributions. Since $\mathsf{g} = \mathsf{g}_\eta + \mathsf{f}_\eta$, this implies

$$0 \le \widehat{\mathsf{g}_\eta} \le \widehat{\mathsf{g}}. \tag{4.5}$$

Moreover, the potential difference satisfies

$$0 \le \mathsf{f}_\eta(x) \le C_\gamma\frac{\eta^{\gamma-\mathsf{s}}}{|x|^\gamma}, \qquad \gamma > \mathsf{s}, \tag{4.6}$$

$$\mathsf{f}_\eta(x) \ge c\mathsf{g}(x), \qquad |x| \le \eta. \tag{4.7}$$

In particular, if $\mu \in L^\infty$, then

$$\|\mathsf{f}_\eta * \mu\|_{L^\infty} \le \|\mu\|_{L^\infty}\int_{\mathbb{R}^{\mathsf{d}}}\mathsf{f}_\eta(x)\,\mathrm{d}x \le C_\mu\eta^{\mathsf{d}-\mathsf{s}}. \tag{4.8}$$

For $\mathsf{s} = 0$, let

$$\mathsf{c}_{\phi,\mathsf{d},0} \coloneqq \phi(0)^{-1}, \qquad C_{\phi,T} \coloneqq \phi(0)\log T - \int_0^\infty \log r\, \phi'(r)\, \mathrm{d}r, \tag{4.9}$$

set, for $0 < \eta < T$,

$$\mathsf{g}_{\eta,T}(x) \coloneqq \mathsf{c}_{\phi,\mathsf{d},0}\left(\int_\eta^T t^{\mathsf{d}}\phi_t(x)\,\frac{\mathrm{d}t}{t} - C_{\phi,T}\right), \qquad \mathsf{f}_\eta(x) \coloneqq \mathsf{g}(x) - \mathsf{g}_\eta(x) = \mathsf{c}_{\phi,\mathsf{d},0}\int_0^\eta t^{\mathsf{d}}\phi_t(x)\,\frac{\mathrm{d}t}{t}, \tag{4.10}$$

where g_η is the limit of $\mathsf{g}_{\eta,T}$ as $T \to \infty$ in the sense of tempered distributions. Then

$$\mathsf{g}_\eta(0) = -\log\eta + O(1), \qquad \widehat{\mathsf{g}_\eta} \geq 0 \quad \text{on zero-mean test functions}, \qquad \widehat{\mathsf{f}_\eta} \geq 0, \tag{4.11}$$

$$0 \leq \mathsf{f}_\eta(x) \leq C_\gamma \min\left\{1 + \left(-\log\frac{|x|}{\eta}\right)_+, \frac{\eta^\gamma}{|x|^\gamma}\right\}, \qquad \gamma > 0, \tag{4.12}$$

and consequently, if $\mu \in L^\infty$,

$$\|\mathsf{f}_\eta * \mu\|_{L^\infty} \leq C_\mu \eta^{\mathsf{d}}. \tag{4.13}$$

For every fixed $n \geq 1$, it also holds that

$$|x|^n |\nabla^{\otimes n}\mathsf{f}_\eta(x)| \leq C_n \mathsf{f}_{c_n\eta}(x) \tag{4.14}$$

for a dimensional constant $c_n \geq 2$.

Proof. We only prove (4.14), as [HCRS25] only considers the $n = 1$ case. To see that the inequality holds, observe that every derivative of the Gaussian is a polynomial times the same Gaussian. Consequently, for each $n \geq 1$ there is $C_n = C_n(\mathsf{d})$ such that

$$|z|^n |\nabla^{\otimes n}\phi(z)| \leq C_n \phi(z/2), \qquad z \in \mathbb{R}^{\mathsf{d}}. \tag{4.15}$$

For $x \neq 0$, differentiating the scale integral in (4.10) and substituting $u = 2t$ therefore gives

$$|x|^n |\nabla^{\otimes n}\mathsf{f}_\eta(x)| \leq C_n \mathsf{c}_{\phi,\mathsf{d},0}\int_0^\eta \phi(x/(2t))\,\frac{\mathrm{d}t}{t} = C_n \mathsf{f}_{2\eta}(x). \tag{4.16}$$

The Gaussian decay as $t \downarrow 0$, locally uniformly for $x \neq 0$, justifies differentiation under the integral. Thus one may take $c_n = 2$. □

4.2. **Fourier truncations and positive-definite diagonal bounds.** We use here the truncation facts already recorded in (4.4)-(4.8): positive definiteness and Fourier order, the diagonal and L^∞ bounds, and the short-range convolution estimate. For the Riesz kernel, with our normalization of Fourier transform,

$$\widehat{\mathsf{g}}(\xi) = c_{\mathsf{d},\mathsf{s}}|\xi|^{-(\mathsf{d}-\mathsf{s})}, \qquad \xi \in \mathbb{R}^{\mathsf{d}} \setminus \{0\}, \tag{4.17}$$

for a positive constant $c_{\mathsf{d},\mathsf{s}}$. Given $K \geq 1$, define the low- and high-frequency pieces of g_η by

$$\widehat{\mathsf{g}_{\eta,\leq K}}(\xi) \coloneqq \mathbf{1}_{|\xi|\leq K}\widehat{\mathsf{g}_\eta}(\xi), \qquad \widehat{\mathsf{g}_{\eta,>K}}(\xi) \coloneqq \mathbf{1}_{|\xi|>K}\widehat{\mathsf{g}_\eta}(\xi). \tag{4.18}$$

By (4.4), both kernels are positive definite.

We record here the elementary deterministic estimate which will later give the improved diagonal bound.

Lemma 4.2 (Positive-definite diagonal bound)**.** *Let q be a bounded positive-definite kernel, and let $\tilde{q}$ denote its μ-centering with respect to a probability measure μ. Then*

$$\|\tilde{q}\|_{L^\infty} \le 4 \sup_x q(x,x), \tag{4.19}$$

and, for every $N \ge 1$ and every configuration $X_N = (x_1, \dots, x_N)$,

$$\sum_{1 \le i,j \le N} \tilde{q}(x_i, x_j) \ge 0. \tag{4.20}$$

Consequently,

$$-\frac{1}{N} \sum_{1 \le i \ne j \le N} \tilde{q}(x_i, x_j) \le \frac{1}{N} \sum_{i=1}^N \tilde{q}(x_i, x_i) \le \sup_x \tilde{q}(x,x) \le 4 \sup_x q(x,x). \tag{4.21}$$

Proof. Since the two-point Gram matrix associated to q is positive semidefinite, the nonnegativity of its determinant gives $|q(x,y)|^2 \le q(x,x)q(y,y)$, so $\|q\|_{L^\infty} \le M := \sup_x q(x,x)$. Expanding the centering identity (2.2) then bounds each of the four terms of $\tilde{q}(x,y)$ by M, which proves (4.19) and the last inequality in (4.21).

Next, by (2.2) and the definition (2.7) of μ_N,

$$\sum_{1 \le i,j \le N} \tilde{q}(x_i, x_j) = N^2 \iint_{(\mathbb{R}^{\mathsf{d}})^2} q(x,y) \, \mathrm{d}(\mu_N - \mu)(x) \, \mathrm{d}(\mu_N - \mu)(y), \tag{4.22}$$

which is nonnegative by positive definiteness of q. This gives

$$-\frac{1}{N} \sum_{1 \le i \ne j \le N} \tilde{q}(x_i, x_j) \le \frac{1}{N} \sum_{i=1}^N \tilde{q}(x_i, x_i) \le \sup_x \tilde{q}(x,x). \tag{4.23}$$

□

For the logarithmic proofs below, we also need a finite-T frequency decomposition. The point is to keep the one-body term in $-\beta N \mathsf{F}_N$ until after the infrared part of the interaction has been estimated. Constants may depend on d, on the fixed truncation profile ϕ, and on the displayed bounds for μ, but never on N, u, η, or the auxiliary infrared cutoff T.

Fix once and for all a radial, radially nonincreasing function $\chi \in C_c^\infty(\mathbb{R}^{\mathsf{d}})$ such that $0 \le \chi \le 1$, $\chi = 1$ on $\{|\xi| \le 1\}$, and $\chi = 0$ on $\{|\xi| \ge 2\}$, and set $\chi_K(\xi) := \chi(\xi/K)$. For $0 < \eta < T$, we assign the renormalizing constant in (4.10) to the low-frequency part and define

$$\widehat{\mathsf{g}}_{\eta,T,\le K} := \chi_K(\xi) \, \mathsf{c}_{\phi,\mathsf{d},0} \int_\eta^T t^{\mathsf{d}} \widehat{\phi}(t\xi) \, \frac{\mathrm{d}t}{t} - \mathsf{c}_{\phi,\mathsf{d},0} C_{\phi,T} \delta_0, \tag{4.24}$$

$$\widehat{\mathsf{g}}_{\eta,T,>K} := (1 - \chi_K(\xi)) \, \mathsf{c}_{\phi,\mathsf{d},0} \int_\eta^T t^{\mathsf{d}} \widehat{\phi}(t\xi) \, \frac{\mathrm{d}t}{t}. \tag{4.25}$$

Thus, $\mathsf{g}_{\eta,T} = \mathsf{g}_{\eta,T,\le K} + \mathsf{g}_{\eta,T,>K}$. Since the Dirac mass at the origin is invisible to zero-average test functions, both pieces are positive definite in every centered quadratic form in which they are used below. We also write

$$\mathsf{g}_T := \mathsf{f}_\eta + \mathsf{g}_{\eta,T}, \qquad h_{\mu,T} := \mathsf{g}_T * \mu, \tag{4.26}$$

noting that g_T is independent of the auxiliary choice of η.

Lemma 4.3 (Frequency-truncation estimates)**.** *Let $K \ge 1$, assume $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$, and recall the $L^\infty L^2(\mu)$ and integral-operator norms of Remark 3.2.*

(i) (Riesz case $0 < \mathsf{s} < \mathsf{d}/2$.) *The centered kernels of the splitting* (4.18) *satisfy, uniformly in* $0 < \eta \le \eta_0$,

$$\|\tilde{\mathsf{g}}_{\eta,>K}\|^2_{L^2(\mu^{\otimes 2})} \le C_\mu \int_{|\xi|>K} \widehat{\mathsf{g}}(\xi)^2 \, \mathrm{d}\xi \le C_\mu K^{2\mathsf{s}-\mathsf{d}}, \tag{4.27}$$

$$\|\tilde{\mathsf{g}}_{\eta,>K}\|^2_{L^\infty L^2(\mu)} \le C_\mu \int_{|\xi|>K} \widehat{\mathsf{g}}(\xi)^2 \, \mathrm{d}\xi \le C_\mu K^{2\mathsf{s}-\mathsf{d}}, \tag{4.28}$$

$$\|\mathcal{T}_{\tilde{\mathsf{g}}_{\eta,>K}}\|_{\mathrm{op}} \le C_\mu \sup_{|\xi|>K} \widehat{\mathsf{g}}(\xi) \le C_\mu K^{-(\mathsf{d}-\mathsf{s})}, \tag{4.29}$$

together with the low-mode diagonal bound

$$\sup_x \tilde{\mathsf{g}}_{\eta,\le K}(x,x) \le 4 \int_{|\xi|\le K} \widehat{\mathsf{g}}(\xi) \, \mathrm{d}\xi \le C_1 K^{\mathsf{s}}, \tag{4.30}$$

for a constant $C_1 = C_1(\mathsf{d},\mathsf{s})$.

(ii) (Logarithmic case $\mathsf{s} = 0$.) *Assume* (2.20). *There is a constant* $C_{\mu,K} < \infty$ *such that, for every* $N \ge 1$, *every* $T \ge 2$, *every* $0 < \eta \le \eta_0$, *and every configuration* X_N,

$$\frac{1}{N}\sum_{i=1}^N h_{\mu,T}(x_i) - \frac{1}{2N} \sum_{1\le i\ne j\le N} \tilde{\mathsf{g}}_{\eta,T,\le K}(x_i,x_j) \le C_{\mu,K}. \tag{4.31}$$

Moreover,

$$\|\tilde{\mathsf{g}}_{\eta,T,>K}\|^2_{L^2(\mu^{\otimes 2})} + \|\tilde{\mathsf{g}}_{\eta,T,>K}\|^2_{L^\infty L^2(\mu)} \le C_\mu K^{-\mathsf{d}}, \qquad \|\mathcal{T}_{\tilde{\mathsf{g}}_{\eta,T,>K}}\|_{\mathrm{op}} \le C_\mu K^{-\mathsf{d}}, \tag{4.32}$$

$$\|\tilde{\mathsf{g}}_{\eta,T,>K}\|_{L^\infty} \le C(1+\log(1/\eta)). \tag{4.33}$$

Proof. (i) The bounds (4.27)–(4.29) follow from Plancherel, the Fourier order (4.5), and $\mu \in L^\infty$; the tail integrals are finite exactly because $2\mathsf{s} < \mathsf{d}$. For (4.30): since $|\phi_{\xi,\mu}| \le 2$ for the centered characters $\phi_{\xi,\mu}(x) \coloneqq e^{2\pi \mathrm{i} x\cdot\xi} - \int_{\mathbb{R}^{\mathsf{d}}} e^{2\pi \mathrm{i} y\cdot\xi} \, \mathrm{d}\mu(y)$, the representation

$$\tilde{\mathsf{g}}_{\eta,\le K}(x,y) = \int_{|\xi|\le K} \widehat{\mathsf{g}_\eta}(\xi)\, \phi_{\xi,\mu}(x)\overline{\phi_{\xi,\mu}(y)} \, \mathrm{d}\xi \tag{4.34}$$

gives $\sup_x \tilde{\mathsf{g}}_{\eta,\le K}(x,x) \le 4\int_{|\xi|\le K} \widehat{\mathsf{g}}(\xi) \, \mathrm{d}\xi \le C_1 K^{\mathsf{s}}$, using (4.5) again.

(ii) Let $q \coloneqq \mathsf{g}_{\eta,T,\le K}$. Since q is conditionally positive definite,

$$\frac{1}{N}\sum_{i=1}^N h_{\mu,T}(x_i) - \frac{1}{2N} \sum_{1\le i\ne j\le N} \tilde{q}(x_i,x_j) \le \frac{1}{N}\sum_{i=1}^N \Big(h_{\mu,T}(x_i) + \frac{1}{2}\tilde{q}(x_i,x_i)\Big). \tag{4.35}$$

Expanding the centered diagonal gives

$$h_{\mu,T}(x) + \frac{1}{2}\tilde{q}(x,x) = (\mathsf{g}_T - q) * \mu(x) + \frac{1}{2}q(0) + \frac{1}{2}\int_{(\mathbb{R}^{\mathsf{d}})^2} q(z-w) \, \mathrm{d}\mu^{\otimes 2}(z,w). \tag{4.36}$$

Here, $\mathsf{g}_T - q = \mathsf{f}_\eta + \mathsf{g}_{\eta,T,>K}$. The contribution of f_η is bounded in L^∞ by (4.13). For the high-frequency term, the multiplier in (4.25) is smooth at $\xi = 0$ and satisfies

$$|\partial_\xi^\alpha \widehat{\mathsf{g}}_{\eta,T,>K}(\xi)| \le C_{\alpha,K}|\xi|^{-\mathsf{d}-|\alpha|}, \qquad \xi \ne 0, \tag{4.37}$$

for $|\alpha| \le \mathsf{d}+2$, uniformly in T and η. A smooth dyadic decomposition of $\{|\xi| \ge K\}$, followed by rescaling and repeated integration by parts on each annulus, therefore gives the absolute kernel bound

$$|\mathsf{g}_{\eta,T,>K}(x)| \le C_K\Big(1 + (-\log|x|)_+\Big)\mathbf{1}_{|x|\le 1} + C_K\langle x\rangle^{-\mathsf{d}-2}. \tag{4.38}$$

Since $\mu \in L^1 \cap L^\infty$, this implies, uniformly in $T \ge 2$ and $0 < \eta \le \eta_0$, that

$$\|\mathsf{g}_{\eta,T,>K} * \mu\|_{L^\infty} \le C_{\mu,K}. \tag{4.39}$$

For the scalar terms in (4.36), the nonconstant Fourier multiplier defining q is real, radial, and nonnegative. Consequently, the nonconstant part of q is a real translation-invariant positive-definite kernel and is therefore maximized at the origin. Since subtracting the constant term in (4.24) does not affect this comparison, $q(x) \le q(0)$ for every $x \in \mathbb{R}^{\mathsf{d}}$, and hence

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} q(z-w)\ \mathrm{d}\mu^{\otimes 2}(z,w) \le q(0). \tag{4.40}$$

Set

$$F_K(t) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \chi_K(\xi) t^{\mathsf{d}} \widehat{\phi}(t\xi)\ \mathrm{d}\xi = \int_{\mathbb{R}^{\mathsf{d}}} \chi_K(z/t)\widehat{\phi}(z)\ \mathrm{d}z. \tag{4.41}$$

Fourier inversion and (4.9) give

$$q(0) = \mathsf{c}_{\phi,\mathsf{d},0}\left(\int_\eta^T F_K(t)\,\frac{\mathrm{d}t}{t} - \phi(0)\log T + \int_0^\infty \log r\,\phi'(r)\ \mathrm{d}r\right). \tag{4.42}$$

Here $0 \le F_K(t) \le \phi(0)$ for every $t > 0$, while $F_K(t) \le C_K t^{\mathsf{d}}$ for $0 < t \le 1$. Splitting at $t = 1$ if $\eta \le 1$, and using $\int_\eta^T F_K(t)\ \mathrm{d}t/t \le \phi(0)\log(T/\eta)$ if $\eta > 1$, gives $q(0) \le C_K$ uniformly in T and η. The preceding pointwise bound therefore yields

$$\frac{1}{2}q(0) + \frac{1}{2}\int_{(\mathbb{R}^{\mathsf{d}})^2} q(z-w)\ \mathrm{d}\mu^{\otimes 2}(z,w) \le C_K \tag{4.43}$$

uniformly in T and η. Combining these bounds with (4.35)–(4.36) proves (4.31).

It remains to record the U-statistic norms. From (4.24), the bounds $0 \le 1 - \chi_K \le 1$, the support condition $1 - \chi_K = 0$ on $\{|\xi| \le K\}$, and the change of variables $\zeta = t\xi$, we have

$$0 \le \widehat{\mathsf{g}}_{\eta,T,>K}(\xi) \le C|\xi|^{-\mathsf{d}}\mathbf{1}_{|\xi|\ge K}. \tag{4.44}$$

Plancherel and $\mu \in L^\infty$ give

$$\|\tilde{\mathsf{g}}_{\eta,T,>K}\|^2_{L^2(\mu^{\otimes 2})} + \|\tilde{\mathsf{g}}_{\eta,T,>K}\|^2_{L^\infty L^2(\mu)} \le C_\mu \int_{|\xi|\ge K} |\xi|^{-2\mathsf{d}}\ \mathrm{d}\xi \le C_\mu K^{-\mathsf{d}}, \tag{4.45}$$

and the operator norm is bounded by the same Fourier multiplier estimate,

$$\|\mathcal{T}_{\tilde{\mathsf{g}}_{\eta,T,>K}}\|_{\mathrm{op}} \le C_\mu \sup_{|\xi|\ge K} |\xi|^{-\mathsf{d}} \le C_\mu K^{-\mathsf{d}}. \tag{4.46}$$

Finally, by positive definiteness and (4.11),

$$\|\tilde{\mathsf{g}}_{\eta,T,>K}\|_{L^\infty} \le C\mathsf{g}_\eta(0) \le C(1 + \log(1/\eta)). \tag{4.47}$$

This completes the proof. □

5. Repulsive partition-function estimates

This section proves Theorem 2.2 and Corollary 5.7. The proof separates into three independent estimates, which we state now as a roadmap. For $\xi_1, \dots, \xi_N$ iid with law μ, define the centered two-body statistic and the one-body average

$$W_N \coloneqq \frac{1}{N}\sum_{1\le i\ne j\le N} \tilde{\mathsf{g}}(\xi_i,\xi_j), \qquad R_N \coloneqq \frac{1}{N}\sum_{i=1}^N \psi_\mu(\xi_i), \qquad \psi_\mu \coloneqq h_\mu - 2I_\mu. \tag{5.1}$$

The exact identity

$$\mathsf{K}_{N,\beta}(\mu) = e^{\beta I_\mu}\,\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N + \beta R_N}\right] \tag{5.2}$$

splits the modulated partition function into a centered two-body part and a one-body remainder. The proof of Theorem 2.2 then consists of:

(a) a *uniform exponential-moment bound* for the centered two-body statistics, proved in Section 5.1 by verifying, separately for the Riesz and logarithmic kernels, the hypotheses of the master layer-cake estimate of Section 3.2 (Proposition 3.5);

(b) a *regularized-determinant limit with rate* for $\mathbb{E}[e^{-\frac{\beta}{2}W_N}]$, proved in Section 5.2 by Hilbert–Schmidt truncation, with the uniform bound of part (a) as the key input;

(c) a *one-body reduction* showing that R_N contributes an $O(N^{-1/2})$ error, proved in Section 5.3.

These are assembled into the proof of Theorem 2.2 at the end of Section 5.3. The temperature-interpolation corollary advertised in the introduction is proved in Section 5.4, and the sharpness of the threshold $\mathsf{s} < \mathsf{d}/2$ in Section 5.5. The attractive logarithmic case is proved in Section 6, immediately following this section.

5.1. **Uniform bound for the repulsive log/Riesz kernels.** We now verify the hypotheses of Proposition 3.5 for the repulsive kernels (2.19). Throughout this subsection, $\beta > 0$ is fixed (the case $\beta = 0$ of the statements below being trivial), $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$ under the standing density convention, and all constants may depend on β, d, s, and the relevant bounds for μ, but never on N, u, η, t, or the infrared cutoff T. For $0 \le \eta \le \eta_0$, let $W_{N,\eta}$ denote the statistic (5.1) with g replaced by the truncation g_η of Lemma 4.1, with the convention $\mathsf{g}_0 \coloneqq \mathsf{g}$, so that $W_{N,0} = W_N$.

Proposition 5.1 (Uniform exponential-moment bounds)**.** *Assume $0 \le \mathsf{s} < \mathsf{d}/2$, $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$, and $\beta \ge 0$ is fixed. If $\mathsf{s} = 0$, assume in addition* (2.20)*. Then*

$$\sup_{N\ge 1} \mathsf{K}_{N,\beta}(\mu) \le C_{\mathrm{ub}}(\beta,\mu) < \infty. \tag{5.3}$$

If moreover, in the case $\mathsf{s} = 0$, $\int_{\mathbb{R}^{\mathsf{d}}} e^{-2\beta h_\mu}\,\mathrm{d}\mu < \infty$, then also

$$\sup_{N\ge 1}\sup_{0\le\eta\le\eta_0} \mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_{N,\eta}}\right] \le C_{\mathrm{cf}}(\beta,\mu) < \infty. \tag{5.4}$$

The proof occupies the remainder of this subsection: the case $0 < \mathsf{s} < \mathsf{d}/2$ is treated first, then the case $\mathsf{s} = 0$. In each case we specify the data (g, w) in (3.10), the threshold U_0, and the kernels $q_{N,u}$, verify (M1)–(M3), and check the growth condition (3.14); throughout, we write S_N for the supremum appearing there. In both cases the cap $\mathsf{m}_{N,u}$ will be an explicit number, at least 1, that also bounds the L^∞ norm, $\|\tilde q_{N,u}\|_{L^\infty} \le \mathsf{m}_{N,u}$; hence $1 + \|\tilde q_{N,u}\|_{L^\infty} \le 2\mathsf{m}_{N,u}$ and $\mathsf{S}_N \le 2\sup_{u\ge U_0} \min\{u, 2\mathsf{m}_{N,u}\}\,\mathsf{m}_{N,u}$.

5.1.1. *The case $0 < \mathsf{s} < \mathsf{d}/2$.*

Proof of Proposition 5.1, case $0 < \mathsf{s} < \mathsf{d}/2$: (5.3) and (5.4). Throughout the proof we assume, as we may, $\eta_0 \le 1$.

Step 1: Reduction to the centered statistics. Both (5.3) and (5.4) follow from the family bound

$$\sup_{N\ge 1}\sup_{0\le\eta\le\eta_0} \mathbb{E}_{X_N\sim\mu^{\otimes N}}\left[e^{\Xi_N^{(\eta)}}\right] \le C_{\beta,\mu}, \qquad \Xi_N^{(\eta)} \coloneqq -\frac{\beta}{2}W_{N,\eta} = -\frac{\beta}{2N}\sum_{1\le i\ne j\le N} \tilde{\mathsf{g}}_\eta(x_i,x_j). \tag{5.5}$$

Since $\mu \in L^\infty$ and $0 < \mathsf{s} < \mathsf{d}$, splitting the integral at unit distance gives $h_\mu = \mathsf{g} * \mu \in L^\infty$ with $\|h_\mu\|_{L^\infty} \le C(1 + \|\mu\|_{L^\infty})$. By (2.4),

$$-\beta N \mathsf{F}_N(X_N,\mu) = -\frac{\beta}{2N}\sum_{1\le i\ne j\le N} \tilde{\mathsf{g}}(x_i,x_j) + \frac{\beta}{N}\sum_{i=1}^N h_\mu(x_i) - \beta I_\mu \le -\frac{\beta}{2}W_N + \beta\|h_\mu\|_{L^\infty}, \tag{5.6}$$

using $I_\mu \ge 0$; hence $\mathsf{K}_{N,\beta}(\mu) \le e^{\beta\|h_\mu\|_{L^\infty}}\mathbb{E}[e^{-\frac{\beta}{2}W_N}]$, so the case $\eta = 0$ of (5.5) gives (5.3), while (5.5) is precisely (5.4).

Step 2: Choice of the data. With C_0 the near-field constant in (4.8), C_μ and C_1 the constants in (4.27)–(4.29) and (4.30) of Lemma 4.3(i), C_2 the diagonal constant in (4.4), and $C_4 \coloneqq \max\{8C_2,\ \beta^{-1}\}$, define the frequency cutoff and the threshold

$$K_{\beta,\mu} \coloneqq \max\Big\{(4L\,\beta^2 C_\mu)^{\frac{1}{\mathsf{d}-2\mathsf{s}}},\ (4L\,\beta C_\mu)^{\frac{1}{\mathsf{d}-\mathsf{s}}},\ 1\Big\}, \tag{5.7}$$

$$U_0 \coloneqq \frac{3\beta}{2}\,C_1 K_{\beta,\mu}^{\mathsf{s}} + 1, \tag{5.8}$$

and, for $N \geq 1$, $u \geq U_0$, and $\eta \in [0,\eta_0]$, the adaptive physical scales, kernels, and caps

$$\delta_{N,u} \coloneqq \min\left\{\eta_0,\ \left(\frac{u}{8C_0\beta N}\right)^{\frac{1}{\mathsf{d}-\mathsf{s}}}\right\}, \qquad \bar\delta \coloneqq \max\{\delta_{N,u},\eta\}, \tag{5.9}$$

$$q_{N,u} \coloneqq 2\beta\,\mathsf{g}_{\bar\delta,>K_{\beta,\mu}}, \qquad \mathsf{m}_{N,u} \coloneqq C_4\,\beta\,\delta_{N,u}^{-\mathsf{s}}, \tag{5.10}$$

the hypotheses of Proposition 3.5 hold for each fixed $\eta \in [0,\eta_0]$, with $g = \beta\mathsf{g}_\eta$ and $w = 0$ (so that $\Xi_N = \Xi_N^{(\eta)}$), and the majorant condition (3.16) of Remark 3.6 holds uniformly in $\eta \in [0,\eta_0]$.
Fix $\eta \in [0,\eta_0]$. We use the three-scale decomposition, valid since $\bar\delta \geq \eta$,

$$\mathsf{g}_\eta = (\mathsf{f}_{\bar\delta} - \mathsf{f}_\eta) + \mathsf{g}_{\bar\delta,\leq K_{\beta,\mu}} + \mathsf{g}_{\bar\delta,>K_{\beta,\mu}}, \tag{5.11}$$

where $\mathsf{f}_{\bar\delta} - \mathsf{f}_\eta = \mathsf{g}_\eta - \mathsf{g}_{\bar\delta} \geq 0$ by (4.3) (and $\mathsf{f}_0 \coloneqq 0$).

(M1). For the near-field part, we distinguish the two cases of $\bar\delta = \max\{\delta_{N,u},\eta\}$. If $\eta \geq \delta_{N,u}$, then $\bar\delta = \eta$, so $\mathsf{f}_{\bar\delta} - \mathsf{f}_\eta = 0$ and the near-field part vanishes. If instead $\eta < \delta_{N,u}$, then $\bar\delta = \delta_{N,u}$, and the nonnegativity of $\mathsf{f}_{\delta_{N,u}} - \mathsf{f}_\eta$, the centering identity (2.2), and (4.8) at scale $\delta_{N,u}$ give, for all x, y,

$$-\big(\tilde{\mathsf{f}}_{\delta_{N,u}} - \tilde{\mathsf{f}}_\eta\big)(x,y) \leq \big(\mathsf{f}_{\delta_{N,u}} - \mathsf{f}_\eta\big) * \mu(x) + \big(\mathsf{f}_{\delta_{N,u}} - \mathsf{f}_\eta\big) * \mu(y) \leq 2C_0\delta_{N,u}^{\mathsf{d}-\mathsf{s}}. \tag{5.12}$$

In both cases, for every configuration,

$$-\frac{\beta}{2N}\sum_{1\leq i\neq j\leq N}\big(\tilde{\mathsf{f}}_{\bar\delta} - \tilde{\mathsf{f}}_\eta\big)(x_i,x_j) \leq C_0\beta N\delta_{N,u}^{\mathsf{d}-\mathsf{s}} \leq \frac{u}{8}, \tag{5.13}$$

the last inequality by the definition (5.9) of $\delta_{N,u}$. For the low-frequency part, $\mathsf{g}_{\bar\delta,\leq K}$ is positive definite by (4.18), so Lemma 4.2 and (4.30) give, for every configuration,

$$-\frac{\beta}{2N}\sum_{1\leq i\neq j\leq N}\tilde{\mathsf{g}}_{\bar\delta,\leq K_{\beta,\mu}}(x_i,x_j) \leq \frac{\beta}{2}C_1K_{\beta,\mu}^{\mathsf{s}} \leq \frac{u}{3}, \tag{5.14}$$

the last inequality by (5.8) and $u \geq U_0$. Combining (5.11)–(5.14), on the event $\{\Xi_N^{(\eta)} \geq u\}$,

$$-\frac{1}{2N}\sum_{1\leq i\neq j\leq N}\tilde q_{N,u}(x_i,x_j) \geq 2\left(u - \frac{u}{8} - \frac{u}{3}\right) = \frac{13u}{12} \geq u, \tag{5.15}$$

which is (M1).

(M2) and (M3). By (4.27)–(4.29), the norms of $\tilde q_{N,u} = 2\beta\tilde{\mathsf{g}}_{\bar\delta,>K_{\beta,\mu}}$ satisfy, using the choice (5.7) of $K_{\beta,\mu}$,

$$\|\tilde q_{N,u}\|_{L^\infty L^2(\mu)}^2,\ \|\tilde q_{N,u}\|_{L^2(\mu^{\otimes 2})}^2 \leq 4\beta^2 C_\mu K_{\beta,\mu}^{2\mathsf{s}-\mathsf{d}} \leq \frac{1}{L}, \qquad \|\mathcal{T}_{\tilde q_{N,u}}\|_{\mathrm{op}} \leq 2\beta C_\mu K_{\beta,\mu}^{-(\mathsf{d}-\mathsf{s})} \leq \frac{1}{2L}. \tag{5.16}$$

This is (M3). Moreover, since $\mathsf{g}_{\bar\delta,>K}$ is positive definite with $\mathsf{g}_{\bar\delta,>K}(0) \leq \mathsf{g}_{\bar\delta}(0) \leq C_2\bar\delta^{-\mathsf{s}} \leq C_2\delta_{N,u}^{-\mathsf{s}}$, the L^∞ bound (4.19) gives $\|\tilde{\mathsf{g}}_{\bar\delta,>K}\|_{L^\infty} \leq 4\,\mathsf{g}_{\bar\delta,>K}(0)$, hence

$$\|\tilde q_{N,u}\|_{L^\infty} \leq 8\beta\,\mathsf{g}_{\bar\delta,>K}(0) \leq 8C_2\,\beta\,\delta_{N,u}^{-\mathsf{s}} \leq \mathsf{m}_{N,u}, \tag{5.17}$$

since $C_4 \geq 8C_2$; note also that $\mathsf{m}_{N,u} \geq C_4\beta \geq 1$, since $\delta_{N,u} \leq \eta_0 \leq 1$. Since $q_{N,u}$ is positive definite, Lemma 4.2 gives $-\frac{1}{2N}\sum_{1\leq i\neq j\leq N}\tilde q_{N,u}(x_i,x_j) \leq \frac{1}{2}\|\tilde q_{N,u}\|_{L^\infty} \leq \mathsf{m}_{N,u}$ for every configuration, which

is (M2). Consequently, only layer heights $u \le \mathsf{m}_{N,u} = C_4\beta\,\delta_{N,u}^{-\mathsf{s}}$ contribute. Combined with the definition (5.9) of $\delta_{N,u}$, this constraint gives, after adjusting $C_{\beta,\mu}$,

$$U_0 \le u \le 2\mathsf{m}_{N,u} \quad \Longrightarrow \quad u \le C_{\beta,\mu}\, N^{\mathsf{s}/\mathsf{d}}. \tag{5.18}$$

Indeed, if $\delta_{N,u} = \eta_0$ this is immediate. If instead the second entry of the minimum in (5.9) is active, then $u \le 2C_4\beta(8C_0\beta N/u)^{\mathsf{s}/(\mathsf{d}-\mathsf{s})}$, that is, $u^{\mathsf{d}/(\mathsf{d}-\mathsf{s})} \le C_{\beta,\mu}N^{\mathsf{s}/(\mathsf{d}-\mathsf{s})}$.

(Sublinear growth (3.14)**).** This is the second place where $\mathsf{s} < \mathsf{d}/2$ enters. We claim that

$$\mathsf{S}_N \le 2 \sup_{u\ge U_0} \ \min\{u, 2\mathsf{m}_{N,u}\}\,\mathsf{m}_{N,u} \le C_{\beta,\mu}\, N^{2\mathsf{s}/\mathsf{d}}, \qquad \text{uniformly in } \eta \in [0,\eta_0]. \tag{5.19}$$

Fix $u \ge U_0$. If $\delta_{N,u} = \eta_0$, then $\mathsf{m}_{N,u} = C_4\beta\,\eta_0^{-\mathsf{s}} = O_{\beta,\mu}(1)$, so $\min\{u, 2\mathsf{m}_{N,u}\}\,\mathsf{m}_{N,u} \le 2\mathsf{m}_{N,u}^2 \le C_{\beta,\mu}$. Suppose instead $\delta_{N,u} = (u/8C_0\beta N)^{1/(\mathsf{d}-\mathsf{s})}$. If $u \le 2\mathsf{m}_{N,u}$, then, using (5.17) and then (5.18),

$$\begin{aligned}\min\{u, 2\mathsf{m}_{N,u}\}\,\mathsf{m}_{N,u} \le u\,\mathsf{m}_{N,u} &= C_4\beta\,u\Big(\frac{8C_0\beta N}{u}\Big)^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}} = C_{\beta,\mu}\,u^{\frac{\mathsf{d}-2\mathsf{s}}{\mathsf{d}-\mathsf{s}}}\,N^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}} \\ &\le C_{\beta,\mu}\big(N^{\mathsf{s}/\mathsf{d}}\big)^{\frac{\mathsf{d}-2\mathsf{s}}{\mathsf{d}-\mathsf{s}}}\,N^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}} = C_{\beta,\mu}\,N^{2\mathsf{s}/\mathsf{d}}.\end{aligned} \tag{5.20}$$

If instead $u > 2\mathsf{m}_{N,u}$, then the algebra of (5.18), run in reverse, gives $u \ge c_{\beta,\mu}\,N^{\mathsf{s}/\mathsf{d}}$, so

$$\min\{u, 2\mathsf{m}_{N,u}\}\,\mathsf{m}_{N,u} \le 2\mathsf{m}_{N,u}^2 = C_{\beta,\mu}\Big(\frac{N}{u}\Big)^{\frac{2\mathsf{s}}{\mathsf{d}-\mathsf{s}}} \le C_{\beta,\mu}\big(N^{1-\mathsf{s}/\mathsf{d}}\big)^{\frac{2\mathsf{s}}{\mathsf{d}-\mathsf{s}}} = C_{\beta,\mu}\,N^{2\mathsf{s}/\mathsf{d}}. \tag{5.21}$$

This proves (5.19); since $2\mathsf{s}/\mathsf{d} < 1$ and neither $\mathsf{m}_{N,u}$ nor (5.19) depends on η, the majorant condition (3.16) holds uniformly in $\eta \in [0,\eta_0]$. Step 2 is proved.

Step 3: Conclusion. By Step 2 and Remark 3.6,

$$\sup_{0\le\eta\le\eta_0}\ \sup_{N\ge1}\ \mathbb{E}\big[e^{\Xi_N^{(\eta)}}\big] < \infty. \tag{5.22}$$

This proves (5.5), hence, by Step 1, both (5.3) and (5.4). □

5.1.2. *The case* $\mathsf{s} = 0$.

Proof of Proposition 5.1, logarithmic case.

Step 1: Finite-T reduction. Let $\mathsf{F}_{N,T}$ denote the modulated energy obtained from F_N by replacing g, h_μ, I_μ with the infrared truncations g_T, $h_{\mu,T}$, $I_{\mu,T}$ introduced in Section 4.2; it satisfies the identity

$$-\beta N\mathsf{F}_{N,T}(X_N,\mu) = \frac{\beta}{N}\sum_{i=1}^N h_{\mu,T}(x_i) - \frac{\beta}{2N}\sum_{1\le i\ne j\le N}\tilde{\mathsf{g}}_T(x_i,x_j) - \beta I_{\mu,T}. \tag{5.23}$$

Then (5.3) follows from

$$\sup_{T\ge2}\ \sup_{N\ge1}\ \mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\beta N\mathsf{F}_{N,T}(X_N,\mu)}\Big] \le C_{\beta,\mu}. \tag{5.24}$$

By (2.20), $\mathsf{F}_{N,T}(X_N,\mu) \to \mathsf{F}_N(X_N,\mu)$ for $\mu^{\otimes N}$-a.e. X_N as $T\to\infty$, so (5.24) implies (5.3) by Fatou's lemma.

Step 2: Choice of the data. Set $C_I(\mu) := \sup_{T\ge2}|I_{\mu,T}|$, finite by (2.20). With C_0' the near-field constant in (4.13), C_μ and C the constants in the high-frequency estimates (4.32)–(4.33), $C_{\mu,K}$ the constant of the large-scale cancellation estimate (4.31), and $C_3 := \max\{2C,\ \beta^{-1}\}$, define the frequency cutoff and the threshold

$$K_{\beta,\mu} := \max\Big\{(4L\,\beta^2 C_\mu)^{1/\mathsf{d}},\ (4L\,\beta C_\mu)^{1/\mathsf{d}},\ 1\Big\}, \tag{5.25}$$

$$U_0 \coloneqq 3\beta\big(C_{\mu,K_{\beta,\mu}} + C_I(\mu)\big) + 1, \tag{5.26}$$

and, for $N \geq 1$, $u \geq U_0$, and $T \geq 2$, the adaptive physical scales, kernels, and caps

$$\delta_{N,u} \coloneqq \min\left\{\eta_0,\ \left(\frac{u}{8C_0'\beta N}\right)^{1/\mathsf{d}}\right\}, \tag{5.27}$$

$$q_{N,u} \coloneqq 2\beta\, \mathsf{g}_{\delta_{N,u},T,>K_{\beta,\mu}}, \qquad \mathsf{m}_{N,u} \coloneqq \beta\, C_3\big(1 + \log(1/\delta_{N,u})\big), \tag{5.28}$$

the hypotheses of Proposition 3.5 hold for each fixed $T \geq 2$, with $g = \beta \mathsf{g}_T$ and $w = \beta(h_{\mu,T} - I_{\mu,T})$ (so that $\Xi_N = -\beta N \mathsf{F}_{N,T}$ by (5.23)), and the majorant condition (3.16) of Remark 3.6 holds uniformly in $T \geq 2$.

Fix $T \geq 2$ and decompose $\mathsf{g}_T = \mathsf{f}_{\delta_{N,u}} + \mathsf{g}_{\delta_{N,u},T,\leq K_{\beta,\mu}} + \mathsf{g}_{\delta_{N,u},T,>K_{\beta,\mu}}$.

(M1). By Lemma 4.3(ii), the one-body term pairs with the low-frequency block:

$$\frac{\beta}{N}\sum_{i=1}^{N} h_{\mu,T}(x_i) - \frac{\beta}{2N}\sum_{1\leq i\neq j\leq N} \tilde{\mathsf{g}}_{\delta_{N,u},T,\leq K_{\beta,\mu}}(x_i,x_j) \leq \beta\, C_{\mu,K_{\beta,\mu}}. \tag{5.29}$$

This pairing, rather than a bound on the two terms separately, is what allows the infrared cutoff T to be removed at the end; neither term is bounded uniformly in T on its own. By (4.13) and the centering argument used in (5.13),

$$-\frac{\beta}{2N}\sum_{1\leq i\neq j\leq N} \tilde{\mathsf{f}}_{\delta_{N,u}}(x_i,x_j) \leq C_0'\beta N \delta_{N,u}^{\mathsf{d}} \leq \frac{u}{8}. \tag{5.30}$$

Combining (5.23)–(5.30) with $\beta|I_{\mu,T}| \leq \beta C_I(\mu)$ and (5.26), for $u \geq U_0$,

$$\big\{-\beta N \mathsf{F}_{N,T} \geq u\big\} \subset \left\{-\frac{\beta}{2N}\sum_{1\leq i\neq j\leq N} \tilde{\mathsf{g}}_{\delta_{N,u},T,>K_{\beta,\mu}}(x_i,x_j) \geq u - \frac{u}{8} - \frac{u}{3} \geq \frac{u}{2}\right\}, \tag{5.31}$$

on which $-\frac{1}{2N}\sum_{1\leq i\neq j\leq N} \tilde{q}_{N,u}(x_i,x_j) \geq u$; this is (M1).

(M2) and (M3). By (4.32) and the choice (5.25) of $K_{\beta,\mu}$,

$$\|\tilde{q}_{N,u}\|_{L^\infty L^2(\mu)}^2,\ \|\tilde{q}_{N,u}\|_{L^2(\mu^{\otimes 2})}^2 \leq 4\beta^2 C_\mu K_{\beta,\mu}^{-\mathsf{d}} \leq \frac{1}{L}, \qquad \|\mathcal{T}_{\tilde{q}_{N,u}}\|_{\mathrm{op}} \leq 2\beta C_\mu K_{\beta,\mu}^{-\mathsf{d}} \leq \frac{1}{2L}, \tag{5.32}$$

which is (M3); moreover, uniformly in T, $\|\tilde{q}_{N,u}\|_{L^\infty} \leq 2\beta\, C\big(1 + \log(1/\delta_{N,u})\big) \leq \mathsf{m}_{N,u}$ by (4.33) and $C_3 \geq 2C$, and $\mathsf{m}_{N,u} \geq \beta C_3 \geq 1$. Since $\mathsf{g}_{\delta,T,>K}$ is positive definite in centered quadratic forms, Lemma 4.2 again gives $-\frac{1}{2N}\sum_{1\leq i\neq j\leq N} \tilde{q}_{N,u}(x_i,x_j) \leq \frac{1}{2}\|\tilde{q}_{N,u}\|_{L^\infty} \leq \mathsf{m}_{N,u}$ for every configuration, which is (M2). Since $u \geq 1$ forces $\delta_{N,u} \geq \min\{\eta_0, (8C_0'\beta N)^{-1/\mathsf{d}}\}$, uniformly in $u \geq U_0$ and $T \geq 2$,

$$\mathsf{m}_{N,u} = \beta C_3\big(1 + \log(1/\delta_{N,u})\big) \leq C_{\beta,\mu}\log(e+N). \tag{5.33}$$

(Sublinear growth (3.14)). Since $1 + \|\tilde{q}_{N,u}\|_{L^\infty} \leq 2\mathsf{m}_{N,u}$, the bound (5.33) gives

$$\mathsf{S}_N \leq \sup_{u\geq U_0}\ 4\mathsf{m}_{N,u}^2 \leq C_{\beta,\mu}\big(\log(e+N)\big)^2, \qquad \text{uniformly in } T \geq 2, \tag{5.34}$$

so, since $\mathsf{m}_{N,u}$ does not depend on T, the majorant condition (3.16) holds uniformly in $T \geq 2$. Step 2 is proved.

Step 3: Conclusion for (5.3). By Step 2 and Remark 3.6,

$$\sup_{T\geq 2}\ \sup_{N\geq 1}\ \mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\beta N \mathsf{F}_{N,T}}\Big] < \infty, \tag{5.35}$$

which is (5.24); Step 1 then gives (5.3).

Step 4: The centered family (5.4)**.** Assume now $\int_{\mathbb{R}^{\mathsf{d}}} e^{-2\beta h_\mu}\,\mathrm{d}\mu<\infty$. Fix $\eta\in[0,\eta_0]$ and let $h_\mu^{(\eta)}:=\mathsf{g}_\eta*\mu=h_\mu-\mathsf{f}_\eta*\mu$ and $I_\mu^{(\eta)}:=\frac12\iint_{(\mathbb{R}^{\mathsf{d}})^2}\mathsf{g}_\eta\,\mathrm{d}\mu^{\otimes 2}$ denote the potential and energy of the truncated kernel. Applying the splitting identity (2.4) to the kernel g_η,

$$-\frac{\beta}{2}W_{N,\eta}=-\beta N\mathsf{F}_N^{(\eta)}(X_N,\mu)-\frac{\beta}{N}\sum_{i=1}^N h_\mu^{(\eta)}(x_i)+\beta I_\mu^{(\eta)},\tag{5.36}$$

where $\mathsf{F}_N^{(\eta)}$ is the modulated energy with kernel g_η. By (2.20) and (4.13), $h_\mu^{(\eta)}\ge h_\mu-C_0'\eta_0^{\mathsf{d}}$ and $|I_\mu^{(\eta)}|\le C_\mu$, both uniformly in η. Since the one-body sum in (5.36) is an empirical average of iid samples, two applications of Jensen's inequality give

$$\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{2\beta}{N}\sum_{i=1}^N h_\mu^{(\eta)}(x_i)}\Big]=\mathbb{E}_\mu\big[e^{-\frac{2\beta}{N}h_\mu^{(\eta)}}\big]^N\le\mathbb{E}_\mu\big[e^{-2\beta h_\mu^{(\eta)}}\big]\le e^{2\beta C_0'\eta_0^{\mathsf{d}}}\,\mathbb{E}_\mu\big[e^{-2\beta h_\mu}\big],\tag{5.37}$$

which is finite by (2.22), uniformly in η and N. Hence, by (5.36), Cauchy–Schwarz, and (5.37),

$$\mathbb{E}_{\mu^{\otimes N}}\big[e^{-\frac{\beta}{2}W_{N,\eta}}\big]\le e^{\beta C_\mu+\beta C_0'\eta_0^{\mathsf{d}}}\,\mathbb{E}_{\mu^{\otimes N}}\big[e^{-2\beta N\mathsf{F}_N^{(\eta)}}\big]^{1/2}\,\mathbb{E}_\mu\big[e^{-2\beta h_\mu}\big]^{1/2},\tag{5.38}$$

and it therefore suffices to bound $\mathbb{E}[e^{-2\beta N\mathsf{F}_N^{(\eta)}}]$ uniformly in N and η.

This follows by repeating Steps 1–3 with the kernel g_η in place of g and 2β in place of β; we indicate the only changes. At finite T, one applies Proposition 3.5 with $g=2\beta\mathsf{g}_{\eta,T}$, where $\mathsf{g}_{\eta,T}=\mathsf{g}_T-\mathsf{f}_\eta$, and $w=2\beta(h_{\mu,T}^{(\eta)}-I_{\mu,T}^{(\eta)})$, with one-body potential $h_{\mu,T}^{(\eta)}=h_{\mu,T}-\mathsf{f}_\eta*\mu$; a.e. convergence $\mathsf{F}_{N,T}^{(\eta)}\to\mathsf{F}_N^{(\eta)}$ again follows from (2.20) since $0\le\mathsf{f}_\eta\le\mathsf{f}_{\eta_0}$. In the proof of Step 2, one uses the effective scale $\bar\delta:=\max\{\delta_{N,u},\eta\}$ and the decomposition

$$\mathsf{g}_{\eta,T}=(\mathsf{f}_{\bar\delta}-\mathsf{f}_\eta)+\mathsf{g}_{\bar\delta,T,\le K_{2\beta,\mu}}+\mathsf{g}_{\bar\delta,T,>K_{2\beta,\mu}}.\tag{5.39}$$

The near-field part obeys (5.30) verbatim (it vanishes if $\eta\ge\delta_{N,u}$). The low-frequency pairing (5.29) is applied with $h_{\mu,T}$ and costs the additional bounded term $\frac{2\beta}{N}\sum_i\mathsf{f}_\eta*\mu(x_i)\le 2\beta C_0'\eta_0^{\mathsf{d}}$, which is absorbed into the constant $C_{\mu,K_{2\beta,\mu}}$ in (5.26). The high-frequency verification of (M2)–(M3) and of the sublinear growth (3.14) is unchanged, with $\bar\delta\ge\delta_{N,u}$ giving $1+\log\Big(1/\bar\delta\Big)\le 1+\log(1/\delta_{N,u})$. Proposition 3.5, in the uniform form of Remark 3.6, and Fatou's lemma then give (5.4). □

Remark 5.2 (Role of the frequency truncation)**.** The role of the frequency cutoff $K_{\beta,\mu}$ is to replace a high-temperature smallness condition on β by the choice of a (β,μ)-dependent parameter: the low frequencies cost a finite deterministic offset, reflected in the thresholds (5.8) and (5.26), while the high frequencies have the mixed $L^\infty L^2(\mu)$ norm, the $L^2(\mu^{\otimes 2})$ norm, and the integral-operator norm small enough for (3.13) exactly because $\widehat{\mathsf{g}}\in L^2(|\xi|>1)$ in the range $\mathsf{s}<\mathsf{d}/2$. The remaining L^∞ norm is controlled by the adaptive physical truncation together with the deterministic diagonal bound of Lemma 4.2, which is available before any Fourier cutoff is introduced. No smallness assumption on β is used. Note also that the restriction $\mathsf{s}<\mathsf{d}/2$ enters the verification exactly twice: in the finiteness of the tail integrals (4.27)–(4.28), and in the exponent $2\mathsf{s}/\mathsf{d}<1$ in (5.19).

We conclude this subsection with the interpolated exponential-moment bound needed for the determinant limit. For $0<\eta\le\eta_0$ and $t\in[0,1]$, set $\mathsf{g}_{\eta,t}:=\mathsf{g}_\eta+t\mathsf{f}_\eta$ and let $W_{N,\eta,t}$ denote the statistic (5.1) with kernel $\mathsf{g}_{\eta,t}$.

Lemma 5.3 (Interpolated kernels)**.** *Under the hypotheses of Proposition 5.1, including* (2.22) *when* $\mathsf{s}=0$*, for every fixed* $\beta\ge 0$*,*

$$\sup_{N\ge1}\ \sup_{0<\eta\le\eta_0}\ \sup_{0\le t\le1}\ \mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\beta W_{N,\eta,t}}\Big]\le C_{\mathrm{cf}}(2\beta,\mu).\tag{5.40}$$

Proof. Since $\mathsf{f}_\eta = \mathsf{g} - \mathsf{g}_\eta$, the interpolated kernel is the convex combination $\mathsf{g}_{\eta,t} = (1-t)\mathsf{g}_\eta + t\mathsf{g}$, and the Hoeffding centering (2.2) is linear in the kernel. Hence

$$W_{N,\eta,t} = (1-t)W_{N,\eta} + tW_N \tag{5.41}$$

pointwise, and Hölder's inequality with exponents $\frac{1}{1-t}$ and $\frac{1}{t}$ gives

$$\mathbb{E}\Big[e^{-\beta W_{N,\eta,t}}\Big] \le \mathbb{E}\Big[e^{-\beta W_{N,\eta}}\Big]^{1-t}\,\mathbb{E}\Big[e^{-\beta W_N}\Big]^{t} \le C_{\mathrm{cf}}(2\beta,\mu), \tag{5.42}$$

by (5.4) applied at inverse temperature 2β. In the logarithmic case, this requires precisely the first condition in (2.22). □

5.2. **Regularized determinant limit.** Having proved uniform integrability of the exponential weights, we now identify the limiting normalization constant for the centered two-body statistic. If $\beta = 0$, then $\mathbb{E}_{\mu^{\otimes N}}[1] = 1 = \det_2(\mathrm{Id})^{-1/2}$, so the remaining argument assumes $\beta > 0$.

The expansion at a fixed truncation scale rests on the following bounded-feature quadratic Laplace estimate, which is applied after truncating the kernel.

5.2.1. *Feature-space quadratic Laplace estimate.* We first record the fixed-kernel estimate used after truncating the interaction. The formulation allows the squared feature norm $x \mapsto \|k(x)\|_{\mathcal{H}}^2$ to be unbounded, provided it satisfies the exponential-moment condition (5.44); this is needed for logarithmic truncations against noncompact backgrounds.

Proposition 5.4 (Feature-space quadratic Laplace estimate)**.** *Assume that $\tilde{\mathsf{g}}$ admits a Hilbert-space feature representation: there are a separable real Hilbert space $\mathcal{H}$ and a strongly measurable map $k : \mathbb{R}^{\mathsf{d}} \to \mathcal{H}$ such that*

$$\tilde{\mathsf{g}}(x,y) = \langle k(x), k(y)\rangle_{\mathcal{H}} \quad \text{for } \mu^{\otimes 2}\text{-a.e. } (x,y), \qquad \int_{\mathbb{R}^{\mathsf{d}}} k \,\mathrm{d}\mu = 0. \tag{5.43}$$

Set

$$D(x) \coloneqq \|k(x)\|_{\mathcal{H}}^2, \qquad \mathcal{M}_\beta(k) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} e^{\beta D(x)/2}\Big(1 + D(x)^4\Big)\,\mathrm{d}\mu(x). \tag{5.44}$$

If $\mathcal{M}_\beta(k) < \infty$, then there is a constant $C_\beta < \infty$, depending only on β, such that, for every $N \ge 1$,

$$\Big|\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2}W_N}\Big] - \det_2(\mathrm{Id} + \beta T_\mu)^{-1/2}\Big| \le \frac{C_\beta}{N}(1 + \mathcal{M}_\beta(k))^3. \tag{5.45}$$

In particular, if $k \in L^\infty(\mu;\mathcal{H})$, then

$$\Big|\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2}W_N}\Big] - \det_2(\mathrm{Id} + \beta T_\mu)^{-1/2}\Big| \le C_\beta \exp\Big(C_\beta\beta\|k\|_{L^\infty(\mu;\mathcal{H})}^2\Big)N^{-1}. \tag{5.46}$$

Proof. The case $\beta = 0$ is immediate, so assume $\beta > 0$. Let Z be an isonormal Gaussian process over $\mathcal{H}$, independent of the ξ_i's, and write $\mathsf{Z}_x \coloneqq \mathsf{Z}(k(x))$. Thus, Z_x is centered Gaussian with variance $D(x)$. Define

$$\bar{D} \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} D \,\mathrm{d}\mu, \qquad Q(\mathsf{Z}) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \mathsf{Z}_x^2 \,\mathrm{d}\mu(x), \tag{5.47}$$

$$B(\mathsf{Z}) \coloneqq \frac{\beta}{2}\Big(\bar{D} - Q(\mathsf{Z})\Big), \qquad R(\mathsf{Z}) \coloneqq \beta^{3/2}\int_{\mathbb{R}^{\mathsf{d}}}\Big(\frac{1}{2}D(x)\mathsf{Z}_x - \frac{1}{6}\mathsf{Z}_x^3\Big)\,\mathrm{d}\mu(x). \tag{5.48}$$

The moment assumption implies $D \in L^p(\mu)$ for every $1 \le p \le 4$, so all of these quantities are well-defined. Moreover,

$$\mathbb{E}_{\mathsf{Z}}[B^2 + R^2] \le C_\beta(1 + \mathcal{M}_\beta(k))^2. \tag{5.49}$$

Indeed, Wick's formula gives

$$\mathrm{Var}_{\mathsf{Z}}(Q) = 2\iint_{(\mathbb{R}^{\mathsf{d}})^2}\langle k(x),k(y)\rangle_{\mathcal{H}}^2 \,\mathrm{d}\mu^{\otimes 2}(x,y) \le 2\bar{D}^2, \tag{5.50}$$

while Jensen's inequality in x and the Gaussian sixth-moment identity give $\mathbb{E}_{\mathsf{Z}}[R^2] \le C_\beta \int_{\mathbb{R}^{\mathsf{d}}} D^3 \, \mathrm{d}\mu$.

Step 1: Oscillatory Hubbard–Stratonovich transformation. With $S_N := N^{-1/2} \sum_{i=1}^N k(\xi_i)$, the feature representation gives

$$W_N = \|S_N\|_{\mathcal{H}}^2 - \frac{1}{N} \sum_{i=1}^N D(\xi_i). \tag{5.51}$$

Consequently,

$$\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2} W_N}\Big] = \mathbb{E}_{\mathsf{Z}}\big[A_N(\mathsf{Z})^N\big], \qquad A_N(\mathsf{Z}) := \int_{\mathbb{R}^{\mathsf{d}}} \exp\left(\mathrm{i}\sqrt{\frac{\beta}{N}} \mathsf{Z}_x + \frac{\beta}{2N} D(x) \right) \mathrm{d}\mu(x). \tag{5.52}$$

To justify Fubini, set

$$a_N := \int_{\mathbb{R}^{\mathsf{d}}} e^{\beta D/(2N)} \, \mathrm{d}\mu, \qquad A_\beta := \int_{\mathbb{R}^{\mathsf{d}}} e^{\beta D/2} \, \mathrm{d}\mu. \tag{5.53}$$

Since $r \mapsto r^{1/N}$ is concave,

$$|A_N(\mathsf{Z})| \le a_N, \qquad a_N^N \le A_\beta, \qquad e^{\beta \bar{D}/2} \le A_\beta. \tag{5.54}$$

The same bounds justify every interchange below. Equivalently, one may first truncate to the sets $\{D \le m\}$ and then pass to the limit by dominated convergence using (5.44).

Step 2: Expansion of A_N. Set

$$f(\tau) := \int_{\mathbb{R}^{\mathsf{d}}} \exp\left(\mathrm{i}\tau\sqrt{\beta}\, \mathsf{Z}_x + \frac{\tau^2 \beta}{2} D(x) \right) \mathrm{d}\mu(x). \tag{5.55}$$

Then $A_N = f(N^{-1/2})$, $f(0) = 1$, and the centering of k gives $f'(0) = 0$. Direct differentiation yields $f''(0)/2 = B$ and $f'''(0)/6 = \mathrm{i}R$. Hence

$$A_N = 1 + \frac{B}{N} + \mathrm{i}\frac{R}{N^{3/2}} + \frac{E_N}{N^2}, \tag{5.56}$$

where B is even and R is odd under $\mathsf{Z} \mapsto -\mathsf{Z}$, and

$$\sup_{N \ge 1} \mathbb{E}_{\mathsf{Z}} |E_N| \le C_\beta \mathcal{M}_\beta(k). \tag{5.57}$$

Indeed, if $g_x(\tau) = \mathrm{i}\tau\sqrt{\beta}\, \mathsf{Z}_x + \tau^2 \beta D(x)/2$, then

$$(e^{g_x})'''' = e^{g_x}\Big((g_x')^4 + 6(g_x')^2 g_x'' + 3(g_x'')^2 \Big), \tag{5.58}$$

with $\operatorname{Re} g_x(\tau) \le \beta D(x)/2$, $|g_x'(\tau)| \le \sqrt{\beta}|\mathsf{Z}_x| + \beta D(x)$, and $g_x'' = \beta D(x)$ for $0 \le \tau \le 1$. Taking the Gaussian expectation and using $\mathbb{E}|\mathsf{Z}_x|^{2p} = C_p D(x)^p$ proves (5.57).

Step 3: Passage to the Gaussian limit. Put $u_N = N^{-1}(B + \mathrm{i}R/\sqrt{N})$. Then

$$A_N^N - e^B = \Big(A_N^N - e^{N u_N} \Big) + e^B \Big(e^{\mathrm{i}R/\sqrt{N}} - 1 - \mathrm{i}R/\sqrt{N} \Big) + \frac{\mathrm{i}}{\sqrt{N}} e^B R. \tag{5.59}$$

The last term has zero expectation by parity and is integrable by (5.54) and (5.49). The middle term is bounded in expectation by $A_\beta \mathbb{E}[R^2]/(2N)$. For the first term, the telescoping identity gives, on $\{|u_N| \le 1\}$,

$$|A_N^N - e^{N u_N}| \le \frac{A_\beta}{N} \Big(|E_N| + e(|B| + |R|)^2 \Big). \tag{5.60}$$

On $\{|u_N| > 1\}$, one has $|B| + |R| > N$, while $|A_N|^N + |e^{N u_N}| \le 2A_\beta$; hence the contribution of this event is at most

$$2A_\beta N^{-2} \mathbb{E}\big[(|B| + |R|)^2\big]. \tag{5.61}$$

Combining these estimates yields

$$\left|\mathbb{E}_{\mathsf{Z}}[A_N^N] - \mathbb{E}_{\mathsf{Z}}[e^B]\right| \le \frac{C_\beta}{N}(1+\mathcal{M}_\beta(k))^3. \tag{5.62}$$

Step 4: Identification of the leading term. The covariance operator

$$\Sigma f \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \langle k(x), f\rangle_{\mathcal{H}} k(x)\ \mathrm{d}\mu(x) \tag{5.63}$$

is positive and trace class, with $\operatorname{Tr}\Sigma = \bar{D}$, and has the same nonzero eigenvalues $(\lambda_j)_{j\ge1}$ as T_μ. Diagonalizing Σ therefore gives

$$\mathbb{E}_{\mathsf{Z}}[e^B] = \prod_{j\ge1} e^{\beta\lambda_j/2}(1+\beta\lambda_j)^{-1/2} = \det{}_2(\mathrm{Id}+\beta T_\mu)^{-1/2}. \tag{5.64}$$

Together with (5.62), this proves (5.45). If $\|k\|_{L^\infty} \le M$, then $\mathcal{M}_\beta(k) \le e^{\beta M^2/2}(1+M^8)$, and (5.46) follows after increasing C_β. □

Proposition 5.5. *Assume $0 \le \mathsf{s} < \mathsf{d}/2$, $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$, and $\beta > 0$ is fixed. If $\mathsf{s} = 0$, assume in addition* (2.20) *and* (2.22)*. Then there are constants $C_{\beta,\mu} < \infty$ and, in the case $\mathsf{s} = 0$, $c_{\beta,\mu} > 0$ such that, for all $N \ge 1$,*

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N}\right] - \det{}_2(\mathrm{Id}+\beta T_\mu)^{-1/2}\right| \le C_{\beta,\mu}\begin{cases}(\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}, & 0<\mathsf{s}<\mathsf{d}/2,\\ (e+N)^{-c_{\beta,\mu}}, & \mathsf{s}=0.\end{cases} \tag{5.65}$$

Proof. Throughout, $W_{N,\eta}$ and $T_{\mu,\eta}$ denote the analogues of (5.1) and (2.13) with g replaced by g_η, using the logarithmic truncation of (4.10) when $\mathsf{s} = 0$. Set

$$\alpha_\eta \coloneqq \begin{cases}\eta^{-\mathsf{s}}, & 0<\mathsf{s}<\mathsf{d}/2,\\ 1+\log(1/\eta), & \mathsf{s}=0,\end{cases} \qquad \rho_\eta = \eta^{\mathsf{d}/2-\mathsf{s}} \tag{5.66}$$

and, with $C_{\mathrm{L}} = C_{\mathrm{L}}(\beta,\mu)$ the constant furnished by Step 1 below, fix the truncation scale

$$\eta(N) \coloneqq \begin{cases}\left((2C_{\mathrm{L}})^{-1}\log N\right)^{-1/\mathsf{s}}, & 0<\mathsf{s}<\mathsf{d}/2,\\ N^{-a}, \quad a \coloneqq \min\{1,\ (2C_{\mathrm{L}})^{-1}\}, & \mathsf{s}=0,\end{cases} \qquad \text{so that} \qquad e^{C_{\mathrm{L}}\alpha_{\eta(N)}} \le e^{C_{\mathrm{L}}} N^{1/2}. \tag{5.67}$$

Step 1: Laplace expansion at fixed scale. We claim that there is $C_{\mathrm{L}} = C_{\mathrm{L}}(\beta,\mu) < \infty$ such that, for all $N \ge 1$ and $0 < \eta \le \eta_0$,

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_{N,\eta}}\right] - \det{}_2(\mathrm{Id}+\beta T_{\mu,\eta})^{-1/2}\right| \le C_{\mathrm{L}}\, e^{C_{\mathrm{L}}\alpha_\eta}\, N^{-1}. \tag{5.68}$$

Assume first $0 < \mathsf{s} < \mathsf{d}/2$. The centered truncated kernel has a bounded feature representation as in Proposition 5.4. Indeed, if

$$\mathsf{g}_\eta(x-y) = \langle \kappa_\eta(x), \kappa_\eta(y)\rangle_{\mathcal{H}_\eta}, \tag{5.69}$$

then

$$\tilde{\mathsf{g}}_\eta(x,y) = \Big\langle \kappa_\eta(x) - \int_{\mathbb{R}^{\mathsf{d}}} \kappa_\eta\ \mathrm{d}\mu, \kappa_\eta(y) - \int_{\mathbb{R}^{\mathsf{d}}} \kappa_\eta\ \mathrm{d}\mu\Big\rangle_{\mathcal{H}_\eta}, \tag{5.70}$$

so centering preserves positive definiteness and

$$\sup_x \Big\|\kappa_\eta(x) - \int_{\mathbb{R}^{\mathsf{d}}} \kappa_\eta\ \mathrm{d}\mu\Big\|_{\mathcal{H}_\eta}^2 \le 4\mathsf{g}_\eta(0) \le C\eta^{-\mathsf{s}} = C\alpha_\eta. \tag{5.71}$$

The bounded specialization (5.46) gives (5.68).

Assume now $\mathsf{s} = 0$. The centered kernel $\tilde{\mathsf{g}}_\eta$ is positive definite; let k_η be its canonical feature map and set

$$D_\eta(x) := \|k_\eta(x)\|^2_{\mathcal{H}_\eta} = \tilde{\mathsf{g}}_\eta(x,x) = \mathsf{g}_\eta(0) - 2h^{(\eta)}_\mu(x) + 2I^{(\eta)}_\mu, \tag{5.72}$$

where $h^{(\eta)}_\mu = \mathsf{g}_\eta * \mu$ and $I^{(\eta)}_\mu = \frac{1}{2}\iint_{(\mathbb{R}^{\mathsf{d}})^2} \mathsf{g}_\eta \, \mathrm{d}\mu^{\otimes 2}$. Since $D_\eta \in L^1(\mu)$ and $\tilde{\mathsf{g}}_\eta$ is canonical, the feature map may be chosen with $\int_{\mathbb{R}^{\mathsf{d}}} k_\eta \, \mathrm{d}\mu = 0$. By $h^{(\eta)}_\mu = h_\mu - \mathsf{f}_\eta * \mu$, (4.13), the diagonal estimate (4.11), and the uniform boundedness of $I^{(\eta)}_\mu$, there is $C_\mu < \infty$ such that

$$0 \le D_\eta(x) \le C_\mu \alpha_\eta - 2h_\mu(x) \qquad \text{for } \mu\text{-a.e. } x. \tag{5.73}$$

Since h_μ is bounded above and $\int_{\mathbb{R}^{\mathsf{d}}} e^{-4\beta h_\mu} \, \mathrm{d}\mu < \infty$, the polynomial factor in (5.44) can be absorbed by this exponential moment, and

$$\mathcal{M}_\beta(k_\eta) = \int_{\mathbb{R}^{\mathsf{d}}} e^{\beta D_\eta/2}\Big(1 + D^4_\eta\Big) \, \mathrm{d}\mu \le C_{\beta,\mu} e^{C_{\beta,\mu}\alpha_\eta}. \tag{5.74}$$

Applying Proposition 5.4 and enlarging the constant to absorb the fixed power in (5.45) proves (5.68).

Step 2: Determinant truncation error. It holds that

$$\|\tilde{\mathsf{f}}_\eta\|_{L^2(\mu^{\otimes 2})} \le C_\mu \, \rho_\eta \qquad \text{and} \qquad \Big|\det\nolimits_2(\mathrm{Id} + \beta T_\mu)^{-1/2} - \det\nolimits_2(\mathrm{Id} + \beta T_{\mu,\eta})^{-1/2}\Big| \le C_{\beta,\mu} \, \rho_\eta. \tag{5.75}$$

The short-range remainder satisfies

$$\|\mathsf{f}_\eta\|_{L^2(\mu^{\otimes 2})} \le C_\mu \, \rho_\eta : \tag{5.76}$$

for $0 < \mathsf{s} < \mathsf{d}/2$, this is the scaling in (4.3), while for $\mathsf{s} = 0$, one has $\mathsf{f}_\eta(x) = F(x/\eta)$ for a fixed $F \in L^2(\mathbb{R}^{\mathsf{d}})$, using (4.12). Since Hoeffding centering is bounded on $L^2(\mu^{\otimes 2})$, the first bound in (5.75) follows.

For the second, the continuity estimate of [Sim05, Theorem 9.2(c)] shows that $A \mapsto \det_2(\mathrm{Id} + \beta A)$ is Lipschitz on Hilbert–Schmidt balls. On the cone of nonnegative self-adjoint Hilbert–Schmidt operators one also has

$$\det\nolimits_2(\mathrm{Id} + \beta A) \ge \exp\left(-\frac{\beta^2}{2}\|A\|^2_{\mathrm{HS}}\right), \tag{5.77}$$

so composing with the inverse square root is locally Lipschitz there. The asserted estimate now follows from

$$\|T_\mu - T_{\mu,\eta}\|_{\mathrm{HS}} = \|\tilde{\mathsf{f}}_\eta\|_{L^2(\mu^{\otimes 2})}. \tag{5.78}$$

Step 3: Finite-N truncation error. It holds that

$$\Big|\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2}W_N}\Big] - \mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2}W_{N,\eta}}\Big]\Big| \le C_{\beta,\mu} \, \rho_\eta. \tag{5.79}$$

With $\mathsf{g}_{\eta,t} = \mathsf{g}_\eta + t\mathsf{f}_\eta$ and $W_{N,\eta,t}$ as in Lemma 5.3, differentiating in t and applying Cauchy–Schwarz gives

$$\Big|\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2}W_{N,\eta,t}}\Big]\Big| \le \frac{\beta}{2}\,\mathbb{E}_{\mu^{\otimes N}}\Big[\Big|\frac{1}{N}\sum_{i\neq j}\tilde{\mathsf{f}}_\eta(\xi_i,\xi_j)\Big|^2\Big]^{1/2}\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\beta W_{N,\eta,t}}\Big]^{1/2}, \tag{5.80}$$

and, since $\tilde{\mathsf{f}}_\eta$ is canonical,

$$\mathbb{E}_{\mu^{\otimes N}}\Big[\Big|\frac{1}{N}\sum_{i\neq j}\tilde{\mathsf{f}}_\eta(\xi_i,\xi_j)\Big|^2\Big] = \frac{2(N-1)}{N}\|\tilde{\mathsf{f}}_\eta\|^2_{L^2(\mu^{\otimes 2})} \le 2\|\tilde{\mathsf{f}}_\eta\|^2_{L^2(\mu^{\otimes 2})}. \tag{5.81}$$

Combining Lemma 5.3, (5.81), and (5.75), and integrating t from 0 to 1, proves (5.79).

Step 4: Conclusion. For $N = 1$ and the finitely many N with $\eta(N) > \eta_0$, the left-hand side of (5.65) is bounded by a finite constant depending on (β, μ), by Lemma 5.3 and the estimate

$$\det_2(\mathrm{Id} + \beta T_\mu)^{-1/2} \le \exp\left(\frac{\beta^2}{4}\|T_\mu\|_{\mathrm{HS}}^2\right), \tag{5.82}$$

which follows from $x - \log(1+x) \le x^2/2$ for $x \ge 0$, and is absorbed into $C_{\beta,\mu}$, the right-hand side of (5.65) being bounded below on finite sets. For the remaining N, summing Steps 1–3 at the scale $\eta = \eta(N)$ and using (5.67),

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N}\right] - \det_2(\mathrm{Id} + \beta T_\mu)^{-1/2}\right| \le C_{\beta,\mu}\,\rho_{\eta(N)} + C_{\mathrm{L}} e^{C_{\mathrm{L}}}\, N^{-1/2}. \tag{5.83}$$

If $0 < \mathsf{s} < \mathsf{d}/2$, then $\rho_{\eta(N)} = \left((2C_{\mathrm{L}})^{-1}\log N\right)^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}} \le C_{\beta,\mu}(\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}$, which dominates $N^{-1/2}$ and gives the first line of (5.65). If $\mathsf{s} = 0$, then $\rho_{\eta(N)} = N^{-a\mathsf{d}/2} \le C(e+N)^{-a\mathsf{d}/2}$, and the second line follows with $c_{\beta,\mu} \coloneqq \min\{a\mathsf{d}/2,\ 1/2\}$. □

5.3. **One-body reduction and proof of the main estimate.**

Lemma 5.6 (One-body reduction)**.** *Assume the hypotheses of Proposition 5.5, and recall* $\psi_\mu \coloneqq h_\mu - 2I_\mu$ *and* $R_N \coloneqq \frac{1}{N}\sum_{i=1}^N \psi_\mu(\xi_i)$ *from* (5.2)*. Then, for all* $N \ge 1$*,*

$$\begin{aligned}\left|\mathsf{K}_{N,\beta}(\mu) - e^{\beta I_\mu}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N}\right]\right| \\ \le \beta\,\|\psi_\mu\|_{L^2(\mu)}\Big(C_{\mathrm{ub}}(2\beta,\mu)^{1/2} + C_{\mathrm{ub}}(4\beta,\mu)^{1/4}\,\mathbb{E}_\mu\big[e^{-4\beta\psi_\mu}\big]^{1/4}\Big)\,N^{-1/2}.\end{aligned} \tag{5.84}$$

Proof. Note first that the hypotheses guarantee $\psi_\mu \in L^2(\mu)$ and $e^{-4\beta\psi_\mu} \in L^1(\mu)$: for $0 < \mathsf{s} < \mathsf{d}/2$ both follow from $h_\mu \in L^\infty$ as in Step 1 of the Riesz case, and for $\mathsf{s} = 0$ from (2.22), since h_μ is bounded above (Remark 2.5) and $x^2 \le C_\beta(1 + e^{-4\beta x})$ for x in a half-line $(-\infty, C]$. By (2.4) and the definition (5.1) of W_N, the identity (5.2) holds:

$$\mathsf{K}_{N,\beta}(\mu) = e^{\beta I_\mu}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N + \beta R_N}\right]. \tag{5.85}$$

Since $\mathbb{E}_\mu[\psi_\mu] = 0$,

$$\mathbb{E}_{\mu^{\otimes N}}[R_N^2] = \frac{\|\psi_\mu\|_{L^2(\mu)}^2}{N}, \tag{5.86}$$

and, since R_N is an empirical average of iid samples, two applications of Jensen's inequality give

$$\mathbb{E}_{\mu^{\otimes N}}\big[e^{-4\beta R_N}\big] = \mathbb{E}_\mu\big[e^{-\frac{4\beta}{N}\psi_\mu}\big]^N \le \mathbb{E}_\mu\big[e^{-4\beta\psi_\mu}\big]. \tag{5.87}$$

Moreover, by Cauchy–Schwarz, (5.85) at inverse temperature 4β, (5.87), and the uniform bound (5.3),

$$\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\beta W_N}\right] \le \mathbb{E}_{\mu^{\otimes N}}\left[e^{-2\beta W_N + 4\beta R_N}\right]^{1/2}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-4\beta R_N}\right]^{1/2} \le e^{-2\beta I_\mu}\,C_{\mathrm{ub}}(4\beta,\mu)^{1/2}\,\mathbb{E}_\mu\big[e^{-4\beta\psi_\mu}\big]^{1/2}. \tag{5.88}$$

Therefore, by $|e^z - 1| \le |z|(1 + e^z)$ with $z = \beta R_N$, followed by Cauchy–Schwarz in each term,

$$\begin{aligned}&\left|\mathsf{K}_{N,\beta}(\mu) - e^{\beta I_\mu}\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\frac{\beta}{2}W_N}\right]\right| \\ &\le e^{\beta I_\mu}\,\beta\left(\mathbb{E}_{\mu^{\otimes N}}\left[|R_N|e^{-\frac{\beta}{2}W_N}\right] + \mathbb{E}_{\mu^{\otimes N}}\left[|R_N|e^{-\frac{\beta}{2}W_N + \beta R_N}\right]\right) \\ &\le e^{\beta I_\mu}\,\beta\,\mathbb{E}_{\mu^{\otimes N}}[R_N^2]^{1/2}\Big(\mathbb{E}_{\mu^{\otimes N}}\left[e^{-\beta W_N}\right]^{1/2} + \mathbb{E}_{\mu^{\otimes N}}\left[e^{-\beta W_N + 2\beta R_N}\right]^{1/2}\Big).\end{aligned} \tag{5.89}$$

By (5.85) at inverse temperature 2β and (5.3), $\mathbb{E}_{\mu^{\otimes N}}[e^{-\beta W_N+2\beta R_N}] \le e^{-2\beta I_\mu} C_{\mathrm{ub}}(2\beta,\mu)$. Combining this with (5.86) and (5.88), the prefactor $e^{\beta I_\mu}$ cancels against the factors $e^{-\beta I_\mu}$ from the two square roots, and (5.84) follows. □

Proof of Theorem 2.2. The uniform bound (2.21) is exactly (5.3) in Proposition 5.1. For the convergence rate, combining Lemma 5.6, the determinant estimate (5.65), and the definition (2.16) of $\mathsf{K}_{\infty,\beta}(\mu)$,

$$\begin{aligned}\Big|\mathsf{K}_{N,\beta}(\mu)-\mathsf{K}_{\infty,\beta}(\mu)\Big| &\le C_{\beta,\mu}N^{-1/2}+C_{\beta,\mu}\begin{cases}(\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}, & 0<\mathsf{s}<\mathsf{d}/2,\\ (e+N)^{-c_{\beta,\mu}}, & \mathsf{s}=0\end{cases}\\ &\le C_{\mathrm{rate}}(\beta,\mu)\begin{cases}(\log(e+N))^{-\frac{\mathsf{d}-2\mathsf{s}}{2\mathsf{s}}}, & 0<\mathsf{s}<\mathsf{d}/2,\\ (e+N)^{-c_{\log}(\beta,\mu)}, & \mathsf{s}=0,\end{cases}\end{aligned} \tag{5.90}$$

which proves (2.23). □

5.4. **Temperature interpolation corollary.** The following corollary records how the fixed-temperature estimate interpolates when the inverse temperature varies with N. It uses the pointwise lower bound for modulated energies from [HCRS25].

Corollary 5.7 (Temperature interpolation)**.** *Assume $0 \le \mathsf{s} < \mathsf{d}/2$ and $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$; if $\mathsf{s}=0$, assume* (2.20)*. Set*

$$a_{N,\mathsf{s}} \coloneqq \begin{cases}\log(e+N), & \mathsf{s}=0,\\ N^{\mathsf{s}/\mathsf{d}}, & 0<\mathsf{s}<\mathsf{d}.\end{cases} \tag{5.91}$$

For every $B>0$, there is a constant $C_B(\mu)<\infty$ such that

$$\sup_{N\ge 1}\sup_{0\le\beta\le B}\mathsf{K}_{N,\beta}(\mu) \le C_B(\mu) \tag{5.92}$$

and, for all $\beta \ge 0$ and $N \ge 1$,

$$\mathsf{K}_{N,\beta}(\mu) \le C_B(\mu)\exp\Big(C_B(\mu)\,(\beta-B)_+\,a_{N,\mathsf{s}}\Big). \tag{5.93}$$

In particular, for every $B_*>0$ and $\gamma\in\mathbb{R}$, there is a constant $C_{B_*,\mu}<\infty$ such that, along the temperature scaling $\beta_N=\beta_* N^\gamma$,

$$\sup_{0\le\beta_*\le B_*}\log \mathsf{K}_{N,\beta_* N^\gamma}(\mu) \le \begin{cases}C_{B_*,\mu}, & \gamma\le 0,\\ C_{B_*,\mu}\,N^\gamma\log(e+N), & \gamma>0,\ \mathsf{s}=0,\\ C_{B_*,\mu}\,N^{\gamma+\mathsf{s}/\mathsf{d}}, & \gamma>0,\ 0<\mathsf{s}<\mathsf{d}/2.\end{cases} \tag{5.94}$$

Remark 5.8 (Microscopic endpoint)**.** At the microscopic temperature scale, namely $\beta_N \sim N$ in the logarithmic case and $\beta_N \sim N^{1-\mathsf{s}/\mathsf{d}}$ in the positive Riesz case, (5.94) gives $\log \mathsf{K}_{N,\beta_N} = O(N\log(e+N))$ for $\mathsf{s}=0$ and $O(N)$ for $0<\mathsf{s}<\mathsf{d}/2$. These are the expected endpoint orders in the standard Coulomb/Riesz-gas normalization; see [Ser26, Chapters 8–13]. This comparison is only a consistency check: Corollary 5.7 is an upper bound for arbitrary bounded backgrounds and neither identifies the next-order constants nor establishes sharpness at intermediate temperature scales.

Proof of Corollary 5.7. We combine the uniform bound (5.3) with the pointwise lower bound for modulated energies from [HCRS25], which, applied under the assumption $\mu\in L^\infty$, gives a constant $C_{\mathrm{pt}}(\mu)<\infty$ such that, for every $N\ge 1$ and every configuration X_N,

$$\mathsf{F}_N(X_N,\mu) \ge -C_{\mathrm{pt}}(\mu)\frac{a_{N,\mathsf{s}}}{N}. \tag{5.95}$$

Set $G_N(X_N) \coloneqq -N\mathsf{F}_N(X_N,\mu)$. By (5.95),

$$G_N(X_N) \le C_{\mathrm{pt}}(\mu)\,a_{N,\mathsf{s}}. \tag{5.96}$$

If $0 \le \beta \le B$, then Hölder's inequality and (5.3) give

$$\mathsf{K}_{N,\beta}(\mu) = \mathbb{E}_{\mu^{\otimes N}}\left[e^{\beta G_N}\right] \le \mathbb{E}_{\mu^{\otimes N}}\left[e^{BG_N}\right]^{\beta/B} = \mathsf{K}_{N,B}(\mu)^{\beta/B} \le C_B(\mu), \tag{5.97}$$

after increasing $C_B(\mu)$ if necessary. This proves (5.92). If $\beta > B$, then (5.96) gives

$$\begin{aligned}\mathsf{K}_{N,\beta}(\mu) &= \mathbb{E}_{\mu^{\otimes N}}\left[e^{BG_N} e^{(\beta-B)G_N}\right] \\ &\le \exp\Big(C_{\mathrm{pt}}(\mu)(\beta-B)a_{N,\mathsf{s}}\Big)\mathsf{K}_{N,B}(\mu) \le C_B(\mu)\exp\Big(C_{\mathrm{pt}}(\mu)(\beta-B)a_{N,\mathsf{s}}\Big),\end{aligned} \tag{5.98}$$

which is (5.93) after increasing $C_B(\mu)$ once more to dominate $C_{\mathrm{pt}}(\mu)$. Taking $\beta = \beta_* N^\gamma$ gives (5.94): for $\gamma \le 0$, use (5.92) with $B = B_*$, while for $\gamma > 0$, use (5.93) with any fixed positive B and absorb B_* and the fixed-temperature constant into $C_{B_*,\mu}$. □

5.5. **Sharpness of the threshold.** This final subsection gives an explicit lower rate, though expected to be nonoptimal, for the divergence beyond the Hilbert–Schmidt threshold. The proof applies conditional Jensen on a dyadic spatial partition whose resolution is coupled to N, and then uses the bounded-feature Laplace estimate from Proposition 5.4.

Lemma 5.9 (Quantitative dyadic coarse-graining)**.** *Let $\mathsf{d}/2 \le \mathsf{s} < \mathsf{d}$ and $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$. There exist constants $c_0, C_0, L_0 > 0$, a dyadic cube Q_0, and, for every sufficiently small scale δ in the dyadic subdivision of Q_0, a finite sub-σ-algebra $\mathcal{G}_\delta$ of the Borel σ-algebra such that, with*

$$h_\delta := \mathbb{E}_{\mu^{\otimes 2}}[\tilde{\mathsf{g}} \mid \mathcal{G}_\delta \otimes \mathcal{G}_\delta], \tag{5.99}$$

the following properties hold:

(i) h_δ is bounded, symmetric, canonical, positive definite, and finite rank.

(ii) If T_δ denotes the integral operator with kernel h_δ, then

$$0 \le T_\delta \le L_0 \,\mathrm{Id} \tag{5.100}$$

in the quadratic-form sense.

(iii) The canonical feature map of h_δ satisfies

$$\sup_x h_\delta(x,x) \le C_0 \delta^{-\mathsf{s}}. \tag{5.101}$$

(iv) Its Hilbert–Schmidt norm satisfies

$$\|h_\delta\|^2_{L^2(\mu^{\otimes 2})} \ge c_0 \begin{cases} \log(1/\delta), & 2\mathsf{s} = \mathsf{d}, \\ \delta^{\mathsf{d}-2\mathsf{s}}, & 2\mathsf{s} > \mathsf{d}. \end{cases} \tag{5.102}$$

Proof. Write $f = \mathrm{d}\mu/\,\mathrm{d}x$. Choose $a > 0$ such that

$$E := \{x \in \mathbb{R}^{\mathsf{d}} : f(x) \ge a\} \tag{5.103}$$

has positive Lebesgue measure, and let x_0 be a Lebesgue density point of E. Since $\mathsf{s} < \mathsf{d}$ and $\mu \in L^1 \cap L^\infty$, the potential $h_\mu = \mathsf{g} * \mu$ is bounded. Hence, after choosing a sufficiently small dyadic cube Q_0 containing x_0, we may arrange that

$$|E \cap Q_0| \ge (1-\varepsilon)|Q_0|, \qquad \mu(Q_0) \le \frac{1}{2}, \tag{5.104}$$

and, for $\mu^{\otimes 2}$-a.e. $x, y \in Q_0$,

$$\tilde{\mathsf{g}}(x,y) = \mathsf{g}(x-y) - h_\mu(x) - h_\mu(y) + 2I_\mu \ge \frac{1}{2}\mathsf{g}(x-y). \tag{5.105}$$

Here $\varepsilon > 0$ is a sufficiently small dimensional constant. Indeed, $I_\mu \ge 0$, while $\mathsf{g}(z) \to +\infty$ as $z \to 0$.

Partition Q_0 into dyadic cubes Q of side length δ. Call Q good if

$$|E \cap Q| \geq \frac{1}{2}|Q|, \tag{5.106}$$

and, for each good cube, set $A_Q \coloneqq E \cap Q$. Since every bad cube contains at least half of its volume in $Q_0 \setminus E$, (5.104) gives

$$\#\{Q : Q \text{ is bad}\}\, \delta^{\mathsf{d}} \leq 2\varepsilon |Q_0|. \tag{5.107}$$

Let R be the complement in $\mathbb{R}^{\mathsf{d}}$ of the union of the good partition sets A_Q, and let $\mathcal{G}_\delta$ be generated by the finitely many sets A_Q and R. Since $\mu(Q_0) \leq 1/2$, one has $\mu(R) \geq 1/2$. Moreover, for every good cube,

$$\frac{a}{2}\delta^{\mathsf{d}} \leq \mu(A_Q) \leq \|\mu\|_{L^\infty}\delta^{\mathsf{d}}. \tag{5.108}$$

Let P_δ denote conditional expectation from $L^2(\mu)$ onto the $\mathcal{G}_\delta$-measurable functions, and let T_μ be the integral operator with kernel $\tilde{\mathsf{g}}$. Then h_δ is the kernel of $P_\delta T_\mu P_\delta$. It is therefore symmetric, canonical, positive definite, and finite rank. Also,

$$|\tilde{\mathsf{g}}(x,y)| \leq \mathsf{g}(x-y) + h_\mu(x) + h_\mu(y) + 2I_\mu, \tag{5.109}$$

so

$$\sup_x \int_{\mathbb{R}^{\mathsf{d}}} |\tilde{\mathsf{g}}(x,y)|\, \mathrm{d}\mu(y) \leq 2\|h_\mu\|_{L^\infty} + 4I_\mu < \infty. \tag{5.110}$$

Schur's test (see, e.g., [HS78, Theorem 5.2]) yields $\|T_\mu\|_{\mathrm{op}} \leq L_0$, and, since P_δ is an orthogonal projection, (5.100) follows.

For a good partition set A_Q, the definition of conditional expectation, (5.108), and $\mathsf{s} < \mathsf{d}$ give

$$\begin{aligned} \forall x \in A_Q, \qquad 0 \leq h_\delta(x,x) &= \frac{1}{\mu(A_Q)^2} \iint_{A_Q \times A_Q} \tilde{\mathsf{g}}(u,v)\, \mathrm{d}\mu(u)\, \mathrm{d}\mu(v) \\ &\leq C\delta^{-2\mathsf{d}} \iint_{Q\times Q} |u-v|^{-\mathsf{s}}\, \mathrm{d}u\, \mathrm{d}v + C \leq C\delta^{-\mathsf{s}}. \end{aligned} \tag{5.111}$$

On the remainder set,

$$0 \leq h_\delta(x,x) \leq \mu(R)^{-2}\|\tilde{\mathsf{g}}\|_{L^1(\mu^{\otimes 2})} \leq 4\|\tilde{\mathsf{g}}\|_{L^1(\mu^{\otimes 2})}. \tag{5.112}$$

This proves (5.101).

It remains to prove (5.102). Write ℓ_0 for the side length of Q_0 and $M = \ell_0/\delta$. Index its dyadic subcubes by the lattice box $\{0,\dots,M-1\}^{\mathsf{d}}$. For every displacement $k \in \mathbb{Z}^{\mathsf{d}}$ with

$$2 \leq |k| \leq cM, \tag{5.113}$$

the number of ordered pairs $(Q, Q+k)$ lying in Q_0 is at least cM^{d}. For fixed k, at most twice the number of bad cubes can occur as an endpoint of such a pair. Thus, by (5.107), after choosing ε sufficiently small, the number of good–good pairs with displacement k is at least $cM^{\mathsf{d}} = c\delta^{-\mathsf{d}}$.

If Q and $Q+k$ are good, then (5.105) and the estimate $|u-v| \leq C\delta(1+|k|)$ give

$$\frac{1}{\mu(A_Q)\mu(A_{Q+k})} \iint_{A_Q \times A_{Q+k}} \tilde{\mathsf{g}}(u,v)\, \mathrm{d}\mu(u)\, \mathrm{d}\mu(v) \geq c\delta^{-\mathsf{s}}(1+|k|)^{-\mathsf{s}}. \tag{5.114}$$

Consequently,

$$\|h_\delta\|^2_{L^2(\mu^{\otimes 2})} \geq c\delta^{\mathsf{d}-2\mathsf{s}} \sum_{2\leq |k| \leq c/\delta} (1+|k|)^{-2\mathsf{s}}. \tag{5.115}$$

The lattice sum is bounded below by $c\log(1/\delta)$ when $2\mathsf{s} = \mathsf{d}$, and by a positive constant when $2\mathsf{s} > \mathsf{d}$. This proves (5.102). □

Proposition 5.10 (Quantitative sharpness of the partition-function threshold)**.** *Let $\beta > 0$, let $\mathsf{d}/2 \le \mathsf{s} < \mathsf{d}$, and assume $\mu \in L^\infty \cap \mathcal{P}(\mathbb{R}^{\mathsf{d}})$. There exist constants $c, C > 0$ and $N_0 \ge 3$, depending on $(\beta, \mu, \mathsf{d}, \mathsf{s})$, such that, for every $N \ge N_0$,*

$$\mathsf{K}_{N,\beta}(\mu) \ge C^{-1} \begin{cases} (\log N)^c, & \mathsf{s} = \mathsf{d}/2, \\ \exp\left\{c(\log N)^{2-\mathsf{d}/\mathsf{s}}\right\}, & \mathsf{d}/2 < \mathsf{s} < \mathsf{d}. \end{cases} \tag{5.116}$$

In particular, $\mathsf{K}_{N,\beta}(\mu) \to +\infty$ as $N \to \infty$ throughout $\mathsf{d}/2 \le \mathsf{s} < \mathsf{d}$.

Proof. Let $\xi_1, \dots, \xi_N$ be iid with law μ, and set

$$W_N \coloneqq \frac{1}{N} \sum_{1 \le i \ne j \le N} \tilde{\mathsf{g}}(\xi_i, \xi_j), \qquad Z_N \coloneqq \mathbb{E}\Big[e^{-\frac{\beta}{2} W_N}\Big]. \tag{5.117}$$

Since h_μ is bounded, the exact one-body decomposition (5.2) gives

$$\begin{aligned} \mathsf{K}_{N,\beta}(\mu) &= e^{\beta I_\mu} \mathbb{E}\left[e^{-\frac{\beta}{2} W_N + \frac{\beta}{N} \sum_{i=1}^N (h_\mu(\xi_i) - 2I_\mu)}\right] \\ &\ge c_{\beta,\mu} Z_N, \qquad c_{\beta,\mu} \coloneqq \exp\left\{\beta I_\mu - \beta \|h_\mu - 2I_\mu\|_{L^\infty}\right\} > 0. \end{aligned} \tag{5.118}$$

Fix a sufficiently small dyadic scale δ from Lemma 5.9, and set

$$W_{N,\delta} \coloneqq \frac{1}{N} \sum_{1 \le i \ne j \le N} h_\delta(\xi_i, \xi_j), \qquad Z_{N,\delta} \coloneqq \mathbb{E}\Big[e^{-\frac{\beta}{2} W_{N,\delta}}\Big]. \tag{5.119}$$

Let $\mathcal{F}_{N,\delta}$ be the σ-algebra generated by the events $\{\xi_i \in A_Q\}$ and $\{\xi_i \in R\}$, for $1 \le i \le N$ and all good cubes Q; equivalently, it records which set in the finite partition $\{A_Q : Q \text{ is good}\} \cup \{R\}$ contains each ξ_i. By (5.99),

$$\mathbb{E}[W_N \mid \mathcal{F}_{N,\delta}] = W_{N,\delta}. \tag{5.120}$$

Conditional Jensen therefore gives

$$Z_N \ge Z_{N,\delta}. \tag{5.121}$$

Let T_δ be the operator with kernel h_δ. Since h_δ is positive definite, the Moore–Aronszajn construction gives a canonical feature map k_δ (see, e.g., [SC08, Theorem 4.21]); by (5.101), it is bounded and satisfies

$$\|k_\delta\|^2_{L^\infty(\mu; \mathcal{H}_\delta)} = \sup_x h_\delta(x, x) \le C\delta^{-\mathsf{s}}. \tag{5.122}$$

The bounded specialization (5.46) of Proposition 5.4 yields

$$\left|Z_{N,\delta} - \det{}_2(\mathrm{Id} + \beta T_\delta)^{-1/2}\right| \le \frac{C}{N} \exp\left\{C\delta^{-\mathsf{s}}\right\}. \tag{5.123}$$

Let $(\lambda_{\delta,j})_j$ denote the nonzero eigenvalues of T_δ. By (5.100), they belong to $[0, L_0]$. Since

$$F_\beta(t) \coloneqq \beta t - \log(1 + \beta t) \tag{5.124}$$

satisfies

$$\inf_{0 \le t \le L_0} \frac{F_\beta(t)}{t^2} > 0, \tag{5.125}$$

with the quotient interpreted continuously at $t = 0$, one has

$$\log \det{}_2(\mathrm{Id} + \beta T_\delta)^{-1/2} = \frac{1}{2} \sum_j F_\beta(\lambda_{\delta,j}) \ge c \sum_j \lambda_{\delta,j}^2 = c\|h_\delta\|^2_{L^2(\mu^{\otimes 2})}. \tag{5.126}$$

Choose $a > 0$ so small that $C2^{\mathsf{s}}a < 1/2$, where C is the constant in the exponential on the right-hand side of (5.123). For every sufficiently large N, choose a scale δ_N in the dyadic subdivision of Q_0 such that

$$a \log N \le \delta_N^{-\mathsf{s}} \le 2^{\mathsf{s}} a \log N. \tag{5.127}$$

After increasing N_0 if necessary, (5.123) and (5.127) give

$$\frac{C}{N} \exp\left\{C\delta_N^{-\mathsf{s}}\right\} \le \frac{1}{2}. \tag{5.128}$$

The determinant in (5.123) is at least one, so

$$Z_{N,\delta_N} \ge \frac{1}{2}\det{}_2(\mathrm{Id} + \beta T_{\delta_N})^{-1/2}. \tag{5.129}$$

Combining this estimate with (5.118), (5.121), (5.126), and (5.102) gives

$$\log \mathsf{K}_{N,\beta}(\mu) \ge c \log(1/\delta_N) - C \ge c \log\log N - C, \qquad 2\mathsf{s} = \mathsf{d}, \tag{5.130}$$

$$\log \mathsf{K}_{N,\beta}(\mu) \ge c\delta_N^{\mathsf{d}-2\mathsf{s}} - C \ge c(\log N)^{2-\mathsf{d}/\mathsf{s}} - C, \qquad 2\mathsf{s} > \mathsf{d}. \tag{5.131}$$

Exponentiating proves (5.116). □

Remark 5.11 (Expected sharp ultraviolet growth)**.** The rates in Proposition 5.10 reflect the exponential dependence on the feature diagonal in (5.123) and are not expected to be optimal. If the density of μ is bounded below by a positive constant on some ball, the natural prediction is

$$\log \mathsf{K}_{N,\beta}(\mu) \asymp \begin{cases} \log N, & \mathsf{s} = \mathsf{d}/2, \\ N^{2-\mathsf{d}/\mathsf{s}}, & \mathsf{d}/2 < \mathsf{s} < \mathsf{d}. \end{cases} \tag{5.132}$$

The heuristic comes from the singular part of the second Mayer correction,

$$N^2 \iint_{(\mathbb{R}^{\mathsf{d}})^2} \left(e^{-\beta \mathsf{g}(x-y)/N} - 1 + \frac{\beta}{N}\mathsf{g}(x-y)\right) \mathrm{d}\mu(x)\,\mathrm{d}\mu(y). \tag{5.133}$$

At fixed inverse temperature, the quadratic Mayer approximation should therefore be cut off at the scale $r \asymp N^{-1/\mathsf{s}}$ at which the single-pair contribution $\beta \mathsf{g}(r)/N$ to the Gibbs exponent becomes of order one and ceases to be perturbative. At $\mathsf{s} = \mathsf{d}/2$, this produces $\int_{N^{-1/\mathsf{s}}}^1 r^{-1}\,\mathrm{d}r \asymp \log N$. If $\mathsf{s} > \mathsf{d}/2$, the change of variables $x - y = N^{-1/\mathsf{s}} z$ gives the scale $N^{2-\mathsf{d}/\mathsf{s}}$. The bounded terms introduced by Hoeffding centering do not affect these ultraviolet powers. The problem of proving matching bounds, identifying the leading constants, and determining whether a renormalized limiting object exists is beyond the scope of the present paper.

6. The attractive logarithmic case

This section proves both assertions of Theorem 2.6. The uniform-bound argument is carried out for a general interaction kernel W with the right growth: it uses only the integrability $W \in L^1(\mu^{\otimes 2})$ of the kernel, the exponential integrability $e^{2\beta h_\mu} \in L^1(\mu)$ of its potential $h_\mu := \int_{\mathbb{R}^{\mathsf{d}}} W(\cdot, y)\,\mathrm{d}\mu(y)$, and the Gaussian moment growth (2.34) of the centered kernel, and therefore applies verbatim to every interaction satisfying these hypotheses. The attractive logarithmic kernel $W(x,y) = \log|x-y|$ is the instance appearing in the theorem statement. In the uniform-bound subsection, we write h_μ, I_μ, F_N, $\mathsf{K}_{N,\beta}$, and $\tilde{W}$ without the superscript att. In the quantitative subsection, where the attractive target quantities and the repulsive logarithmic truncations occur simultaneously, we restore the superscript only on the genuinely attractive quantities. The quantitative determinant argument in Section 6.2 uses the additional logarithmic truncation and bounded-density hypotheses. The section concludes with the periodic uniform-background phase diagram, whose purpose is to illustrate, already in the simplest homogeneous setting, why an N-uniform partition-function bound cannot be expected for every inverse temperature. Under the corresponding small-β and

integrability hypotheses, the commutator, mean-field, and fluctuation consequences follow by the same Donsker–Varadhan, Gronwall, and Gaussian-tilting arguments as in the repulsive case; the commutator input may alternatively be obtained from the argument of Jabin and Wang [JW18].

6.1. **Uniform bound at small inverse temperature.** The uniform-bound proof has two ingredients. A one-body reduction bounds $\mathsf{K}_{N,\beta}(\mu)$ by the centered partition function

$$Z_{N,\beta}(\mu) \coloneqq \mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{2N}S_N}\Big], \qquad S_N \coloneqq \sum_{1\le i\ne j\le N} \tilde W(\xi_i,\xi_j), \qquad \xi_1,\dots,\xi_N \overset{\text{iid}}{\sim} \mu. \tag{6.1}$$

A Taylor expansion then reduces the uniform bound on $Z_{N,\beta}$ to Gaussian-factorial growth of the even moments of S_N, which is proved by decoupling and counting: after the decoupled expectation is expanded, only index patterns in which every active index is repeated survive the centering.

Proof of the uniform-bound assertion in Theorem 2.6. **Step 1: One-body reduction.** For every $\beta \ge 0$ with $e^{2\beta h_\mu} \in L^1(\mu)$,

$$\mathsf{K}_{N,\beta}(\mu) \le e^{\beta|I_\mu|}\,\mathbb{E}_\mu\big[e^{2\beta h_\mu}\big]^{1/2}\, Z_{N,2\beta}(\mu)^{1/2}. \tag{6.2}$$

From the definition of F_N, with $|I_\mu| \le \frac12\|\mathsf{g}_{\mathrm{att}}\|_{L^1(\mu^{\otimes 2})} < \infty$,

$$-\beta N \mathsf{F}_N(X_N,\mu) = -\frac{\beta}{2N}\sum_{i\ne j}\tilde W(x_i,x_j) + \frac{\beta}{N}\sum_{i=1}^N h_\mu(x_i) - \beta I_\mu. \tag{6.3}$$

Since the one-body term is an empirical average of iid samples, Jensen's inequality gives $\mathbb{E}_{\mu^{\otimes N}}[e^{\frac{2\beta}{N}\sum_i h_\mu(\xi_i)}] = \mathbb{E}_\mu[e^{\frac{2\beta}{N}h_\mu}]^N \le \mathbb{E}_\mu[e^{2\beta h_\mu}]$. Therefore, by Cauchy–Schwarz,

$$\mathsf{K}_{N,\beta}(\mu) \le e^{\beta|I_\mu|}\,\mathbb{E}_{\mu^{\otimes N}}\Big[e^{-\frac{\beta}{N}S_N}\Big]^{1/2}\mathbb{E}_{\mu^{\otimes N}}\Big[e^{\frac{2\beta}{N}\sum_{i=1}^N h_\mu(\xi_i)}\Big]^{1/2} \le e^{\beta|I_\mu|}\, Z_{N,2\beta}(\mu)^{1/2}\,\mathbb{E}_\mu\big[e^{2\beta h_\mu}\big]^{1/2}. \tag{6.4}$$

Step 2: Gaussian-factorial moment bound. There is a universal constant $C_* < \infty$ such that, for all $k \ge 1$ and $N \ge 1$,

$$\mathbb{E}_{\mu^{\otimes N}}\big[|S_N|^{2k}\big] \le (C_* C_{\mathrm{mom}})^{2k} N^{2k} (2k)!. \tag{6.5}$$

The case $N = 1$ is trivial, since $S_1 = 0$; assume $N \ge 2$. The kernel $\tilde W$ is canonical by (2.2) and Fubini, and integrable by the case $p = 1$ of (2.34), so the moment decoupling of Remark 3.4, applied with $\Phi(x) = x^{2k}$, and the expansion of the $2k$-th power give

$$\mathbb{E}_{\mu^{\otimes N}}\big[|S_N|^{2k}\big] \le 4^{2k}\,\mathbb{E}\Big[\Big|\sum_{i\ne j}\tilde W(\xi_i,\xi'_j)\Big|^{2k}\Big] \le 4^{2k}\sum_{\substack{1\le i_\ell,j_\ell\le N\\ i_\ell\ne j_\ell,\ 1\le \ell\le 2k}}\Big|\mathbb{E}\Big[\prod_{\ell=1}^{2k}\tilde W(\xi_{i_\ell},\xi'_{j_\ell})\Big]\Big|, \tag{6.6}$$

where $\xi'_1,\dots,\xi'_N$ is an independent copy of $\xi_1,\dots,\xi_N$. Fix a multi-index and set $a_m \coloneqq \#\{\ell : i_\ell = m\}$ and $b_m \coloneqq \#\{\ell : j_\ell = m\}$. By the same centering, the expectation vanishes unless every active a_m and b_m is at least 2; call such multi-indices *surviving*. For a surviving multi-index, condition on $\xi'_1,\dots,\xi'_N$: by the independence of the unprimed variables, the conditional expectation factorizes over the active indices m into factors $\mathbb{E}_{\xi_m}[\prod_{\ell:i_\ell=m}\tilde W(\xi_m,\xi'_{j_\ell})]$ with a_m terms each. Hölder's inequality in ξ_m with the a_m equal exponents $(a_m,\dots,a_m)$, followed by the moment hypothesis (2.34) with $p = a_m$, applied uniformly in the frozen variables ξ'_{j_ℓ}, gives

$$\Big|\mathbb{E}_{\xi_m}\Big[\prod_{\ell:i_\ell=m}\tilde W(\xi_m,\xi'_{j_\ell})\Big]\Big| \le \prod_{\ell:i_\ell=m}\Big(\int_{\mathbb{R}^{\mathsf{d}}}|\tilde W(x,\xi'_{j_\ell})|^{a_m}\,\mathrm{d}\mu(x)\Big)^{1/a_m} \le C_{\mathrm{mom}}^{a_m}\,a_m!. \tag{6.7}$$

Multiplying over the active m and using $\sum_m a_m = 2k$,

$$\Big|\mathbb{E}\Big[\prod_{\ell=1}^{2k} \tilde{W}(\xi_{i_\ell}, \xi'_{j_\ell})\Big]\Big| \le C_{\mathrm{mom}}^{2k} \prod_{m=1}^{N} a_m!. \tag{6.8}$$

Since the i-words with multiplicity vector $\alpha \in \mathbb{N}_0^N$, $\sum_m \alpha_m = 2k$, number $(2k)!/\prod_m \alpha_m!$ and each contributes $\prod_m \alpha_m!$, dropping the coupling $i_\ell \neq j_\ell$ factorizes the count:

$$\sum_{\text{surviving } (i,j)} \prod_{m=1}^{N} a_m! \le (2k)! \cdot \#\{\text{surviving } \alpha\} \cdot \#\{\text{surviving } j\text{-words}\}. \tag{6.9}$$

If $N < 4k$, we discard the survival constraints: there are at most N^{2k} words and $\binom{2k+N-1}{N-1} \le 2^{6k}$ multiplicity vectors, so (6.9) is at most $2^{6k}N^{2k}(2k)!$. If $N \ge 4k$, a surviving word or multiplicity vector has at most k active letters, all repeated; choosing the $r \le k$ active letters and assigning positions crudely,

$$\#\{\text{surviving } j\text{-words}\} \le \sum_{r=1}^{k} \binom{N}{r} r^{2k} \le C^k N^k k^k, \tag{6.10}$$

$$\#\{\text{surviving } \alpha\} \le \sum_{r=1}^{k} \binom{N}{r} \binom{2k-r-1}{r-1} \le C^k \frac{N^k}{k^k}. \tag{6.11}$$

Indeed, writing $r = \theta k$ and $N = xk$ with $0 < \theta \le 1$ and $x \ge 4$, and using $\binom{N}{r} \le (eN/r)^r$, $y^\theta \le y$ for $y \ge 1$, and $\sup_{(0,1]} \theta^{-2\theta} < \infty$, the summands are bounded by $(ex/\theta)^{\theta k} k^{2k} \le C^k N^k k^k$ and by $(eN/r)^r (2ek/r)^r = (2e^2 Nk/r^2)^r \le (Cx)^k = C^k N^k/k^k$, respectively, and the factors k from the sums are absorbed into C^k. The factors $k^{\pm k}$ cancel in (6.9), which is again at most $C^k N^{2k}(2k)!$. Taking $C_* \coloneqq 4\max\{2^3, \sqrt{C}\}$ proves (6.5).

Step 3: Summation. Since $e^x \le e^x + e^{-x} = 2\sum_{k\ge 0} x^{2k}/(2k)!$, the moment bound (6.5) gives, for every $0 \le \beta < \beta_0$, with $\beta_0 = 1/(C_* C_{\mathrm{mom}})$ as in (2.35) and C_* the constant of Step 2, and uniformly in N,

$$Z_{N,2\beta}(\mu) \le 2\sum_{k=0}^{\infty} \frac{\beta^{2k}}{N^{2k}(2k)!} \mathbb{E}_{\mu^{\otimes N}}\big[|S_N|^{2k}\big] \le 2\sum_{k=0}^{\infty} \big(\beta C_* C_{\mathrm{mom}}\big)^{2k} = \frac{2}{1 - \big(\beta C_* C_{\mathrm{mom}}\big)^2}, \tag{6.12}$$

which, combined with (6.2), proves the theorem with

$$C_{\mathrm{ub}}^{\mathrm{att}}(\beta, \mu) = e^{\beta |I_\mu|}\, \mathbb{E}_\mu\big[e^{2\beta h_\mu}\big]^{1/2} \left(\frac{2}{1 - \big(\beta C_* C_{\mathrm{mom}}\big)^2}\right)^{1/2}. \tag{6.13}$$

□

6.2. **Quantitative convergence to the determinant.** To prove the quantitative attractive result, we need a fixed-kernel estimate for the positive quadratic exponential $\mathbb{E}[\exp\{\beta \mathsf{W}_N(q)/2\}]$. This is the counterpart of the bounded-feature estimate in Proposition 5.4, with $\mathrm{Id} + \beta T_q$ replaced by $\mathrm{Id} - \beta T_q$. The proof requires the stronger subcritical operator bound $2\beta \|T_q\|_{\mathrm{op}} < 1$, which provides the second exponential moment used below.

Let $q \in L^2(\mu^{\otimes 2})$ be symmetric, canonical, and nonnegative definite, and let

$$q(x,y) = \langle \mathsf{k}(x), \mathsf{k}(y)\rangle_{\mathcal{H}}, \qquad \int_{\mathbb{R}^{\mathsf{d}}} \mathsf{k}(x)\, \mathrm{d}\mu(x) = 0, \tag{6.14}$$

$$D(x) \coloneqq \|\mathsf{k}(x)\|_{\mathcal{H}}^2, \qquad \mathsf{W}_N(q) \coloneqq \frac{1}{N} \sum_{1 \le i \ne j \le N} q(\xi_i, \xi_j), \tag{6.15}$$

be a centered feature representation on a separable real Hilbert space $\mathcal{H}$, where $\xi_1, \dots, \xi_N$ are independent with law μ. Let T_q denote the Hilbert–Schmidt integral operator with kernel q.

Proposition 6.1 (Feature-space Laplace estimate for a positive quadratic exponential)**.** *Fix $\beta > 0$ and assume that*

$$2\beta\|T_q\|_{\mathrm{op}} < 1, \qquad \sup_{M\geq 1} \mathbb{E}\Big[e^{\beta \mathsf{W}_M(q)}\Big] < \infty, \qquad \int_{\mathbb{R}^{\mathsf{d}}} e^{aD(x)}\, \mathrm{d}\mu(x) < \infty \tag{6.16}$$

for some $a > 0$. Then there is a finite constant C such that, for every $N \geq 1$,

$$\Big|\mathbb{E}\Big[e^{\frac{\beta}{2}\mathsf{W}_N(q)}\Big] - \det_2(\mathrm{Id} - \beta T_q)^{-1/2}\Big| \leq CN^{-1/2}. \tag{6.17}$$

For fixed β and a, the constant may be chosen to depend polynomially on $1 + \int_{\mathbb{R}^{\mathsf{d}}} e^{aD(x)}\, \mathrm{d}\mu(x)$, with coefficients depending only on the subcritical margin $1 - 2\beta\|T_q\|_{\mathrm{op}}$, $\|T_q\|_{\mathrm{HS}}$, and the uniform exponential moment in (6.16)*.*

Proof. Let Z be an isonormal Gaussian process over $\mathcal{H}$, independent of $\xi_1, \dots, \xi_N$, and write $\mathsf{Z}_x \coloneqq \mathsf{Z}(\mathsf{k}(x))$. Set

$$A_N(\mathsf{Z}) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \exp\left\{\sqrt{\frac{\beta}{N}}\, \mathsf{Z}_x - \frac{\beta}{2N} D(x)\right\} \mathrm{d}\mu(x). \tag{6.18}$$

Since

$$\mathsf{W}_N(q) = \left\| \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \mathsf{k}(\xi_i) \right\|_{\mathcal{H}}^2 - \frac{1}{N} \sum_{i=1}^{N} D(\xi_i), \tag{6.19}$$

the Hubbard–Stratonovich transformation and Fubini's theorem give

$$\mathbb{E}\Big[e^{\frac{\beta}{2}\mathsf{W}_N(q)}\Big] = \mathbb{E}_{\mathsf{Z}}\Big[A_N(\mathsf{Z})^N\Big]. \tag{6.20}$$

For fixed Z, define

$$f(\tau) \coloneqq \int_{\mathbb{R}^{\mathsf{d}}} \exp\left\{\tau\sqrt{\beta}\, \mathsf{Z}_x - \frac{\tau^2\beta}{2} D(x)\right\} \mathrm{d}\mu(x). \tag{6.21}$$

Then $A_N(\mathsf{Z}) = f(N^{-1/2})$, $f(0) = 1$, and $f'(0) = 0$. Introduce

$$B \coloneqq \frac{\beta}{2} \int_{\mathbb{R}^{\mathsf{d}}} (\mathsf{Z}_x^2 - D(x))\, \mathrm{d}\mu(x), \tag{6.22}$$

$$R \coloneqq \frac{\beta^{3/2}}{6} \int_{\mathbb{R}^{\mathsf{d}}} (\mathsf{Z}_x^3 - 3D(x)\mathsf{Z}_x)\, \mathrm{d}\mu(x). \tag{6.23}$$

Taylor's formula with integral remainder yields

$$A_N(\mathsf{Z}) = 1 + \frac{B}{N} + \frac{R}{N^{3/2}} + \frac{E_N}{N^2}, \tag{6.24}$$

where

$$E_N \coloneqq \frac{1}{6} \int_0^1 (1-u)^3 f^{(4)}(uN^{-1/2})\, \mathrm{d}u. \tag{6.25}$$

The exponential moment of D implies, after discarding finitely many N and enlarging the final constant, that

$$\sup_N \mathbb{E}_{\mathsf{Z}}\Big[|B|^4 + |R|^4 + |E_N|^4\Big] < \infty. \tag{6.26}$$

Indeed, the bounds for B and R follow from Gaussian moment identities and Minkowski's inequality. For the remainder, write

$$g_x(\tau) \coloneqq \tau\sqrt{\beta}\,\mathsf{Z}_x - \frac{\tau^2\beta}{2}D(x). \tag{6.27}$$

Then

$$(e^{g_x})^{(4)} = e^{g_x}\big((g_x')^4 + 6(g_x')^2 g_x'' + 3(g_x'')^2\big), \qquad g_x' = \sqrt{\beta}\,\mathsf{Z}_x - \tau\beta D(x), \qquad g_x'' = -\beta D(x). \tag{6.28}$$

For all sufficiently large N, Gaussian moment estimates give

$$\sup_{0\le\tau\le N^{-1/2}} \mathbb{E}_{\mathsf{Z}}\Big[\Big|(e^{g_x})^{(4)}(\tau)\Big|^4\Big] \le C_{\beta,a}\, e^{aD(x)/2}(1+D(x)^{16}). \tag{6.29}$$

The right-hand side is integrable after decreasing a if necessary and absorbing the polynomial factor into the exponential. Jensen's inequality in x then gives the asserted uniform fourth-moment bound for E_N.

Write $A_N = A_N(\mathsf{Z})$ and set $G_N \coloneqq \{|A_N - 1| \le 1/2\}$. Using $|\log(1+y) - y| \le 2y^2$ for $|y| \le 1/2$, together with (6.24), gives, on G_N,

$$|N\log A_N - B| \le \frac{|R|}{\sqrt{N}} + \frac{|E_N|}{N} + 2N|A_N - 1|^2. \tag{6.30}$$

Consequently, (6.26) implies

$$\|(N\log A_N - B)\mathbf{1}_{G_N}\|_{L^2(\mathsf{Z})} \le CN^{-1/2}. \tag{6.31}$$

Moreover, Markov's inequality and (6.24)–(6.26) yield

$$\mathbb{P}_{\mathsf{Z}}(G_N^c) \le 16\mathbb{E}_{\mathsf{Z}}\Big[|A_N - 1|^4\Big] \le CN^{-4}. \tag{6.32}$$

We next prove the estimate

$$\Big|\mathbb{E}_{\mathsf{Z}}\Big[A_N^N\Big] - \mathbb{E}_{\mathsf{Z}}\Big[e^B\Big]\Big| \le CN^{-1/2}. \tag{6.33}$$

The two second moments used below are uniformly bounded:

$$\mathbb{E}_{\mathsf{Z}}\Big[A_N^{2N}\Big] = \mathbb{E}\Big[e^{\beta \mathsf{W}_{2N}(q)}\Big] \le C, \tag{6.34}$$

$$\mathbb{E}_{\mathsf{Z}}\Big[e^{2B}\Big] = \det{}_2(\mathrm{Id} - 2\beta T_q)^{-1/2} < \infty. \tag{6.35}$$

The first identity follows by introducing $2N$ independent samples in A_N^{2N} and applying the Hubbard–Stratonovich transformation; the bound is (6.16). The second follows by diagonalizing the covariance operator of $\mathsf{k}(\xi)$.

On G_N, one has $|A_N^N - e^B| \le |N\log A_N - B|(A_N^N + e^B)$. Hence, by Cauchy–Schwarz and the preceding estimates,

$$\begin{aligned}\mathbb{E}_{\mathsf{Z}}\Big[|A_N^N - e^B|\mathbf{1}_{G_N}\Big] &\le \|(N\log A_N - B)\mathbf{1}_{G_N}\|_{L^2(\mathsf{Z})}\left(\mathbb{E}_{\mathsf{Z}}\Big[A_N^{2N}\Big]^{1/2} + \mathbb{E}_{\mathsf{Z}}\Big[e^{2B}\Big]^{1/2}\right)\\ &\le CN^{-1/2}.\end{aligned} \tag{6.36}$$

On G_N^c, Cauchy–Schwarz and (6.32) give

$$\begin{aligned}\mathbb{E}_{\mathsf{Z}}\Big[(A_N^N + e^B)\mathbf{1}_{G_N^c}\Big] &\le \left(\mathbb{E}_{\mathsf{Z}}\Big[A_N^{2N}\Big]^{1/2} + \mathbb{E}_{\mathsf{Z}}\Big[e^{2B}\Big]^{1/2}\right)\mathbb{P}_{\mathsf{Z}}(G_N^c)^{1/2}\\ &\le CN^{-2}.\end{aligned} \tag{6.37}$$

This proves (6.33).

Finally, if $(\lambda_j)_{j\ge1}$ are the nonzero eigenvalues of T_q, then

$$B=\frac{\beta}{2}\sum_{j\ge1}\lambda_j(Z_j^2-1) \tag{6.38}$$

in L^2, and $\beta\|T_q\|_{\mathrm{op}}<1$ gives

$$\mathbb{E}_{\mathsf{Z}}\Big[e^B\Big]=\prod_{j\ge1}e^{-\beta\lambda_j/2}(1-\beta\lambda_j)^{-1/2}=\det{}_2(\mathrm{Id}-\beta T_q)^{-1/2}. \tag{6.39}$$

Combining (6.20), (6.33), and (6.39) completes the proof. □

Proof of the quantitative-convergence assertion in Theorem 2.6. Write $\mathsf{g}=-\mathsf{g}_{\mathrm{att}}=-\log|\cdot|$ for the repulsive logarithmic kernel, so that

$$\tilde{\mathsf{g}}=-\tilde{\mathsf{g}}_{\mathrm{att}} \tag{6.40}$$

is canonical and nonnegative definite. In this proof, the superscript att is restored on the target attractive quantities, while g, g_η, f_η, $\mathsf{g}_{\eta,t}$, W_N, $W_{N,\eta}$, and $W_{N,\eta,t}$ retain the repulsive logarithmic notation established in Sections 4 and 5.1. Let T_μ, $T_{\mu,\eta}$, and $T_{\mu,\eta,t}$ denote the operators with kernels $\tilde{\mathsf{g}}$, $\tilde{\mathsf{g}}_\eta$, and $\tilde{\mathsf{g}}_{\eta,t}$, respectively. By linearity of Hoeffding centering,

$$\tilde{\mathsf{g}}_{\eta,t}=\tilde{\mathsf{g}}_\eta+t\tilde{\mathsf{f}}_\eta=\tilde{\mathsf{g}}-(1-t)\tilde{\mathsf{f}}_\eta,\qquad 0\le t\le1. \tag{6.41}$$

Uniform exponential moments along the interpolation. The logarithmic truncation satisfies

$$\sup_{0<\eta\le\eta_0}\sup_{y\in\mathbb{R}^{\mathsf{d}}}\int_{\mathbb{R}^{\mathsf{d}}}|\tilde{\mathsf{f}}_\eta(x,y)|^p\,\mathrm{d}\mu(x)\le C_0^p p!\qquad(p\ge1). \tag{6.42}$$

Indeed, by the centering identity,

$$\tilde{\mathsf{f}}_\eta(x,y)=\mathsf{f}_\eta(x-y)-(\mathsf{f}_\eta*\mu)(x)-(\mathsf{f}_\eta*\mu)(y)+\iint_{(\mathbb{R}^{\mathsf{d}})^2}\mathsf{f}_\eta(z-w)\,\mathrm{d}\mu(z)\,\mathrm{d}\mu(w). \tag{6.43}$$

The bounded-density assumption and the logarithmic truncation estimate give

$$\sup_{y\in\mathbb{R}^{\mathsf{d}}}\int_{\mathbb{R}^{\mathsf{d}}}\mathsf{f}_\eta(x-y)^p\,\mathrm{d}\mu(x)\le\|\mu\|_{L^\infty}\int_{\mathbb{R}^{\mathsf{d}}}\mathsf{f}_\eta(z)^p\,\mathrm{d}z\le C^p p!. \tag{6.44}$$

For $|z|\le\eta$, the last estimate follows from

$$\int_0^1(1+|\log r|)^p r^{\mathsf{d}-1}\,\mathrm{d}r\le C_{\mathsf{d}}^p p!, \tag{6.45}$$

while the polynomial decay in (4.12) gives the bound on $|z|>\eta$. The convolution terms are uniformly bounded by $C\eta^{\mathsf{d}}$ by (4.13), proving (6.42). Since (6.41) gives $\tilde{\mathsf{g}}_{\eta,t}=\tilde{\mathsf{g}}-(1-t)\tilde{\mathsf{f}}_\eta$, combining (6.42) with (2.34) yields

$$\sup_{0<\eta\le\eta_0}\sup_{0\le t\le1}\sup_{y\in\mathbb{R}^{\mathsf{d}}}\int_{\mathbb{R}^{\mathsf{d}}}|\tilde{\mathsf{g}}_{\eta,t}(x,y)|^p\,\mathrm{d}\mu(x)\le C_{\mathrm{tr}}^p p!\qquad(p\ge1) \tag{6.46}$$

for a finite C_{tr} depending only on the displayed data.

The factorial-moment argument used for the uniform-bound assertion applies uniformly to the kernels $-\mathsf{g}_{\eta,t}$. Decreasing the threshold β_{q} in (2.37), if necessary, we may therefore arrange that, for every $0<\beta<\beta_{\mathrm{q}}$,

$$\sup_{N\ge1}\sup_{0<\eta\le\eta_0}\sup_{0\le t\le1}\mathbb{E}_{\mu^{\otimes N}}\Big[e^{\beta W_{N,\eta,t}}\Big]<\infty. \tag{6.47}$$

The case $p = 2$ of (6.46) also bounds the Hilbert–Schmidt, hence operator, norm of $T_{\mu,\eta,t}$ uniformly in η and t. Decreasing β_{q} once more if needed gives the common subcritical operator-norm bound

$$\sup_{0<\eta\le\eta_0}\sup_{0\le t\le 1} 2\beta\|T_{\mu,\eta,t}\|_{\mathrm{op}} < 1. \tag{6.48}$$

These choices may be made with $\beta_{\mathrm{q}} \le \beta_0/2$, as required in (2.37).

Finite-particle truncation error. Using the uniform exponential-moment bound (6.47), differentiation in t and Cauchy–Schwarz give

$$\left|\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_{\mu^{\otimes N}}\left[e^{\frac{\beta}{2}W_{N,\eta,t}}\right]\right| \le \frac{\beta}{2}\mathbb{E}_{\mu^{\otimes N}}\left[|W_N - W_{N,\eta}|e^{\frac{\beta}{2}W_{N,\eta,t}}\right] \le C\mathbb{E}_{\mu^{\otimes N}}\left[(W_N - W_{N,\eta})^2\right]^{1/2}. \tag{6.49}$$

Canonicality gives the exact identity

$$\mathbb{E}_{\mu^{\otimes N}}\left[(W_N - W_{N,\eta})^2\right] = \frac{2(N-1)}{N}\|\tilde{\mathsf{f}}_\eta\|^2_{L^2(\mu^{\otimes 2})}, \tag{6.50}$$

and the logarithmic truncation estimate gives

$$\|\tilde{\mathsf{f}}_\eta\|_{L^2(\mu^{\otimes 2})} \le C\eta^{\mathsf{d}/2}. \tag{6.51}$$

Integrating (6.49) over $t \in [0,1]$ therefore yields

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{\frac{\beta}{2}W_N}\right] - \mathbb{E}_{\mu^{\otimes N}}\left[e^{\frac{\beta}{2}W_{N,\eta}}\right]\right| \le C\eta^{\mathsf{d}/2}. \tag{6.52}$$

Determinant truncation error. The operators T_μ and $T_{\mu,\eta}$ are nonnegative and satisfy the common operator-norm bound (6.48). For a nonnegative self-adjoint Hilbert–Schmidt operator T satisfying this bound,

$$-\frac{1}{2}\log\det{}_2(\mathrm{Id} - \beta T) = \frac{1}{2}\sum_{m=2}^{\infty}\frac{\beta^m}{m}\operatorname{Tr}(T^m). \tag{6.53}$$

The telescoping identity for $T^m - S^m$, together with the uniform Hilbert–Schmidt bounds, shows that the left-hand side is locally Lipschitz in Hilbert–Schmidt norm on the set of nonnegative self-adjoint Hilbert–Schmidt operators satisfying the common operator-norm bound. Exponentiating on the resulting bounded range gives

$$\left|\det{}_2(\mathrm{Id} - \beta T_\mu)^{-1/2} - \det{}_2(\mathrm{Id} - \beta T_{\mu,\eta})^{-1/2}\right| \le C\|T_\mu - T_{\mu,\eta}\|_{\mathrm{HS}} \le C\eta^{\mathsf{d}/2}. \tag{6.54}$$

Quantitative fixed-truncation limit. Let k_η be the canonical feature map of $\tilde{\mathsf{g}}_\eta$ and set $D_\eta(x) \coloneqq \|\mathsf{k}_\eta(x)\|^2$. The diagonal identity is

$$D_\eta(x) = \mathsf{g}_\eta(0) - 2(\mathsf{g}_\eta * \mu)(x) + 2I_\mu^{(\eta)}, \tag{6.55}$$

where $I_\mu^{(\eta)} \coloneqq \frac{1}{2}\iint_{(\mathbb{R}^{\mathsf{d}})^2}\mathsf{g}_\eta(z-w)\,\mathrm{d}\mu(z)\,\mathrm{d}\mu(w)$. Since $\mathsf{g}_\eta = \mathsf{g} - \mathsf{f}_\eta$ and $\mathsf{g} = -\mathsf{g}_{\mathrm{att}}$, the truncation bounds imply

$$0 \le D_\eta(x) \le C\big(1 + \log(1/\eta)\big) + 2h_\mu^{\mathrm{att}}(x). \tag{6.56}$$

The assumption $e^{4\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$ therefore supplies the feature-diagonal exponential moment required in Proposition 6.1, with a bound $C\eta^{-A}$ for some finite $A = A(\beta,\mu)$. Applying Proposition 6.1 and using (6.47) and (6.48) gives

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{\frac{\beta}{2}W_{N,\eta}}\right] - \det{}_2(\mathrm{Id} - \beta T_{\mu,\eta})^{-1/2}\right| \le C\eta^{-A}N^{-1/2}. \tag{6.57}$$

Combining (6.52), (6.54), and (6.57) gives

$$\left|\mathbb{E}_{\mu^{\otimes N}}\left[e^{\frac{\beta}{2}W_N}\right] - \det{}_2(\mathrm{Id} - \beta T_\mu)^{-1/2}\right| \le C\left(\eta^{\mathsf{d}/2} + \eta^{-A}N^{-1/2}\right). \tag{6.58}$$

Choose $\eta = N^{-1/(2A+\mathsf{d})}$ when this is at most η_0, and enlarge the constant to absorb the remaining finite set of N. Then the right-hand side of (6.58) is bounded by $C(e+N)^{-c}$, where

$$c := \frac{\mathsf{d}}{2(2A+\mathsf{d})} > 0. \tag{6.59}$$

Restoring the one-body term. Set

$$\psi_\mu^{\mathrm{att}} := h_\mu^{\mathrm{att}} - 2I_\mu^{\mathrm{att}}, \qquad R_N := \frac{1}{N}\sum_{i=1}^N \psi_\mu^{\mathrm{att}}(\xi_i). \tag{6.60}$$

The exact splitting identity is

$$\mathsf{K}_{N,\beta}^{\mathrm{att}}(\mu) = e^{\beta I_\mu^{\mathrm{att}}} \mathbb{E}_{\mu^{\otimes N}}\Big[e^{\frac{\beta}{2}W_N + \beta R_N}\Big]. \tag{6.61}$$

The bounded-density assumption gives the uniform lower bound

$$h_\mu^{\mathrm{att}}(x) \geq \|\mu\|_{L^\infty} \int_{|z|<1} \log|z| \, \mathrm{d}z > -\infty, \tag{6.62}$$

because the contribution from $|x-y| \geq 1$ is nonnegative. Together with $e^{4\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$, this implies $\psi_\mu^{\mathrm{att}} \in L^2(\mu)$ and

$$\mathbb{E}_{\mu^{\otimes N}}\Big[R_N^2\Big] = \frac{\|\psi_\mu^{\mathrm{att}}\|_{L^2(\mu)}^2}{N}. \tag{6.63}$$

Using $|e^z - 1| \leq |z|(1+e^z)$ and Cauchy–Schwarz, we obtain

$$\begin{aligned}
&\Big|\mathsf{K}_{N,\beta}^{\mathrm{att}}(\mu) - e^{\beta I_\mu^{\mathrm{att}}} \mathbb{E}_{\mu^{\otimes N}}\Big[e^{\frac{\beta}{2}W_N}\Big]\Big| \\
&\qquad \leq C\mathbb{E}_{\mu^{\otimes N}}\Big[R_N^2\Big]^{1/2}\Big(\mathbb{E}_{\mu^{\otimes N}}\Big[e^{\beta W_N}\Big]^{1/2} + \mathbb{E}_{\mu^{\otimes N}}\Big[e^{\beta W_N + 2\beta R_N}\Big]^{1/2}\Big) \leq CN^{-1/2}.
\end{aligned} \tag{6.64}$$

The first exponential moment is controlled by (6.47). Up to the deterministic factor $e^{-2\beta I_\mu^{\mathrm{att}}}$, the second is the attractive modulated partition function at inverse temperature 2β, which is uniformly bounded by the first assertion of Theorem 2.6, since $\beta_{\mathrm{q}} \leq \beta_0/2$ and $e^{4\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$ is exactly the required one-body integrability at that temperature.

Finally, $T_\mu = -T_\mu^{\mathrm{att}}$, and therefore

$$\det_2(\mathrm{Id} - \beta T_\mu) = \det_2\Big(\mathrm{Id} + \beta T_\mu^{\mathrm{att}}\Big). \tag{6.65}$$

Combining (6.58), (6.64), and (6.65) proves (2.38). □

Remark 6.2. The rate in (2.38) is deliberately stated as an unspecified power. The proof tracks an explicit exponent through the exponential moment of the truncated feature diagonal, but that exponent is not expected to be sharp. For a periodic uniform background, the one-body term vanishes and the hypotheses involving h_μ^{att} are automatic.

6.3. **Periodic uniform backgrounds: phase diagram and lower bounds.** We now specialize to the periodic attractive logarithmic interaction with uniform background. Let μ_{unif} denote normalized Lebesgue measure on $\mathbb{T}^{\mathsf{d}}$, and let g_{d} be the zero-average repulsive periodic logarithmic kernel normalized by

$$(-\Delta)^{\mathsf{d}/2}\mathsf{g}_{\mathsf{d}} = \mathsf{c}_{\mathsf{d}}(\delta_0 - 1), \qquad \mathsf{c}_{\mathsf{d}} := \frac{\Gamma(\mathsf{d}/2)(4\pi)^{\mathsf{d}/2}}{2}, \tag{6.66}$$

so that

$$\widehat{\mathsf{g}_{\mathsf{d}}}(k) = \frac{\mathsf{c}_{\mathsf{d}}}{(2\pi|k|)^{\mathsf{d}}}, \qquad k \in \mathbb{Z}^{\mathsf{d}} \setminus \{0\}. \tag{6.67}$$

The attractive interaction is $-\mathsf{g}_{\mathsf{d}}$, and its modulated partition function around μ_{unif} is

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \coloneqq \int_{(\mathbb{T}^{\mathsf{d}})^N} \exp\left\{\frac{\beta}{2N}\sum_{1\le i\neq j\le N}\mathsf{g}_{\mathsf{d}}(x_i-x_j)\right\}\mathrm{d}\mu_{\mathrm{unif}}^{\otimes N}(X_N). \tag{6.68}$$

Let T_{d} be convolution by g_{d} on the mean-zero subspace of $L^2(\mu_{\mathrm{unif}})$. If a probability measure μ is absolutely continuous with respect to μ_{unif}, we continue the standing abuse of notation and also write μ for its density. For such densities with finite relative entropy, set

$$\mathcal{J}_{\beta,\mathsf{d}}(\mu) \coloneqq \mathsf{H}(\mu|\mu_{\mathrm{unif}}) - \frac{\beta}{2}\left\langle \mu-1, T_{\mathsf{d}}(\mu-1)\right\rangle_{L^2(\mu_{\mathrm{unif}})}. \tag{6.69}$$

Here, the quadratic form denotes $\iint_{(\mathbb{T}^{\mathsf{d}})^2}\mathsf{g}_{\mathsf{d}}(x-y)\,\mathrm{d}\mu^{\otimes 2}(x,y)$, which is finite for probability densities with finite entropy and agrees with the displayed pairing for L^2 densities. All infima involving $\mathcal{J}_{\beta,\mathsf{d}}$ below are taken over probability measures of finite relative entropy with respect to μ_{unif}. We denote by

$$\beta_{\mathrm{gm}}(\mathsf{d}) \coloneqq \sup\left\{\beta\in[0,2\mathsf{d}) : \mathcal{J}_{\beta,\mathsf{d}}(\mu)\ge 0 \text{ for every } \mu\ll\mu_{\mathrm{unif}} \text{ of finite relative entropy}\right\} \tag{6.70}$$

the global-minimization threshold of the uniform phase. The largest eigenvalue of T_{d} equals $(\beta_{\mathrm{s}}(\mathsf{d}))^{-1}$, where

$$\beta_{\mathrm{s}}(\mathsf{d}) \coloneqq \frac{(2\pi)^{\mathsf{d}}}{\mathsf{c}_{\mathsf{d}}}, \tag{6.71}$$

and its eigenspace is

$$E_1 \coloneqq \operatorname{span}\left\{\cos(2\pi x_j), \sin(2\pi x_j) : 1\le j\le \mathsf{d}\right\}. \tag{6.72}$$

Thus, E_1 is the $2\mathsf{d}$-dimensional lowest nonzero Fourier eigenspace, corresponding to the frequencies $k=\pm e_j$. The value $\beta_{\mathrm{s}}(\mathsf{d})$ is the nonlinear stability threshold identified in [CdCRS25a]; moreover, $\beta_{\mathrm{s}}(\mathsf{d})<2\mathsf{d}$ exactly in integer dimensions $\mathsf{d}\ge 11$.

For the formulation of part (ii) below, decompose a density near the uniform state as $\mu-1=u+w$, with $u\in E_1$ and $w\in E_1^{\perp}$. Solving the $E_1^{\perp}$ component of the Euler–Lagrange equation for $w=w(u)$ near the origin gives the finite-dimensional reduced free energy

$$\mathcal{J}^{\mathrm{red}}_{\beta,\mathsf{d}}(u) \coloneqq \mathcal{J}_{\beta,\mathsf{d}}(1+u+w(u)); \tag{6.73}$$

at $\beta=\beta_{\mathrm{s}}(\mathsf{d})$, its first nonconstant term is quartic.

In dimension one, the circular Selberg–Dyson integral [Dys62] gives the exact formula

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) = \frac{\Gamma(1-\beta/2)}{\Gamma(1-\beta/(2N))^N}, \qquad 0\le\beta<2. \tag{6.74}$$

Since $\widehat{\mathsf{g}_1}(k)=(2|k|)^{-1}$ for $k\neq 0$, it follows that

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \longrightarrow e^{-\gamma_{\mathrm{E}}\beta/2}\Gamma(1-\beta/2) = \det{}_2(\mathrm{Id}-\beta T_1)^{-1/2}. \tag{6.75}$$

Here γ_{E} denotes Euler's constant. At the opposite end of the presently understood phase diagram, Lei and the third author [LR26] prove, by means of an explicit wrapped Cauchy competitor, the strict dimension-eleven bound

$$\beta_{\mathrm{gm}}(11) < \frac{8653}{8673}\,\beta_{\mathrm{s}}(11) = \frac{8653}{8673}\frac{64\pi^5}{945} < \beta_{\mathrm{s}}(11). \tag{6.76}$$

Thus, in dimension eleven, the uniform phase loses global minimality strictly before it loses local stability.

The sharp inequalities of Lin and Lucia in dimension two [LL07] and Lei and the third author in dimensions $\mathsf{d}\ge 3$ [LR26] identify $\beta_{\mathrm{gm}}(\mathsf{d})=2\mathsf{d}$ for $2\le\mathsf{d}\le 10$ and $\beta_{\mathrm{gm}}(\mathsf{d})=\beta_{\mathrm{s}}(\mathsf{d})$ for $\mathsf{d}\ge 12$; in the

latter dimensions the macroscopic transition is continuous. Combined with the transfer theorem of [DGR26a] and the exact one-dimensional formula (6.74), these results give

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \longrightarrow \det{}_2(\mathrm{Id}-\beta T_{\mathsf{d}})^{-1/2}, \qquad 0 \le \beta < 2\mathsf{d}, \quad 1 \le \mathsf{d} \le 10. \tag{6.77}$$

In dimension eleven, the determinant limit holds throughout $0 \le \beta < \beta_{\mathrm{gm}}(11)$. For $\mathsf{d} \ge 12$,

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \longrightarrow \det{}_2(\mathrm{Id}-\beta T_{\mathsf{d}})^{-1/2}, \qquad 0 \le \beta < \beta_{\mathrm{s}}(\mathsf{d}). \tag{6.78}$$

Conjecture 6.3 (Critical and supercritical partition asymptotics)**.** (i) In dimension $\mathsf{d} = 11$,

$$\frac{1}{N}\log \mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \longrightarrow -\inf_{\mu \ll \mu_{\mathrm{unif}}} \mathcal{J}_{\beta,11}(\mu) > 0, \qquad \beta_{\mathrm{gm}}(11) < \beta < 22. \tag{6.79}$$

No finite-particle partition asymptotic is asserted here at the coexistence point $\beta = \beta_{\mathrm{gm}}(11)$.

(ii) If $\mathsf{d} \ge 12$, the leading quartic term of $\mathcal{J}^{\mathrm{red}}_{\beta_{\mathrm{s}}(\mathsf{d}),\mathsf{d}}$ is strictly positive away from the origin in E_1. Moreover,

$$\mathsf{K}_{N,\beta_{\mathrm{s}}(\mathsf{d})}(\mu_{\mathrm{unif}}) \sim C_{\mathsf{d}} N^{\mathsf{d}/2}, \qquad C_{\mathsf{d}} \in (0,\infty), \tag{6.80}$$

$$\frac{1}{N}\log \mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \longrightarrow -\inf_{\mu \ll \mu_{\mathrm{unif}}} \mathcal{J}_{\beta,\mathsf{d}}(\mu) > 0, \qquad \beta_{\mathrm{s}}(\mathsf{d}) < \beta < 2\mathsf{d}. \tag{6.81}$$

The subcritical determinant limits are thus established in every dimension. The remaining assertions concern the positivity of the reduced quartic term, the critical normalization, and the matching supercritical pressure asymptotics. We next record lower bounds supporting these assertions.

Proposition 6.4 (Critical and supercritical lower bounds)**.** *Assume* $\mathsf{d} \ge 11$. *Then:*

(i) At the stability threshold,

$$\liminf_{N\to\infty} N^{-\mathsf{d}/2}\mathsf{K}_{N,\beta_{\mathrm{s}}(\mathsf{d})}(\mu_{\mathrm{unif}}) \ge \left(\frac{\sqrt{\pi}}{e}\right)^{\mathsf{d}}. \tag{6.82}$$

(ii) For every $\beta_{\mathrm{s}}(\mathsf{d}) < \beta < 2\mathsf{d}$,

$$\liminf_{N\to\infty} \frac{1}{N}\log \mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) > 0. \tag{6.83}$$

(iii) For every $\beta \ge 2\mathsf{d}$ *and every* $N \ge 2$,

$$\mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) = +\infty. \tag{6.84}$$

Proof of Proposition 6.4. **Part (i): Critical lower bound at the stability threshold.** For $t > 0$, let $\mathsf{g}_{\mathsf{d},t} = e^{t\Delta}\mathsf{g}_{\mathsf{d}}$ be the heat regularization and let $\mathsf{K}^{(t)}_{N,\beta}(\mu_{\mathrm{unif}})$ denote (6.68) with g_{d} replaced by $\mathsf{g}_{\mathsf{d},t}$. The local comparison

$$\mathsf{g}_{\mathsf{d},t}(x) \le \mathsf{g}_{\mathsf{d}}(x) + C_{\mathsf{d}}, \qquad 0 < t \le 1, \tag{6.85}$$

follows from $\mathsf{g}_{\mathsf{d}}(x) = -\log|x| + O(1)$ near the origin by splitting into $|x| \le 2\sqrt{t}$ and its complement. If a block of k particles collapses on scale R, then the relative-coordinate volume element contributes $R^{\mathsf{d}(k-1)-1}\,\mathrm{d}R$, while the attractive pair factor contributes at worst $R^{-\beta k(k-1)/(2N)}$. The corresponding radial integral is therefore bounded by

$$\int_0^{\delta} R^{(k-1)(\mathsf{d}-\beta k/(2N))-1}\,\mathrm{d}R, \tag{6.86}$$

which is finite for every $2 \le k \le N$ when $\beta < 2\mathsf{d}$. Decomposing a neighborhood of the collision set according to the finitely many set partitions of $\{1,\dots,N\}$, and applying (6.86) successively to each

collapsing block, proves local integrability near every collision stratum. Hence, since $\beta_{\mathrm{s}}(\mathsf{d}) < 2\mathsf{d}$, dominated convergence gives

$$\mathsf{K}^{(t)}_{N,\beta_{\mathrm{s}}(\mathsf{d})}(\mu_{\mathrm{unif}}) \longrightarrow \mathsf{K}_{N,\beta_{\mathrm{s}}(\mathsf{d})}(\mu_{\mathrm{unif}}) \qquad (t \downarrow 0) \tag{6.87}$$

for each fixed N.

Let X_t be the centered Gaussian field with covariance $\mathsf{g}_{\mathsf{d},t}$. The Hubbard–Stratonovich transformation gives

$$\mathsf{K}^{(t)}_{N,\beta}(\mu_{\mathrm{unif}}) = \mathbb{E}\left[\left(\int_{\mathbb{T}^{\mathsf{d}}} \exp\left\{\sqrt{\frac{\beta}{N}} X_t(x) - \frac{\beta}{2N}\mathsf{g}_{\mathsf{d},t}(0)\right\} \mathrm{d}\mu_{\mathrm{unif}}(x)\right)^N\right]. \tag{6.88}$$

Decompose $X_t = X_{t,1} + X_{t,\perp}$ according to $E_1 \oplus E_1^{\perp}$. Conditional Jensen in $X_{t,\perp}$ removes the latter from (6.88). The eigenvalue on E_1 is

$$\lambda_t = \frac{e^{-4\pi^2 t}}{\beta_{\mathrm{s}}(\mathsf{d})}, \tag{6.89}$$

and one may write

$$X_{t,1}(x) = \sqrt{2\lambda_t} \sum_{j=1}^{\mathsf{d}} \left(A_j \cos(2\pi x_j) + B_j \sin(2\pi x_j)\right), \tag{6.90}$$

where the A_j, B_j are independent standard Gaussians. Passing to polar coordinates in each pair (A_j, B_j) therefore yields

$$\mathsf{K}^{(t)}_{N,\beta}(\mu_{\mathrm{unif}}) \geq e^{-\beta \mathsf{d} \lambda_t} \left[\int_0^\infty 2re^{-r^2} I_0\left(2\sqrt{\frac{\beta\lambda_t}{N}}\, r\right)^N \mathrm{d}r\right]^{\mathsf{d}}, \tag{6.91}$$

where

$$I_0(z) \coloneqq \int_0^1 e^{z\cos(2\pi x)} \,\mathrm{d}x \tag{6.92}$$

is the modified Bessel function. Taking $\beta = \beta_{\mathrm{s}}(\mathsf{d})$ and then $t \downarrow 0$ in (6.91), using (6.87), gives

$$\mathsf{K}_{N,\beta_{\mathrm{s}}(\mathsf{d})}(\mu_{\mathrm{unif}}) \geq e^{-\mathsf{d}} J_N^{\mathsf{d}}, \qquad J_N \coloneqq \int_0^\infty 2re^{-r^2} I_0\left(\frac{2r}{\sqrt{N}}\right)^N \mathrm{d}r. \tag{6.93}$$

The power series of I_0 gives

$$\log I_0(z) = \frac{z^2}{4} - \frac{z^4}{64} + O(z^6) \qquad (z \to 0). \tag{6.94}$$

After the change of variables $r = N^{1/4}u$,

$$N^{-1/2} J_N = \int_0^\infty 2u \exp\left\{-N^{1/2}u^2 + N \log I_0\left(\frac{2u}{N^{1/4}}\right)\right\} \mathrm{d}u. \tag{6.95}$$

The integrand converges pointwise to $2ue^{-u^4/4}$. Fatou's lemma and (6.93) imply

$$\liminf_{N\to\infty} N^{-1/2} J_N \geq \int_0^\infty 2ue^{-u^4/4} \,\mathrm{d}u = \sqrt{\pi}, \tag{6.96}$$

which proves (6.82).

Part (ii): Exponential lower bound above the stability threshold. Let $\mu \ll \mu_{\mathrm{unif}}$ be a probability measure. The Donsker–Varadhan lemma, applied with the product trial $\mu^{\otimes N}$, gives

$$\log \mathsf{K}_{N,\beta}(\mu_{\mathrm{unif}}) \geq \frac{\beta(N-1)}{2} \left\langle \mu - 1, T_{\mathsf{d}}(\mu - 1)\right\rangle_{L^2(\mu_{\mathrm{unif}})} - N\mathsf{H}(\mu|\mu_{\mathrm{unif}}). \tag{6.97}$$

For $|\varepsilon| < 1$, take $\mu_\varepsilon(x) = 1 + \varepsilon\cos(2\pi x_1)$. By (6.67) and (6.71),

$$\langle \mu_\varepsilon - 1, T_{\mathsf{d}}(\mu_\varepsilon - 1)\rangle_{L^2(\mu_{\mathrm{unif}})} = \frac{\varepsilon^2}{2\beta_{\mathrm{s}}(\mathsf{d})}, \qquad \mathsf{H}(\mu_\varepsilon|\mu_{\mathrm{unif}}) = \frac{\varepsilon^2}{4} + O(\varepsilon^4). \tag{6.98}$$

Consequently, for every $\beta > \beta_{\mathrm{s}}(\mathsf{d})$ and every sufficiently small nonzero ε,

$$\frac{\beta}{2}\langle \mu_\varepsilon - 1, T_{\mathsf{d}}(\mu_\varepsilon - 1)\rangle_{L^2(\mu_{\mathrm{unif}})} - \mathsf{H}(\mu_\varepsilon|\mu_{\mathrm{unif}}) > 0. \tag{6.99}$$

Divide (6.97) by N and let $N \to \infty$ to obtain (6.83).

Part (iii): Divergence at and above the collision threshold. Fix a small coordinate cube $Q \subset \mathbb{T}^{\mathsf{d}}$ and restrict the integral in (6.68) to configurations with $x_1 \in Q$ and $x_i = x_1 + y_i$, $2 \le i \le N$, where the relative vector $y = (y_2, \dots, y_N) \in \mathbb{R}^{\mathsf{d}(N-1)}$ has sufficiently small norm $R = |y|$. Since $\mathsf{g}_{\mathsf{d}}(z) = -\log|z| + O(1)$ near the origin and every pair distance is at most $2R$ on this set, the integrand is bounded from below by $C_{N,\beta,\mathsf{d}} R^{-\beta(N-1)/2}$. The corresponding radial integral is bounded from below by

$$C_{N,\beta,\mathsf{d}} \int_0^\delta R^{(N-1)(\mathsf{d}-\beta/2)-1}\, \mathrm{d}R. \tag{6.100}$$

The integral diverges logarithmically when $\beta = 2\mathsf{d}$ and by a power when $\beta > 2\mathsf{d}$, proving (6.84). □

Remark 6.5 (The dimension-eleven critical point)**.** The lower bound (6.82) holds in dimension eleven but is not rate-sharp there. Indeed, the strict inequality (6.76) and the product trial (6.97) imply

$$\liminf_{N\to\infty} \frac{1}{N} \log \mathsf{K}_{N,\beta_{\mathrm{s}}(11)}(\mu_{\mathrm{unif}}) > 0. \tag{6.101}$$

Thus the conjectured sharp role of the polynomial factor $N^{\mathsf{d}/2}$ begins in dimensions $\mathsf{d} \ge 12$.

7. Entropic commutators and diffusive mean-field estimates

We now turn to the proof of Theorem 2.8, which has three steps. The Donsker–Varadhan reduction in Section 7.1 reduces the desired entropic inequality to an exponential moment under $\mathbb{Q}_{N,\beta}(\mu)$, the local numerator estimate in Section 7.3 proves this moment for small inverse temperature, and the Hölder bootstrap in Section 7.4 removes the smallness restriction. After proving the theorem, Section 7.5 proves Corollary 7.3, which combines the fixed-temperature entropic estimate with the deterministic pointwise commutator bound and is adapted to inverse temperatures growing with N. The section concludes with the proof of the diffusive mean-field application, Theorem 2.9, in Section 7.6.

7.1. Donsker–Varadhan reduction.

The case $\|\nabla v\|_{L^\infty} = 0$ has already been disposed of in the statement, so we assume $\|\nabla v\|_{L^\infty} > 0$. By assumption, $\mathsf{H}(f_N|\mathbb{Q}_{N,\beta}(\mu)) < \infty$. Once (2.42) is proved, it gives $\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu)}[e^{\varepsilon_{\beta,n}|N\mathsf{A}_n|}] \le 2C_{\mathrm{em}}$. Applying the entropy inequality to $\min\{|N\mathsf{A}_n|, M\}$ and letting $M \to \infty$ proves absolute integrability under f_N. By the Donsker–Varadhan lemma, for any $\varepsilon > 0$ and $\sigma \in \{\pm 1\}$,

$$\sigma\mathbb{E}_{X_N\sim f_N}\Big[N\mathsf{A}_n(v, X_N, \mu)\Big] \le \frac{1}{\varepsilon}\left(\mathsf{H}(f_N|\mathbb{Q}_{N,\beta}(\mu)) + \log \mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\sigma\varepsilon N\mathsf{A}_n(v,X_N,\mu)}\Big]\right). \tag{7.1}$$

Using the entropy rewriting (2.11), together with (2.33), it remains to prove (2.42) for the choice $\varepsilon = \varepsilon_{\beta,n}$ in (A.8).

Fix $\sigma \in \{\pm 1\}$. Unpacking the definition of $\mathbb{Q}_{N,\beta}(\mu)$, we have

$$\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu)}\Big] = \frac{1}{\mathsf{K}_{N,\beta}(\mu)}\mathbb{E}_{X_N\sim\mu^{\otimes N}}\Big[e^{\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu)-\beta N\mathsf{F}_N(X_N,\mu)}\Big]. \tag{7.2}$$

In view of (2.12), it suffices to prove the product-space numerator estimate

$$\sup_{N\geq 1}\mathbb{E}_{X_N\sim\mu^{\otimes N}}\left[e^{\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu)-\beta N\mathsf{F}_N(X_N,\mu)}\right]<\infty \tag{7.3}$$

uniformly in N and in $\sigma\in\{\pm 1\}$.

7.2. **Commutator truncation estimates.** The next lemma converts the potential truncations of Section 4.1 into the derivative and short-range estimates needed for the commutator argument.

Lemma 7.1 (Commutator truncation estimates)**.** *Let $0<\mathsf{s}<\mathsf{d}$, let $n\geq 1$, and let $v:\mathbb{R}^{\mathsf{d}}\to\mathbb{R}^{\mathsf{d}}$ be globally Lipschitz. For a differentiable kernel ϕ, let*

$$\mathsf{k}_{n,\phi}(x,y)\coloneqq\nabla^{\otimes n}\phi(x-y):(v(x)-v(y))^{\otimes n}, \tag{7.4}$$

which is symmetric whenever ϕ is even. There are constants C_n and $c_n\geq 2$, depending only on d,s,n, such that the truncations in (4.3) *satisfy, for $x\neq y$,*

$$|\mathsf{k}_{n,\mathsf{g}_\eta}(x,y)|\leq C_n\|\nabla v\|_{L^\infty}^n\mathsf{g}_\eta(x-y), \tag{7.5}$$

$$|\mathsf{k}_{n,\mathsf{f}_\eta}(x,y)|\leq C_n\|\nabla v\|_{L^\infty}^n\mathsf{f}_{c_n\eta}(x-y). \tag{7.6}$$

If $\mu\in L^\infty$ and $\sigma\in\{\pm1\}$, then the short-range deterministic estimate holds for configurations with distinct points:

$$\frac{\sigma}{N^2}\sum_{i\neq j}\tilde{\mathsf{k}}_{n,\mathsf{f}_\eta}(x_i,x_j)\leq\frac{C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n}{N^2}\sum_{i\neq j}\mathsf{f}_{c_n\eta}(x_i-x_j)+C_n\|\nabla v\|_{L^\infty}^n\|\mu\|_{L^\infty}\eta^{\mathsf{d}-\mathsf{s}}, \tag{7.7}$$

where $C_{\mathrm{sr},n}$ depends only on d,s,n.

Proof. Integrating the Gaussian comparison from the proof of (4.14) against $\mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}t^{-\mathsf{s}}\,\mathrm{d}t/t$, and substituting $u=2t$, gives, for $z\neq 0$,

$$|z|^n|\nabla^{\otimes n}\mathsf{f}_\eta(z)|\leq C_n2^{\mathsf{s}}\mathsf{f}_{2\eta}(z), \tag{7.8}$$

$$|z|^n|\nabla^{\otimes n}\mathsf{g}_\eta(z)|\leq C_n2^{\mathsf{s}}\mathsf{g}_{2\eta}(z)\leq C_n2^{\mathsf{s}}\mathsf{g}_\eta(z). \tag{7.9}$$

Gaussian decay at zero scale and the integrable weight at infinity justify differentiation under the integral. Taking the tensor inner product with $(v(x)-v(y))^{\otimes n}$ and using $|v(x)-v(y)|\leq\|\nabla v\|_{L^\infty}|x-y|$ proves (7.5)–(7.6) with $c_n=2$ and enlarged C_n.

The short-range bound (7.6) gives $|\mathsf{k}_{n,\mathsf{f}_\eta}(x,y)|\leq C_n\|\nabla v\|_{L^\infty}^n\mathsf{f}_{2\eta}(x-y)$. Each integral with respect to one variable and the double integral subtracted or added in the centering formula (2.2) is bounded in absolute value by $C_n\|\nabla v\|_{L^\infty}^n\|\mu\|_{L^\infty}\|\mathsf{f}_{2\eta}\|_{L^1(\mathbb{R}^{\mathsf{d}})}$. Hence, for either sign $\sigma\in\{\pm1\}$,

$$\sigma\tilde{\mathsf{k}}_{n,\mathsf{f}_\eta}(x,y)\leq C_n\|\nabla v\|_{L^\infty}^n\mathsf{f}_{2\eta}(x-y)+3C_n\|\nabla v\|_{L^\infty}^n\|\mu\|_{L^\infty}\|\mathsf{f}_{2\eta}\|_{L^1(\mathbb{R}^{\mathsf{d}})}. \tag{7.10}$$

Since $\|\mathsf{f}_{2\eta}\|_{L^1(\mathbb{R}^{\mathsf{d}})}=\mathsf{c}_{\phi,\mathsf{d},\mathsf{s}}(2\eta)^{\mathsf{d}-\mathsf{s}}/(\mathsf{d}-\mathsf{s})$ by Fubini–Tonelli in (4.3), summing over the $N(N-1)$ ordered off-diagonal pairs and using $N(N-1)/N^2\leq 1$ proves (7.7), after enlarging the constants. We may take $C_{\mathrm{sr},n}$ to dominate both occurrences of C_n, as required below. □

7.3. **Local numerator estimate.** This lemma is the main analytic estimate in the commutator proof. It repeats the partition-function layer-cake mechanism with the commutator observable inserted, choosing the coefficient so that the short-range commutator contribution is absorbed by the modulated energy.

Lemma 7.2 (Local numerator estimate)**.** *There exists $\beta_{\mathrm{loc}}=\beta_{\mathrm{loc}}(\mu,v,n)>0$ such that for every $0<\gamma\leq\beta_{\mathrm{loc}}$ and every $\sigma\in\{\pm1\}$,*

$$\sup_{N\geq1}\mathbb{E}_{\mu^{\otimes N}}\left[\exp\left(\sigma\frac{\gamma}{4C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n}N\mathsf{A}_n(v,X_N,\mu)-\gamma N\mathsf{F}_N(X_N,\mu)\right)\right]\leq e+L. \tag{7.11}$$

Proof. We use the truncation and commutator-kernel notation of Lemma 7.1, and set

$$\varepsilon_{\gamma,n} \coloneqq \frac{\gamma}{4C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n}. \tag{7.12}$$

The one-body terms in the expansion of $\mathsf{A}_n(v, X_N, \mu)$ are bounded by a constant depending on μ, v, n. More precisely,

$$\mathsf{A}_n(v, X_N, \mu) = \frac{1}{N^2} \sum_{1\le i\neq j\le N} \tilde{\mathsf{k}}_{n,\mathsf{g}}(x_i, x_j) + R_{n,N}(X_N), \qquad |NR_{n,N}(X_N)| \le C_{\mu,v,n}. \tag{7.13}$$

We treat first the positive Riesz case $0 < \mathsf{s} < \mathsf{d}/2$; the logarithmic case is Step 4.

Step 1: Reduction to the centered statistics. For every $\sigma \in \{\pm 1\}$ and $0 < \gamma \le \beta_{\mathrm{loc}}$, the bound (7.11) follows from

$$\sup_{N\ge 1} \mathbb{E}_{\mu^{\otimes N}}\big[e^{\Xi_N^\sigma}\big] \le e + L, \qquad \Xi_N^\sigma \coloneqq \varepsilon_{\gamma,n} C_{\mu,v,n} + \frac{\sigma\varepsilon_{\gamma,n}}{N} \sum_{1\le i\neq j\le N} \tilde{\mathsf{k}}_{n,\mathsf{g}}(x_i, x_j) - \gamma N \mathsf{F}_N(X_N, \mu), \tag{7.14}$$

and, by (2.4), Ξ_N^σ is of the master form (3.10), with $g = \gamma\mathsf{g} - 2\sigma\varepsilon_{\gamma,n}\mathsf{k}_{n,\mathsf{g}}$ and $w = \gamma(h_\mu - I_\mu) + \varepsilon_{\gamma,n}C_{\mu,v,n}$.

By (7.13), the exponent in (7.11) is $\sigma\varepsilon_{\gamma,n}N\mathsf{A}_n - \gamma N\mathsf{F}_N \le \Xi_N^\sigma$ pointwise, so (7.14) implies (7.11). The second assertion is the splitting identity (2.4), the constant $\varepsilon_{\gamma,n}C_{\mu,v,n}$ being part of the one-body function w.

Step 2: Choice of the data. With c_n and C_n the constants in (7.5)–(7.6) and (7.7), and C_0 the near-field constant in (4.8), set

$$C_{\mathrm{abs},n} \coloneqq \frac{C_n\|\mu\|_{L^\infty}}{4C_{\mathrm{sr},n}} + c_n^{\mathsf{d}-\mathsf{s}}\, C_0. \tag{7.15}$$

Define $U_0 \coloneqq 1$ and, for $N \ge 1$, $u \ge 1$, and $0 < \eta \le \eta_0$, the kernels, adaptive scales, and caps

$$\tilde{h}_{n,\eta}^\sigma \coloneqq \sigma\varepsilon_{\gamma,n}\tilde{\mathsf{k}}_{n,\mathsf{g}_\eta} - \frac{\gamma}{2}\tilde{\mathsf{g}}_{c_n\eta}, \qquad \eta(u,N) \coloneqq \min\left\{\eta_0, \Big(\frac{u}{4C_{\mathrm{abs},n}\gamma N}\Big)^{\frac{1}{\mathsf{d}-\mathsf{s}}}\right\}, \tag{7.16}$$

$$q_{N,u} \coloneqq -4\,\tilde{h}_{n,\eta(u,N)}^\sigma, \qquad \mathsf{m}_{N,u} \coloneqq \frac{N}{2}\,\|\tilde{q}_{N,u}\|_{L^\infty}. \tag{7.17}$$

Assuming, as we may, $C_{\mathrm{sr},n} \ge C_n$, define the uncentered kernel

$$r_{N,u}^\sigma \coloneqq -4\sigma\varepsilon_{\gamma,n}\mathsf{k}_{n,\mathsf{g}_{\eta(u,N)}} + 2\gamma\,\mathsf{g}_{c_n\eta(u,N)}. \tag{7.18}$$

Then $q_{N,u}$ is the μ-centering of $r_{N,u}^\sigma$ and, by (7.5), (7.12), and $0 \le \mathsf{g}_{c_n\eta} \le \mathsf{g}_\eta$,

$$|r_{N,u}^\sigma| \le 3\gamma\,\mathsf{g}_{\eta(u,N)}. \tag{7.19}$$

Since $\mathsf{k}_{n,\mathsf{g}_\eta}$ is not Fourier sign-definite, Lemma 4.2 does not apply, and the full L^∞ norm has to be used in the cap of (7.17).

Then there are constants $M_{\mu,v,n} < \infty$ and $\beta_{\mathrm{loc}} = \beta_{\mathrm{loc}}(\mu, v, n) > 0$ such that, for every $0 < \gamma \le \beta_{\mathrm{loc}}$ and every $\sigma \in \{\pm 1\}$, these data satisfy the hypotheses (M1)–(M3) of Proposition 3.5 for Ξ_N^σ, together with

$$\sup_{u\ge 1} \min\{u,\ \mathsf{m}_{N,u}\}(1 + \|\tilde{q}_{N,u}\|_{L^\infty}) \le M_{\mu,v,n}\,\gamma\,N \qquad \text{and} \qquad 2L^2 M_{\mu,v,n}\gamma \le 1. \tag{7.20}$$

In particular, $2L^2\,\mathsf{S}_N \le 2L^2 M_{\mu,v,n}\gamma N \le N$ for every $N \ge 1$, so the growth condition (3.14) holds with $N_0 = 1$.

(M2). Hypothesis (M2) holds by the definition of $\mathsf{m}_{N,u}$. Indeed, for every configuration,

$$-\frac{1}{2N}\sum_{1\le i\neq j\le N}\tilde{q}_{N,u}(x_i,x_j)\le\frac{N}{2}\|\tilde{q}_{N,u}\|_{L^\infty}. \tag{7.21}$$

(M1). Using the decomposition (2.4) with $\mathsf{g}=\mathsf{g}_{c_n\eta}+\mathsf{f}_{c_n\eta}$, and bounding the one-body term by $\gamma\|h_\mu\|_{L^\infty}$ and $-\gamma I_\mu\le 0$,

$$\Xi_N^\sigma\le C'_{\mu,v,n}\gamma+\frac{1}{N}\sum_{1\le i\neq j\le N}\tilde{h}^\sigma_{n,\eta}(x_i,x_j) + \frac{\sigma\varepsilon_{\gamma,n}}{N}\sum_{1\le i\neq j\le N}\tilde{\mathsf{k}}_{n,\mathsf{f}_\eta}(x_i,x_j)-\frac{\gamma}{2N}\sum_{1\le i\neq j\le N}\tilde{\mathsf{f}}_{c_n\eta}(x_i,x_j), \tag{7.22}$$

where $C'_{\mu,v,n}\coloneqq C_{\mu,v,n}/(4C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n)+\|h_\mu\|_{L^\infty}$. By (7.7), (4.6), (4.8), and the choice (7.12),

$$\frac{\sigma\varepsilon_{\gamma,n}}{N}\sum_{1\le i\neq j\le N}\tilde{\mathsf{k}}_{n,\mathsf{f}_\eta}(x_i,x_j)-\frac{\gamma}{2N}\sum_{1\le i\neq j\le N}\tilde{\mathsf{f}}_{c_n\eta}(x_i,x_j)\le C_{\mathrm{abs},n}\gamma N\eta^{\mathsf{d}-\mathsf{s}}; \tag{7.23}$$

indeed, multiplying (7.7) by $\varepsilon_{\gamma,n}N$ contributes $\frac{\gamma}{4N}\sum\mathsf{f}_{c_n\eta}+\frac{C_n\|\mu\|_{L^\infty}}{4C_{\mathrm{sr},n}}\gamma N\eta^{\mathsf{d}-\mathsf{s}}$, the total coefficient of $\sum\mathsf{f}_{c_n\eta}$ is then nonpositive, and the centered one-body terms of $-\frac{\gamma}{2N}\sum\tilde{\mathsf{f}}_{c_n\eta}$ are bounded by $\gamma N\|\mathsf{f}_{c_n\eta}*\mu\|_{L^\infty}\le c_n^{\mathsf{d}-\mathsf{s}}C_0\,\gamma N\eta^{\mathsf{d}-\mathsf{s}}$, by (4.8), matching (7.15). Combining (7.22) and (7.23) gives, for every $0<\eta\le\eta_0$,

$$\Xi_N^\sigma\le C'_{\mu,v,n}\gamma+\frac{1}{N}\sum_{1\le i\neq j\le N}\tilde{h}^\sigma_{n,\eta}(x_i,x_j)+C_{\mathrm{abs},n}\gamma N\eta^{\mathsf{d}-\mathsf{s}}. \tag{7.24}$$

Assume $4C'_{\mu,v,n}\gamma\le 1$, and let $u\ge 1$. Taking $\eta=\eta(u,N)$ in (7.24), the last term is at most $u/4$ by (7.16), while $C'_{\mu,v,n}\gamma\le 1/4\le u/4$; hence

$$\Xi_N^\sigma\ge u\quad\Longrightarrow\quad\frac{1}{N}\sum_{1\le i\neq j\le N}\tilde{h}^\sigma_{n,\eta(u,N)}(x_i,x_j)\ge\frac{u}{2}\quad\Longleftrightarrow\quad-\frac{1}{2N}\sum_{1\le i\neq j\le N}\tilde{q}_{N,u}(x_i,x_j)\ge u, \tag{7.25}$$

which is (M1).

(M3) and the L^∞ norm. Since the centering costs at most a factor 4 in each of the norms of Remark 3.2, the majorant (7.19) bounds every norm of $\tilde{q}_{N,u}$ by 12γ times the corresponding norm of $\mathsf{g}_{\eta(u,N)}$. These are bounded uniformly in the scale: $0\le\mathsf{g}_\eta\le\mathsf{g}$, the $L^2(\mu^{\otimes 2})$ and $L^\infty L^2(\mu)$ norms of g against μ are finite exactly because $2\mathsf{s}<\mathsf{d}$, as in the partition-function case, and the operator norm follows from the Hilbert–Schmidt bound; only the L^∞ norm sees the scale, through $\|\mathsf{g}_\eta\|_{L^\infty}\le C_2\eta^{-\mathsf{s}}$ from (4.4). Altogether, for $0<\gamma\le 1$, there is $C_{\mathsf{K},\mu,v,n}<\infty$, independent of u, N, γ, and σ, with

$$\max\{\|\tilde{q}_{N,u}\|^2_{L^2(\mu^{\otimes 2})},\ \|\tilde{q}_{N,u}\|^2_{L^\infty L^2(\mu)},\ \|\mathcal{T}_{\tilde{q}_{N,u}}\|_{\mathrm{op}}\}\le\gamma\,C_{\mathsf{K},\mu,v,n}, \tag{7.26}$$

$$\|\tilde{q}_{N,u}\|_{L^\infty}\le\gamma\,C_{\mathsf{K},\mu,v,n}\,\eta(u,N)^{-\mathsf{s}}, \tag{7.27}$$

in the norms of Remark 3.2; hence (M3) holds as soon as $\gamma C_{\mathsf{K},\mu,v,n}\le 1/L$.

Growth bound (7.20)**.** Assume $0<\gamma\le 1$ and let $u\ge 1$. If $\eta(u,N)=\eta_0$, then, by (7.27) and the definition of $\mathsf{m}_{N,u}$,

$$\min\{u,\mathsf{m}_{N,u}\}(1+\|\tilde{q}_{N,u}\|_{L^\infty})\le\frac{\gamma}{2}\,C_{\mathsf{K},\mu,v,n}\eta_0^{-\mathsf{s}}(1+C_{\mathsf{K},\mu,v,n}\eta_0^{-\mathsf{s}})N. \tag{7.28}$$

If instead the second entry of the minimum in (7.16) is selected, then necessarily $u\le 4C_{\mathrm{abs},n}\eta_0^{\mathsf{d}-\mathsf{s}}\,\gamma N$, and

$$\min\{u,\mathsf{m}_{N,u}\}(1+\|\tilde{q}_{N,u}\|_{L^\infty})\le u+\gamma C_{\mathsf{K},\mu,v,n}\left(\frac{4C_{\mathrm{abs},n}\gamma N}{u}\right)^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}}u$$

$$= u + C_{\mathsf{K},\mu,v,n}(4C_{\mathrm{abs},n})^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}}\gamma^{\frac{\mathsf{d}}{\mathsf{d}-\mathsf{s}}}u^{\frac{\mathsf{d}-2\mathsf{s}}{\mathsf{d}-\mathsf{s}}}N^{\frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}}}, \tag{7.29}$$

which is at most $M_{\mu,v,n}\gamma N$ by $u \le C\gamma N$, $\frac{\mathsf{d}-2\mathsf{s}}{\mathsf{d}-\mathsf{s}} + \frac{\mathsf{s}}{\mathsf{d}-\mathsf{s}} = 1$, $\frac{\mathsf{d}}{\mathsf{d}-\mathsf{s}} + \frac{\mathsf{d}-2\mathsf{s}}{\mathsf{d}-\mathsf{s}} = 2$, and $\gamma \le 1$. Enlarging $M_{\mu,v,n}$ to cover the case $\eta(u,N) = \eta_0$, this proves the first bound in (7.20), and it remains to set

$$\beta_{\mathrm{loc}} \coloneqq \min\Big\{1,\ \frac{1}{4C'_{\mu,v,n}},\ \frac{1}{L\,C_{\mathsf{K},\mu,v,n}},\ \frac{1}{2L^2M_{\mu,v,n}}\Big\}, \tag{7.30}$$

so that, for $0 < \gamma \le \beta_{\mathrm{loc}}$, the smallness assumptions used in (M1) and (M3) hold and $2L^2M_{\mu,v,n}\gamma \le 1$. Step 2 is proved.

Step 3: Conclusion, positive Riesz case. By Steps 1 and 2, Proposition 3.5 applies to Ξ_N^σ with $U_0 = 1$ and $N_0 = 1$, so (3.15) gives

$$\sup_{N\ge1}\ \mathbb{E}_{\mu^{\otimes N}}\big[e^{\Xi_N^\sigma}\big] \le e^{U_0} + L = e + L, \tag{7.31}$$

which is (7.14). This proves (7.11) in the positive Riesz case.

Step 4: The logarithmic case $\mathsf{s} = 0$. After further decreasing β_{loc}, the bound (7.11) holds in the logarithmic case as well. Decrease the free cutoff η_0 by the fixed factor c_n, if necessary, so that the dilated scales $c_n\eta$ remain in the original scale range of Lemma 4.3(ii) and below $T \ge 2$.

We shall use the following elementary consequence of the logarithmic truncation. For each fixed $n \ge 1$,

$$|\mathsf{k}_{n,\mathsf{g}_\eta}(x,y)| \le C_n\|\nabla v\|_{L^\infty}^n, \qquad |\mathsf{k}_{n,\mathsf{f}_\eta}(x,y)| \le C_n\|\nabla v\|_{L^\infty}^n \mathsf{f}_{c_n\eta}(x-y); \tag{7.32}$$

the first inequality follows from the homogeneity of $\nabla^{\otimes n}\log|x|$ together with the small-scale truncation, and the second is (4.14).

By (2.20), $\mathsf{F}_{N,T} \to \mathsf{F}_N$ $\mu^{\otimes N}$-a.e. as $T \to \infty$, so, by Step 1 and Fatou's lemma, it suffices to prove the bound (7.14), uniformly in $T \ge 2$, for

$$\Xi_{N,T}^\sigma \coloneqq \varepsilon_{\gamma,n}C_{\mu,v,n} + \frac{\sigma\varepsilon_{\gamma,n}}{N}\sum_{1\le i\ne j\le N}\tilde{\mathsf{k}}_{n,\mathsf{g}}(x_i,x_j) - \gamma N\mathsf{F}_{N,T}(X_N,\mu), \tag{7.33}$$

which, by (5.23), is of the master form (3.10) with $g = \gamma\mathsf{g}_T - 2\sigma\varepsilon_{\gamma,n}\mathsf{k}_{n,\mathsf{g}}$ and $w = \gamma(h_{\mu,T} - I_{\mu,T}) + \varepsilon_{\gamma,n}C_{\mu,v,n}$.

Fix $K = 1$, set $C_I(\mu) \coloneqq \sup_{T\ge2}|I_{\mu,T}| < \infty$, and enlarge the logarithmic constant $C'_{\mu,v,n}$ to include $C_I(\mu)$. Set $C'_{\mathrm{abs},n} \coloneqq \frac{C_n\|\mu\|_{L^\infty}}{4C_{\mathrm{sr},n}} + c_n^{\mathsf{d}}\,C'_0$ with C'_0 the near-field constant in (4.13), and repeat Steps 2 and 3 at each fixed $T \ge 2$ with the data

$$\tilde h_{n,\eta,T}^\sigma \coloneqq \sigma\varepsilon_{\gamma,n}\tilde{\mathsf{k}}_{n,\mathsf{g}_\eta} - \frac{\gamma}{2}\tilde{\mathsf{g}}_{c_n\eta,T,>K}, \qquad \eta(u,N) \coloneqq \min\left\{\eta_0, \left(\frac{u}{4C'_{\mathrm{abs},n}\gamma N}\right)^{1/\mathsf{d}}\right\}, \tag{7.34}$$

$q_{N,u} \coloneqq -4\,\tilde h_{n,\eta(u,N),T}^\sigma$, and $\mathsf{m}_{N,u} \coloneqq \frac{N}{2}\|\tilde q_{N,u}\|_{L^\infty}$, so that (M2) again holds by definition. The remaining verifications change as follows.

Verification of (M1). Decompose $\mathsf{g} = \mathsf{g}_\eta + \mathsf{f}_\eta$ in the commutator term and $\mathsf{g}_T = \mathsf{g}_{c_n\eta,T,\le K} + \mathsf{g}_{c_n\eta,T,>K} + \mathsf{f}_{c_n\eta}$ in the energy term; combining the one-body term in $-\gamma N\mathsf{F}_{N,T}$ with the low-frequency energy by Lemma 4.3(ii), and using (7.32), (4.13), and the choice (7.12), gives the analogue of (7.24),

$$\Xi_{N,T}^\sigma \le C'_{\mu,v,n}\gamma + \frac{1}{N}\sum_{1\le i\ne j\le N}\tilde h_{n,\eta,T}^\sigma(x_i,x_j) + C'_{\mathrm{abs},n}\gamma N\eta^{\mathsf{d}}, \tag{7.35}$$

with η^{d} in place of $\eta^{\mathsf{d}-\mathsf{s}}$, uniformly in $T \ge 2$; (M1) then follows exactly as in (7.25).

Verification of (M3) and the L^∞ norm. (7.32) and (4.32)–(4.33) give, uniformly in $T \geq 2$, the analogue of (7.26)–(7.27) with $\eta(u,N)^{-\mathsf{s}}$ replaced by $1 + \log(1/\eta(u,N))$, for a constant $C'_{\mathsf{K},\mu,v,n}$; hence (M3) holds after further decreasing β_{loc}.

Verification of the growth bound (7.20)**.** The case $\eta(u,N) = \eta_0$ is as before. If the second entry of the minimum in (7.34) is selected, then $r \coloneqq u/(4C'_{\mathrm{abs},n}\gamma N) \leq \eta_0^{\mathsf{d}} \leq 1$ and $1 + \log(1/\eta(u,N)) = 1 + \frac{1}{\mathsf{d}}\log(1/r)$, so

$$\min\{u, \mathsf{m}_{N,u}\}(1 + \|\tilde{q}_{N,u}\|_{L^\infty}) \leq u + \gamma C'_{\mathsf{K},\mu,v,n}\Big(1 + \tfrac{1}{\mathsf{d}}\log(1/r)\Big)\, u \leq M_{\mu,v,n}\gamma N, \tag{7.36}$$

after enlarging $M_{\mu,v,n}$, because $u = 4C'_{\mathrm{abs},n}\gamma N\, r$ and $r \mapsto r\big(1 + \log(1/r)\big)$ is bounded on $(0,1]$.

Hence, for $0 < \gamma \leq \beta_{\mathrm{loc}}$, Proposition 3.5, applied with $U_0 = 1$ and $N_0 = 1$, gives, uniformly in $T \geq 2$, $\sup_{N\geq 1} \mathbb{E}_{\mu^{\otimes N}}[e^{\Xi^\sigma_{N,T}}] \leq e + L$, and Fatou's lemma concludes. □

7.4. **Removal of the smallness assumption.** The preceding lemma proves the required numerator estimate only for inverse temperatures at most β_{loc}. We now complete the proof of Theorem 2.8 by retaining one factor at this local inverse temperature and controlling the remaining Gibbs factor by Hölder's inequality and the partition-function theorem.

Proof of Theorem 2.8. In the local range $0 < \beta \leq \beta_{\mathrm{loc}}$, applying Lemma 7.2 with $\gamma = \beta$ gives (7.3). Dividing by $\mathsf{K}_{N,\beta}(\mu)$ in (7.2) and using (2.12) gives (2.42); in the positive Riesz case, one may take $C_{\mathrm{em}}(\beta,\mu,v,n) = e + L$, while in the logarithmic case, one may take $C_{\mathrm{em}}(\beta,\mu,v,n) = e^{-\beta I_\mu}(e + L)$.

Assume next that $\beta > \beta_{\mathrm{loc}}$. Then

$$\varepsilon_{\beta,n} = \frac{\beta_{\mathrm{loc}}}{8C_{\mathrm{sr},n}\|\nabla v\|^n_{L^\infty}}. \tag{7.37}$$

For the positive Riesz energy, and for the finite-T logarithmic energy before passage to the limit, we have the exact decomposition

$$\begin{aligned}
&\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu) - \beta N\mathsf{F}_N(X_N,\mu)\\
&\quad= \frac{1}{2}\left[\sigma\frac{\beta_{\mathrm{loc}}}{4C_{\mathrm{sr},n}\|\nabla v\|^n_{L^\infty}}N\mathsf{A}_n(v,X_N,\mu) - \beta_{\mathrm{loc}}N\mathsf{F}_N(X_N,\mu)\right] - \left(\beta - \frac{\beta_{\mathrm{loc}}}{2}\right)N\mathsf{F}_N(X_N,\mu).
\end{aligned} \tag{7.38}$$

Hölder's inequality with exponents $2, 2$ gives

$$\begin{aligned}
&\mathbb{E}_{\mu^{\otimes N}}\Big[e^{\sigma\varepsilon_{\beta,n}N\mathsf{A}_n(v,X_N,\mu) - \beta N\mathsf{F}_N(X_N,\mu)}\Big]\\
&\quad\leq \left(\mathbb{E}_{\mu^{\otimes N}}\Big[e^{\sigma\frac{\beta_{\mathrm{loc}}}{4C_{\mathrm{sr},n}\|\nabla v\|^n_{L^\infty}}N\mathsf{A}_n(v,X_N,\mu) - \beta_{\mathrm{loc}}N\mathsf{F}_N(X_N,\mu)}\Big]\right)^{1/2}\mathsf{K}_{N,2\beta-\beta_{\mathrm{loc}}}(\mu)^{1/2}\\
&\quad\leq (e+L)^{1/2}C_{\mathrm{ub}}(2\beta - \beta_{\mathrm{loc}},\mu)^{1/2}.
\end{aligned} \tag{7.39}$$

In the logarithmic case, the same estimate is first applied with $\mathsf{F}_{N,T}$ in place of F_N, using the finite-T local numerator estimate and the uniform-in-T partition-function bound (5.24) from Section 5.1. Fatou's lemma then gives (7.39) for F_N.

Dividing (7.39) by $\mathsf{K}_{N,\beta}(\mu)$ and using the Jensen lower bound (2.12) proves (2.42) for $\beta > \beta_{\mathrm{loc}}$ with the constant displayed in (A.9). This proves the exponential-moment assertion for all $\beta > 0$.

Finally, inserting (2.42) and (2.33) into (7.1) and using the entropy rewriting (2.11) with $\varepsilon = \varepsilon_{\beta,n}$, gives, for every $\sigma \in \{\pm 1\}$,

$$\begin{aligned}
\sigma\mathbb{E}_{X_N\sim f_N}\Big[N\mathsf{A}_n(v,X_N,\mu)\Big] &\leq \frac{\beta}{\varepsilon_{\beta,n}}N\mathsf{E}_N(f_N,\mu) + \frac{1}{\varepsilon_{\beta,n}}(\log\mathsf{K}_{N,\beta}(\mu) + \log C_{\mathrm{em}}(\beta,\mu,v,n))\\
&\leq \frac{\beta}{\varepsilon_{\beta,n}}N\mathsf{E}_N(f_N,\mu) + \frac{1}{\varepsilon_{\beta,n}}(\log C_{\mathrm{pf}}(\beta,\mu) + \log C_{\mathrm{em}}(\beta,\mu,v,n)).
\end{aligned} \tag{7.40}$$

Taking the maximum over $\sigma \in \{\pm 1\}$ yields (2.43). The constants are exactly those displayed in (2.44), which completes the proof. □

7.5. **Thermal interpolation corollary.** The following corollary records the better of the fixed-temperature entropic estimate and the deterministic pointwise commutator estimate. The latter becomes useful when the inverse temperature grows with N.

Corollary 7.3. *Assume the hypotheses of Theorem 2.8 and moreover that the sharp pointwise commutator bound holds with the scale $a_{N,\mathsf{s}}$ from* (5.91)*: there are constants $C_{\mathrm{pw}}(\mu,v,n), C_{\mathrm{pt}}(\mu) < \infty$ such that, for every $N \geq 1$ and every configuration X_N,*

$$|\mathsf{A}_n(v, X_N, \mu)| \leq C_{\mathrm{pw}}(\mu,v,n)\Big(\mathsf{F}_N(X_N,\mu) + C_{\mathrm{pt}}(\mu)\frac{a_{N,\mathsf{s}}}{N}\Big). \tag{7.41}$$

Then, for every $f_N \in \mathcal{P}((\mathbb{R}^{\mathsf{d}})^N)$ with finite relative entropy with respect to $\mathbb{Q}_{N,\beta}(\mu)$,

$$\begin{aligned}\Big|\mathbb{E}_{X_N\sim f_N}\Big[N\mathsf{A}_n(v,X_N,\mu)\Big]\Big| \leq \min\Big\{&C_{\beta,n,v}N\mathsf{E}_N(f_N,\mu) + C_{\beta,\mu,v,n},\\ &C_{\mathrm{pw}}(\mu,v,n)N\mathsf{E}_N(f_N,\mu) + C'_{\mathrm{pw}}(\mu,v,n)a_{N,\mathsf{s}}\Big\},\end{aligned} \tag{7.42}$$

where one may take $C'_{\mathrm{pw}}(\mu,v,n) = C_{\mathrm{pw}}(\mu,v,n)C_{\mathrm{pt}}(\mu)$.

Proof of Corollary 7.3. The entropic estimate is precisely Theorem 2.8. For the pointwise estimate, (7.41) gives

$$\begin{aligned}\Big|\mathbb{E}_{X_N\sim f_N}\Big[N\mathsf{A}_n(v,X_N,\mu)\Big]\Big| &\leq \mathbb{E}_{X_N\sim f_N}\Big[N|\mathsf{A}_n(v,X_N,\mu)|\Big]\\ &\leq C_{\mathrm{pw}}(\mu,v,n)N\mathbb{E}_{X_N\sim f_N}\Big[\mathsf{F}_N(X_N,\mu)\Big] + C'_{\mathrm{pw}}(\mu,v,n)a_{N,\mathsf{s}}\\ &\leq C_{\mathrm{pw}}(\mu,v,n)N\mathsf{E}_N(f_N,\mu) + C'_{\mathrm{pw}}(\mu,v,n)a_{N,\mathsf{s}},\end{aligned} \tag{7.43}$$

using the nonnegativity of the relative entropy in the definition (2.10) of E_N. Combining the two estimates proves (7.42). □

7.6. **Diffusive mean-field convergence.** We have all the ingredients to prove Theorem 2.9. The dissipation identity for the modulated free energy is standard in the modulated-free-energy method; in the notation used here, it is the dynamical counterpart of the entropy rewriting (2.11) and may be read, for instance, from [RS25]. After discarding the nonpositive relative Fisher-information term, this identity gives the abstract differential inequality (2.49).

Applying Theorem 2.8 with $n = 1$, $\mu = \mu^t$, and $v = u^t$ gives, for a.e. t,

$$\Big|\mathbb{E}_{X_N\sim f_N^t}\Big[N\mathsf{A}_1(u^t,X_N,\mu^t)\Big]\Big| \leq C_{\beta,1,u^t}N\mathsf{E}_N^t + C_{\beta,\mu^t,u^t,1}, \tag{7.44}$$

where the constants are those in Theorem 2.8. Combining (2.49) and (7.44), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{E}_N^t \leq C_{\beta,1,u^t}\mathsf{E}_N^t + \frac{C_{\beta,\mu^t,u^t,1}}{N} \tag{7.45}$$

for a.e. t. Multiplying by the integrating factor $\exp\Big(-\int_0^t C_{\beta,1,u^\tau}(\mu^\tau)\,\mathrm{d}\tau\Big)$ and integrating from 0 to t gives

$$\mathsf{E}_N^t \leq \exp\Big(\int_0^t C_{\beta,1,u^\tau}(\mu^\tau)\,\mathrm{d}\tau\Big)\mathsf{E}_N^0 + \frac{1}{N}\int_0^t \exp\Big(\int_\tau^t C_{\beta,1,u^r}(\mu^r)\,\mathrm{d}r\Big)C_{\beta,\mu^\tau,u^\tau,1}\,\mathrm{d}\tau, \tag{7.46}$$

which is exactly (2.52) in view of (2.50). The uniform-in-time claim in Remark 2.10 follows from the same formula when $\sup_{t\geq 0}\mathcal{A}^t < \infty$ and $\sup_{t\geq 0}\mathcal{B}^t < \infty$.

8. Static and dynamical fluctuations

This section proves Proposition 2.11 and Theorem 2.13. Section 8.1 proves the bounded-test moment-generating-function formula in Proposition 2.11 by inserting a linear tilt into the modulated partition function and combining the joint linear/quadratic fluctuation limit with the determinant normalization; a uniform $L^2(\mu^{\otimes N})$ density bound and truncation then extend the weak central limit theorem to arbitrary $L^2(\mu)$ tests. Section 8.2 proves Lemma 8.1, which supplies the uniform entropy, commutator, and bounded-linear-statistic estimates needed for the dynamical argument. Finally, Section 8.3 first proves the finite-N adjoint-duality identity Proposition 8.2 and then uses it to prove the dynamical central limit theorem Theorem 2.13.

8.1. **Static linear statistics.**

Proof of Proposition 2.11. **Bounded tests and moment generating functions.** Suppose first that $\phi_1, \dots, \phi_k \in L^\infty(\mu)$. Fix $\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k$ and set

$$\phi_\theta \coloneqq \sum_{i=1}^{k} \theta_i \bar{\phi}_i. \tag{8.1}$$

For $X_N = (x_1, \dots, x_N) \sim \mu^{\otimes N}$, set

$$U_N \coloneqq \frac{1}{N} \sum_{1 \le i \ne j \le N} \tilde{\mathsf{g}}(x_i, x_j). \tag{8.2}$$

Let Z be the centered isonormal Gaussian process over $L^2_0(\mu)$. If $(\lambda_\ell, e_\ell)_{\ell \ge 1}$ are the nonzero spectral data of T_μ, set $G_\ell \coloneqq \mathsf{Z}(e_\ell)$ and

$$U_\infty \coloneqq \sum_{\ell \ge 1} \lambda_\ell (G_\ell^2 - 1), \tag{8.3}$$

where the series converges in L^2. The general degenerate U-statistic limit theorem [Ser80, Section 5.5.2], applied jointly with the linear statistic, gives[4]

$$(S_N(\phi_\theta), U_N) \Longrightarrow (\mathsf{Z}(\phi_\theta), U_\infty). \tag{8.4}$$

From (2.4),

$$-\beta N \mathsf{F}_N(X_N, \mu) = -\frac{\beta}{2} U_N + \beta \int_{\mathbb{R}^{\mathsf{d}}} h_\mu \, \mathrm{d}\mu_N - \beta I_\mu. \tag{8.5}$$

In the logarithmic case, (2.20) implies $h_\mu \in L^1(\mu)$; in the positive Riesz case, $h_\mu \in L^\infty$. Hence the weak law of large numbers gives $\int_{\mathbb{R}^{\mathsf{d}}} h_\mu \, \mathrm{d}\mu_N \to 2I_\mu$ in probability. Combining this with (8.4),

$$S_N(\phi_\theta) - \beta N \mathsf{F}_N(X_N, \mu) \Longrightarrow \mathsf{Z}(\phi_\theta) - \frac{\beta}{2} U_\infty + \beta I_\mu. \tag{8.6}$$

To pass to exponential moments, choose $r > 1$ and conjugate exponents $p, q > 1$. Hölder's inequality gives

$$\mathbb{E}_{\mu^{\otimes N}}\Big[e^{rS_N(\phi_\theta) - r\beta N \mathsf{F}_N(X_N, \mu)}\Big] \le \mathbb{E}_{\mu^{\otimes N}}\Big[e^{prS_N(\phi_\theta)}\Big]^{1/p} \mathsf{K}_{N, qr\beta}(\mu)^{1/q}. \tag{8.7}$$

The first factor is bounded uniformly in N because ϕ_θ is bounded and centered, and the second is bounded by Theorem 2.2(i). Thus, the exponentials are uniformly integrable, and

$$\mathbb{E}_{\mu^{\otimes N}}\Big[e^{S_N(\phi_\theta) - \beta N \mathsf{F}_N(X_N, \mu)}\Big] \longrightarrow e^{\beta I_\mu} \mathbb{E}\Big[e^{\mathsf{Z}(\phi_\theta) - \frac{\beta}{2} U_\infty}\Big]. \tag{8.8}$$

[4]The cited result is stated for the quadratic degenerate U-statistic. The joint formulation follows from the same orthogonal-expansion argument after adjoining the linear statistic.

For a centered $f \in L^2(\mu)$, let f_0 be its orthogonal projection onto $\ker T_\mu$ and set $f_\ell := \langle f, e_\ell\rangle_{L^2(\mu)}$. Writing $\mathsf{Z}(f)$ and U_∞ in their orthogonal-series representations and using the independence of the Gaussian coordinates,

$$\begin{aligned}
&\mathbb{E}\Big[\exp\Big(\mathsf{Z}(f) - \frac{\beta}{2}U_\infty\Big)\Big] \\
&\quad= \exp\Big(\frac{1}{2}\|f_0\|^2_{L^2(\mu)}\Big)\prod_{\ell\ge 1} e^{\beta\lambda_\ell/2}\mathbb{E}\Big[\exp\Big(f_\ell G_\ell - \frac{\beta\lambda_\ell}{2}G_\ell^2\Big)\Big] \\
&\quad= \exp\Big(\frac{1}{2}\|f_0\|^2_{L^2(\mu)}\Big)\prod_{\ell\ge 1} e^{\beta\lambda_\ell/2}(1+\beta\lambda_\ell)^{-1/2}\exp\Big(\frac{f_\ell^2}{2(1+\beta\lambda_\ell)}\Big) \\
&\quad= \det{}_2(\mathrm{Id}+\beta T_\mu)^{-1/2}\exp\Big(\frac{1}{2}\langle f, (\mathrm{Id}+\beta T_\mu)^{-1} f\rangle_{L^2(\mu)}\Big).
\end{aligned} \tag{8.9}$$

The product converges because $(\lambda_\ell)_\ell \in \ell^2$, $(f_\ell)_\ell \in \ell^2$, and

$$\frac{\beta\lambda}{2} - \frac{1}{2}\log(1+\beta\lambda) = O(\lambda^2) \tag{8.10}$$

as $\lambda \to 0$. Dividing by the partition-function limit in Theorem 2.2(ii) and taking $f = \phi_\theta$ proves (2.56). Since the limiting moment generating function is finite in a neighborhood of the origin, the continuity theorem for moment generating functions gives the asserted Gaussian convergence for bounded tests (see, e.g., [Bil95, Section 30]).

Extension to $L^2(\mu)$. Define the density

$$L_N(X_N) := \frac{\mathrm{d}\mathbb{Q}_{N,\beta}(\mu)}{\mathrm{d}\mu^{\otimes N}}(X_N) = \frac{\exp\{-\beta N \mathsf{F}_N(X_N,\mu)\}}{\mathsf{K}_{N,\beta}(\mu)}. \tag{8.11}$$

By the definition of the partition function,

$$\mathbb{E}_{\mu^{\otimes N}}\Big[L_N^2\Big] = \frac{\mathsf{K}_{N,2\beta}(\mu)}{\mathsf{K}_{N,\beta}(\mu)^2}. \tag{8.12}$$

At $\beta = 0$, one has $L_N \equiv 1$. For $\beta > 0$, Theorem 2.2(i), applied at 2β, and the Jensen lower bound (2.12) give the required bound. Thus, in either case, there is a constant $C_{\beta,\mu} < \infty$ such that

$$\sup_{N\ge 1}\|L_N\|_{L^2(\mu^{\otimes N})} \le C_{\beta,\mu}. \tag{8.13}$$

In the logarithmic case, only the uniform-bound part of Theorem 2.2 is used at 2β, so no doubled version of (2.22) is required.

Fix $\theta = (\theta_1,\dots,\theta_k) \in \mathbb{R}^k$ and set

$$f := \sum_{i=1}^k \theta_i\phi_i, \qquad f^{(M)} := (-M)\vee(f\wedge M). \tag{8.14}$$

Since S_N centers its argument,

$$S_N(f) - S_N(f^{(M)}) = S_N(f - f^{(M)}). \tag{8.15}$$

Cauchy–Schwarz, the preceding density bound, and the exact product-law variance identity give

$$\begin{aligned}
&\sup_{N\ge1}\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[|S_N(f) - S_N(f^{(M)})|\Big] \\
&\quad\le C_{\beta,\mu}\sup_{N\ge1}\mathbb{E}_{\mu^{\otimes N}}\Big[|S_N(f-f^{(M)})|^2\Big]^{1/2} \\
&\quad= C_{\beta,\mu}\,\mathrm{Var}_\mu(f-f^{(M)})^{1/2} \le C_{\beta,\mu}\|f-f^{(M)}\|_{L^2(\mu)} \longrightarrow 0
\end{aligned} \tag{8.16}$$

as $M\to\infty$.

For each fixed M, the bounded-test Gaussian convergence just proved gives

$$\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\mathrm{i}S_N(f^{(M)})}\Big] \longrightarrow \exp\Big(-\frac{1}{2}\Sigma_{\beta,\mu}(f^{(M)},f^{(M)})\Big). \tag{8.17}$$

Moreover, using $|e^{\mathrm{i}u}-e^{\mathrm{i}v}|\le|u-v|$ for $u,v\in\mathbb{R}$,

$$\Big|\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\mathrm{i}S_N(f)}\Big]-\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[e^{\mathrm{i}S_N(f^{(M)})}\Big]\Big| \le \mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu)}\Big[|S_N(f)-S_N(f^{(M)})|\Big]. \tag{8.18}$$

Since $f^{(M)}\to f$ in $L^2(\mu)$, the continuity of $\Sigma_{\beta,\mu}$ yields

$$\Sigma_{\beta,\mu}(f^{(M)},f^{(M)}) \longrightarrow \Sigma_{\beta,\mu}(f,f). \tag{8.19}$$

First letting $N\to\infty$ and then $M\to\infty$ proves (2.55) for the chosen θ. Since $\theta\in\mathbb{R}^k$ was arbitrary, Lévy's continuity theorem gives the joint Gaussian convergence (see, e.g., [Dur19, Theorem 3.3.17]). □

8.2. **Entropy transfers.** Before deriving the finite-N adjoint-duality identity, we collect three uniform estimates used to control the dynamical error terms: a relative-entropy bound, an L^1 estimate for the transport commutator, and an L^1 estimate for bounded linear statistics of the empirical measure.

Lemma 8.1 (Finite-time entropy transfers)**.** *Assume the hypotheses of Theorem 2.13. Then:*

(i)

$$\sup_{N\ge1}\sup_{0\le t\le T}\mathsf{H}\Big(f_N^t|\mathbb{Q}_{N,\beta}(\mu^t)\Big)<\infty. \tag{8.20}$$

(ii) For each terminal pair (t,ϕ), with $f^r=\mathsf{U}^{r,t}\phi$,

$$\sup_{0\le r\le t}\mathbb{E}_{X_N\sim f_N^r}[|\mathsf{A}_1(\nabla f^r,X_N,\mu^r)|]\le\frac{C_{t,\phi,T}}{N}. \tag{8.21}$$

(iii) If $(q^r)_{0\le r\le T}$ is a deterministic family of bounded measurable functions with $\sup_r\operatorname{osc}(q^r)<\infty$, then

$$\sup_{0\le r\le T}\mathbb{E}_{X_N\sim f_N^r}\Big[\Big|\int_{\mathbb{R}^{\mathsf{d}}}q^r\ \mathrm{d}(\mu_N^r-\mu^r)\Big|\Big]\le\frac{C_{q,T}}{\sqrt{N}}. \tag{8.22}$$

Proof. Since $f_N^0=\mathbb{Q}_{N,\beta}(\mu^0)$, (2.11) gives

$$\mathsf{E}_N(f_N^0,\mu^0)=-\frac{1}{\beta N}\log\mathsf{K}_{N,\beta}(\mu^0). \tag{8.23}$$

Combining (2.52) with (2.11),

$$\mathsf{H}\Big(f_N^t|\mathbb{Q}_{N,\beta}(\mu^t)\Big)\le-\mathcal{A}^t\log\mathsf{K}_{N,\beta}(\mu^0)+\beta\mathcal{B}^t+\log\mathsf{K}_{N,\beta}(\mu^t), \tag{8.24}$$

which is uniformly bounded by hypotheses (i) and (ii) of Theorem 2.13. This proves (i).

For (ii), fix a terminal pair (t,ϕ) and set $f^r=\mathsf{U}^{r,t}\phi$. By hypotheses (i) and (iii),

$$M_T\coloneqq\sup_{0\le r\le T}\|\mu^r\|_{L^\infty}<\infty,\qquad \Lambda_{t,\phi}\coloneqq\sup_{0\le r\le t}\|\nabla^{\otimes2}f^r\|_{L^\infty}<\infty, \tag{8.25}$$

$$\text{when } \mathsf{s}=0,\qquad J_T\coloneqq\max\Big\{0,-\inf_{0\le r\le T}I_{\mu^r}\Big\}<\infty. \tag{8.26}$$

If $\Lambda_{t,\phi}=0$, then $\mathsf{A}_1(\nabla f^r,X_N,\mu^r)=0$ for every r. Otherwise, the locally bounded dependence recorded in Remark A.1, together with hypothesis (ii), shows that Theorem 2.8 supplies constants $\varepsilon_{t,\phi}>0$ and $C_{t,\phi}<\infty$, independent of r and N, for both signs in

$$\sup_{0\le r\le t}\sup_{N\ge1}\sup_{\sigma\in\{-1,1\}}\mathbb{E}_{X_N\sim\mathbb{Q}_{N,\beta}(\mu^r)}[\exp(\sigma\varepsilon_{t,\phi}N\mathsf{A}_1(\nabla f^r,X_N,\mu^r))]\le C_{t,\phi}. \tag{8.27}$$

Consequently, $e^{|z|} \le e^z + e^{-z}$ gives a uniform exponential moment of $N|\mathsf{A}_1(\nabla f^r, X_N, \mu^r)|$ under the time-r modulated Gibbs measure $\mathbb{Q}_{N,\beta}(\mu^r)$. The Donsker–Varadhan lemma, together with (8.20), now yields (8.21).

For (iii), put

$$S_N^r(q^r) \coloneqq \sqrt{N} \int_{\mathbb{R}^{\mathsf{d}}} q^r \, \mathrm{d}(\mu_N^r - \mu^r). \tag{8.28}$$

For every fixed $\lambda > 0$, Cauchy–Schwarz and the definition of $\mathbb{Q}_{N,\beta}(\mu^r)$ give

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}(\mu^r)}\Big[e^{\lambda|S_N^r(q^r)|}\Big] \le \mathsf{K}_{N,\beta}(\mu^r)^{-1}\mathsf{K}_{N,2\beta}(\mu^r)^{1/2}\mathbb{E}_{(\mu^r)^{\otimes N}}\Big[e^{2\lambda|S_N^r(q^r)|}\Big]^{1/2}. \tag{8.29}$$

By Hoeffding's lemma (see, e.g., [BLM13, Lemma 2.2]) and $e^{|z|} \le e^z + e^{-z}$,

$$\mathbb{E}_{(\mu^r)^{\otimes N}}\Big[e^{2\lambda|S_N^r(q^r)|}\Big] \le 2\exp\left(\frac{\lambda^2}{2}\operatorname{osc}(q^r)^2\right). \tag{8.30}$$

The uniform oscillation bound and the partition-function bounds therefore make the preceding exponential moment uniform in N and r. A second application of the Donsker–Varadhan lemma gives

$$\sup_{0\le r\le T} \mathbb{E}_{X_N\sim f_N^r}[|S_N^r(q^r)|] \le C_{q,T}, \tag{8.31}$$

which is (8.22) after division by $\sqrt{N}$. □

8.3. **Adjoint duality and characteristic functions.** The next proposition gives an exact finite-N decomposition of a terminal fluctuation observable into its backward-transported initial value, a continuous martingale, and the transport-commutator remainder controlled by Lemma 8.1.

Proposition 8.2 (Finite-N adjoint-duality identity)**.** *For $t \in [0,T]$ and $\phi \in \mathscr{D}_T$, let $f^r = \mathsf{U}^{r,t}\phi$ denote the classical solution of the backward adjoint problem* (2.64)*. Then*

$$S_N^t(\phi) = S_N^0(\mathsf{U}^{0,t}\phi) + \sqrt{N}\, M_N^t[f] - \frac{\sqrt{N}}{2}\int_0^t \mathsf{A}_1(\nabla f^r, X_N^r, \mu^r)\, \mathrm{d}r, \tag{8.32}$$

where

$$M_N^\tau[f] \coloneqq \frac{1}{N}\sqrt{\frac{2}{\beta}}\sum_{i=1}^N \int_0^\tau \nabla f^r(x_i^r)\cdot \mathrm{d}B_i^r, \qquad 0 \le \tau \le t. \tag{8.33}$$

By the boundedness of ∇f, $(M_N^\tau[f])_{0\le\tau\le t}$ is a continuous square-integrable martingale. The last term in (8.32) *converges to zero in L^1 at rate $O(N^{-1/2})$.*

Proof. For a fixed C_b^2 test φ, Itô's formula in (2.45) gives

$$\begin{aligned}\mathrm{d}\int_{\mathbb{R}^{\mathsf{d}}} \varphi \, \mathrm{d}\mu_N^r = &-\frac{1}{N^2}\sum_{i\ne j}\nabla\varphi(x_i^r)\cdot\nabla\mathsf{g}(x_i^r - x_j^r)\, \mathrm{d}r + \frac{1}{\beta}\int_{\mathbb{R}^{\mathsf{d}}}\Delta\varphi\, \mathrm{d}\mu_N^r\, \mathrm{d}r \\ &+ \sqrt{\frac{2}{\beta}}\frac{1}{N}\sum_{i=1}^N \nabla\varphi(x_i^r)\cdot \mathrm{d}B_i^r.\end{aligned} \tag{8.34}$$

Since g is even and $\nabla\mathsf{g}$ is odd,

$$\frac{1}{N^2}\sum_{i\ne j}\nabla\varphi(x_i)\cdot\nabla\mathsf{g}(x_i - x_j) = \frac{1}{2}\int_{(\mathbb{R}^{\mathsf{d}})^2\setminus\triangle}\nabla\mathsf{g}(x-y)\cdot(\nabla\varphi(x) - \nabla\varphi(y))\, \mathrm{d}\mu_N^{\otimes 2}(x,y). \tag{8.35}$$

Testing the mean-field equation (2.46) against φ, integrating by parts, and symmetrizing the interaction term give

$$\frac{\mathrm{d}}{\mathrm{d}r}\int_{\mathbb{R}^{\mathsf{d}}}\varphi\, \mathrm{d}\mu^r = -\frac{1}{2}\int_{(\mathbb{R}^{\mathsf{d}})^2}\nabla\mathsf{g}(x-y)\cdot(\nabla\varphi(x) - \nabla\varphi(y))\, \mathrm{d}\mu^r(x)\, \mathrm{d}\mu^r(y) + \frac{1}{\beta}\int_{\mathbb{R}^{\mathsf{d}}}\Delta\varphi\, \mathrm{d}\mu^r. \tag{8.36}$$

Because μ^r has a bounded density, it has no atoms, and the difference-of-gradients kernel is locally integrable. Hence

$$\int_{(\mathbb{R}^{\mathsf{d}})^2\setminus\triangle} \nabla \mathsf{g}(x-y)\cdot(\nabla\varphi(x)-\nabla\varphi(y))\,\mathrm{d}\Big((\mu_N^r)^{\otimes 2}-(\mu^r)^{\otimes 2}\Big)(x,y)$$
$$= 2\int_{(\mathbb{R}^{\mathsf{d}})^2} \nabla \mathsf{g}(x-y)\cdot(\nabla\varphi(x)-\nabla\varphi(y))\,\mathrm{d}(\mu_N^r-\mu^r)(x)\,\mathrm{d}\mu^r(y)+\mathsf{A}_1(\nabla\varphi,X_N^r,\mu^r). \tag{8.37}$$

Subtracting (8.36) from (8.34), multiplying by $\sqrt{N}$, and using (2.59), we obtain

$$\mathrm{d}S_N^r(\varphi)=S_N^r(\mathcal{L}_{\beta,\mu^r}\varphi)\,\mathrm{d}r-\frac{\sqrt{N}}{2}\mathsf{A}_1(\nabla\varphi,X_N^r,\mu^r)\,\mathrm{d}r+\sqrt{\frac{2}{\beta N}}\sum_{i=1}^N\nabla\varphi(x_i^r)\cdot\mathrm{d}B_i^r. \tag{8.38}$$

For the time-dependent test f^r, the additional drift $S_N^r(\partial_r f^r)\,\mathrm{d}r$ cancels the linear term by (2.64). Integrating this identity with respect to r over $[0,t]$, using $f^t=\phi$ and the backward adjoint equation, proves (8.32). Finally, (8.21) and Fubini's theorem give

$$\mathbb{E}\left[\left|\frac{\sqrt{N}}{2}\int_0^t\mathsf{A}_1(\nabla f^r,X_N^r,\mu^r)\,\mathrm{d}r\right|\right]\le\frac{tC_{t,\phi,T}}{2\sqrt{N}}, \tag{8.39}$$

which proves the stated remainder estimate. □

Proof of Theorem 2.13. After relabeling, assume $0\le t_1\le\cdots\le t_k\le T$. Fix $\theta=(\theta_1,\dots,\theta_k)\in\mathbb{R}^k$, set $f_a^r=\mathsf{U}^{r,t_a}\phi_a$, and extend ∇f_a^r by zero for $r>t_a$. Define

$$\phi_\theta\coloneqq\sum_{a=1}^k\theta_a f_a^0, \tag{8.40}$$
$$v_\theta^r\coloneqq\sum_{a:\,r\le t_a}\theta_a\nabla f_a^r, \tag{8.41}$$
$$M_{N,\theta}^r\coloneqq\sqrt{\frac{2}{\beta N}}\sum_{i=1}^N\int_0^r v_\theta^u(x_i^u)\cdot\mathrm{d}B_i^u. \tag{8.42}$$

The process $(M_{N,\theta}^r)_{0\le r\le t_k}$ is a continuous square-integrable martingale. By Proposition 8.2,

$$\sum_{a=1}^k\theta_a S_N^{t_a}(\phi_a)=S_N^0(\phi_\theta)+M_{N,\theta}^{t_k}+R_{N,\theta},\qquad\mathbb{E}[|R_{N,\theta}|]\longrightarrow 0. \tag{8.43}$$

Its predictable quadratic variation is

$$\mathrm{d}\langle M_{N,\theta}\rangle_r=\frac{2}{\beta}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\,\mathrm{d}\mu_N^r\,\mathrm{d}r. \tag{8.44}$$

By the adjoint regularity, $(|v_\theta^r|^2)_{0\le r\le t_k}$ is a deterministic family satisfying

$$\sup_{0\le r\le t_k}\big\||v_\theta^r|^2\big\|_{L^\infty}<\infty. \tag{8.45}$$

Applying (8.22) with $q^r=|v_\theta^r|^2$ gives

$$\int_0^{t_k}\mathbb{E}\left[\left|\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\,\mathrm{d}(\mu_N^r-\mu^r)\right|\right]\mathrm{d}r=O(N^{-1/2}). \tag{8.46}$$

Set

$$\chi_N(r)\coloneqq\mathbb{E}\Big[\exp\Big(\mathrm{i}\Big(S_N^0(\phi_\theta)+M_{N,\theta}^r\Big)\Big)\Big],\qquad 0\le r\le t_k. \tag{8.47}$$

Itô's formula gives the exact identity

$$\chi_N(r)=\chi_N(0)-\frac{1}{\beta}\int_0^r\mathbb{E}\left[e^{\mathrm{i}(S_N^0(\phi_\theta)+M_{N,\theta}^u)}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^u|^2\,\mathrm{d}\mu_N^u\right]\mathrm{d}u. \tag{8.48}$$

The complex exponential has modulus one, so

$$\left|\mathbb{E}\left[e^{\mathrm{i}(S_N^0(\phi_\theta)+M_{N,\theta}^r)}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\ \mathrm{d}(\mu_N^r-\mu^r)\right]\right| \le \mathbb{E}\left[\left|\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\ \mathrm{d}(\mu_N^r-\mu^r)\right|\right]. \tag{8.49}$$

Using (8.46) and (8.49),

$$\sup_{0\le r\le t_k}\left|\chi_N(r)-\chi_N(0)+\frac{1}{\beta}\int_0^r\left(\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^u|^2\ \mathrm{d}\mu^u\right)\chi_N(u)\ \mathrm{d}u\right|\longrightarrow 0. \tag{8.50}$$

Comparison with the corresponding scalar Volterra equation and Gronwall's inequality yield

$$\chi_N(t_k)-\chi_N(0)\exp\left(-\frac{1}{\beta}\int_0^{t_k}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\ \mathrm{d}\mu^r\ \mathrm{d}r\right)\longrightarrow 0. \tag{8.51}$$

By Proposition 2.11,

$$\chi_N(0)\longrightarrow\exp\left(-\frac{1}{2}\Sigma_{\beta,\mu^0}(\phi_\theta,\phi_\theta)\right). \tag{8.52}$$

Moreover,

$$\left|\mathbb{E}\left[\exp\left(\mathrm{i}\sum_{a=1}^k\theta_a S_N^{t_a}(\phi_a)\right)\right]-\chi_N(t_k)\right|\le\mathbb{E}[|R_{N,\theta}|]\longrightarrow 0. \tag{8.53}$$

Consequently,

$$\mathbb{E}\left[\exp\left(\mathrm{i}\sum_{a=1}^k\theta_a S_N^{t_a}(\phi_a)\right)\right]\longrightarrow\exp\left(-\frac{1}{2}\Sigma_{\beta,\mu^0}(\phi_\theta,\phi_\theta)-\frac{1}{\beta}\int_0^{t_k}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\ \mathrm{d}\mu^r\ \mathrm{d}r\right). \tag{8.54}$$

The first quadratic expression in the exponent is

$$\Sigma_{\beta,\mu^0}(\phi_\theta,\phi_\theta)=\sum_{a,b=1}^k\theta_a\theta_b\Sigma_{\beta,\mu^0}\left(\mathsf{U}^{0,t_a}\phi_a,\mathsf{U}^{0,t_b}\phi_b\right), \tag{8.55}$$

while the second satisfies

$$\frac{2}{\beta}\int_0^{t_k}\int_{\mathbb{R}^{\mathsf{d}}}|v_\theta^r|^2\ \mathrm{d}\mu^r\ \mathrm{d}r=\sum_{a,b=1}^k\theta_a\theta_b\frac{2}{\beta}\int_0^{t_a\wedge t_b}\int_{\mathbb{R}^{\mathsf{d}}}\nabla\mathsf{U}^{r,t_a}\phi_a\cdot\nabla\mathsf{U}^{r,t_b}\phi_b\ \mathrm{d}\mu^r\ \mathrm{d}r. \tag{8.56}$$

Their sum is exactly the covariance in (2.67). The Cramér–Wold device and Lévy's continuity theorem now give the joint convergence (see, e.g., [Dur19, Theorems 3.10.6 and 3.3.17]). □

9. Consequences and open problems

We close by recording several directions suggested by the preceding estimates. The first is to go beyond the leading determinant normalization. In the Hilbert–Schmidt regime, the present work identifies the limiting modulated partition function as the Carleman–Fredholm determinant associated with the centered interaction operator. It remains natural to ask for sharper asymptotics, including the first correction term and whether rare near-collision configurations, with pair separations below the typical interparticle scale $N^{-1/\mathsf{d}}$, produce additional local contributions not encoded by the limiting Gaussian second chaos and its Carleman–Fredholm determinant.

A second direction is to determine the sharp ultraviolet growth beyond the Hilbert–Schmidt threshold. Proposition 5.10 gives a polylogarithmic lower bound at $\mathsf{s}=\mathsf{d}/2$ and a stretched-exponential-in-log N lower bound when $\mathsf{d}/2<\mathsf{s}<\mathsf{d}$. The larger rates predicted under the local nondegeneracy hypothesis in Remark 5.11 remain open, as do the leading constants and the question whether a renormalized limiting object governs the corresponding modulated Gibbs ensembles.

For the periodic attractive logarithmic interaction with uniform background, the remaining questions include determining the exact value of $\beta_{\mathrm{gm}}(11)$ and the structure of its coexistence phases, establishing the partition asymptotics at coexistence, proving the critical asymptotic of order $N^{\mathsf{d}/2}$

in dimensions $\mathsf{d} \geq 12$ and the matching supercritical pressure limits, and identifying the determinant and fluctuation normalizations around nonuniform minimizing phases.

Another open problem is to pass from the fixed diffusive regime to growing inverse temperature. Corollary 5.7 gives an upper bound as $\beta = \beta_N$ increases, up to $\beta_N \sim N^{1-\mathsf{s}/\mathsf{d}}$ for $0 < \mathsf{s} < \mathsf{d}/2$, where the factor β_N/N compensates the size $N^{\mathsf{s}/\mathsf{d}}$ of the Riesz kernel at the typical spacing $N^{-1/\mathsf{d}}$; in the logarithmic case, the corresponding scale is $\beta_N \sim N$. The sharpness of this interpolation is not addressed here. This regime is also connected to forthcoming work by the third author and Serfaty [RS], which proves quantitative Gaussian CLTs for linear statistics of logarithmic and Riesz gases across the range from $\beta_N = O(1)$ to the microscopic Riesz scale.

The finite-dimensional fluctuation theorem Theorem 2.13 leaves open the corresponding path-space tightness problem. As explained in Remark 2.15, tightness in a suitable negative-regularity path space, together with trajectory regularity of the limiting Gaussian field, would upgrade the identification proved here to a path-space central limit theorem.

Finally, as described in the introduction, the estimates proved here serve as inputs for the authors' companion works [DGR26d, DG26] on sharp size-of-chaos bounds and on quantitative fluctuations of the canonical ensemble. Together, these applications suggest that the modulated partition-function estimates provide a flexible normalization input for fluctuation, commutator, and quantitative-chaos problems for singular mean-field Gibbs measures.

Appendix A. Dependence of constants

This appendix records how the constants in the main estimates may be chosen from the quantities appearing in the proofs. The point is not to optimize universal numerical constants, but to make clear which norms of the background measure and transport field enter the estimates. Throughout the appendix, universal constants may depend on d, s and, in the commutator estimates, on the order n, as well as on the fixed choices in the GLZ and truncation estimates. For the reference measure μ, it is convenient to introduce the notation

$$\mathsf{M}_{\beta,\mu} \coloneqq 1 + \|\mu\|_{L^\infty} + |I_\mu| + \|\tilde{\mathsf{g}}\|_{L^2(\mu^{\otimes 2})} + \|\psi_\mu\|_{L^2(\mu)} + \mathbb{E}_\mu\big[e^{-4\beta\psi_\mu}\big], \qquad \psi_\mu \coloneqq h_\mu - 2I_\mu. \tag{A.1}$$

In the logarithmic case, one also keeps track of

$$\mathsf{L}_{\beta,\mu} \coloneqq 1 + \int_{(\mathbb{R}^{\mathsf{d}})^2} |\log|x-y||\, \mathrm{d}\mu^{\otimes 2}(x,y) + \int_{\mathbb{R}^{\mathsf{d}}} e^{-4\beta h_\mu}\, \mathrm{d}\mu + \|\tilde{\mathsf{g}}\|_{L^2(\mu^{\otimes 2})}. \tag{A.2}$$

These are exactly the additional logarithmic inputs used for the determinant limit and its consequences. The commutator theorem itself requires only (2.20) in the logarithmic case.

For the attractive logarithmic case, fix any admissible constant C_{mom} in (2.34). The threshold β_0 is given by (2.35), with C_* the universal constant of the moment bound (6.5). For every $0 \leq \beta < \beta_0$ with $e^{2\beta h_\mu^{\mathrm{att}}} \in L^1(\mu)$, one admissible attractive uniform-bound constant is

$$C_{\mathrm{ub}}^{\mathrm{att}}(\beta,\mu) = e^{\beta|I_\mu|}\, \mathbb{E}_\mu\big[e^{2\beta h_\mu^{\mathrm{att}}}\big]^{1/2} \left(\frac{2}{1-\big(\beta C_* C_{\mathrm{mom}}\big)^2}\right)^{1/2}. \tag{A.3}$$

For the quantitative conclusion in (2.38), the threshold β_{q} is chosen smaller than $\beta_0/2$ so that the factorial-moment estimate and the operator spectral gap hold uniformly along the logarithmic truncation interpolation. The constants $C_{\mathrm{rate}}^{\mathrm{att}}$ and c_{att} may depend on d, $\|\mu\|_{L^\infty}$, C_{mom}, β, the logarithmic truncation constants, and $\int_{\mathbb{R}^{\mathsf{d}}} e^{4\beta h_\mu^{\mathrm{att}}}\, \mathrm{d}\mu$. No uniformity over a class of backgrounds specified only by $\|\mu\|_{L^\infty}$ is asserted.

For $0 < \mathsf{s} < \mathsf{d}/2$ and fixed $\beta > 0$, an admissible uniform partition-function constant can be obtained as follows. Choose $K_{\beta,\mu} \geq 1$ satisfying

$$\frac{1}{\beta^2 C_\mu K_{\beta,\mu}^{2\mathsf{s}-\mathsf{d}}} \geq 4L, \qquad \frac{1}{\beta C_\mu K_{\beta,\mu}^{-(\mathsf{d}-\mathsf{s})}} \geq 4L, \tag{A.4}$$

where C_μ denotes the constants in (4.27)–(4.29). With the layer-cake threshold

$$U_{\beta,\mu} = 4C\beta K_{\beta,\mu}^{\mathsf{s}} + 1, \tag{A.5}$$

one may take $C_{\mathrm{ub}}(\beta,\mu)$ to dominate the corresponding layer-cake bound

$$e^{\beta\|h_\mu\|_{L^\infty}}\left(e^{U_{\beta,\mu}} + L\int_{U_{\beta,\mu}}^{\infty} e^{-u}\,\mathrm{d}u + \max_{1\le N<N_{\beta,\mu}} \mathsf{K}_{N,\beta}(\mu)\right) \tag{A.6}$$

after increasing $N_{\beta,\mu}$ so that (5.19) holds. In the logarithmic case, the same recipe uses the cutoff in (5.25), the finite-T cancellation estimate (4.31), and the threshold in (5.33); the resulting constant is a function of β and the logarithmic package (A.2).

For the convergence rate, let $C_{\mathrm{det}}(\beta,\mu)$ denote any constant for the determinant and truncation error in (5.65). Then the one-body reduction (5.84) shows that $C_{\mathrm{rate}}(\beta,\mu)$ may be chosen to dominate

$$C_{\mathrm{det}}(\beta,\mu) + \beta\,\|\psi_\mu\|_{L^2(\mu)}\Big(C_{\mathrm{ub}}(2\beta,\mu)^{1/2} + C_{\mathrm{ub}}(4\beta,\mu)^{1/4}\,\mathbb{E}_\mu\big[e^{-4\beta\psi_\mu}\big]^{1/4}\Big). \tag{A.7}$$

Thus, the only new one-body information needed after the determinant estimate is the $L^2(\mu)$ size and the exponential moment $\mathbb{E}_\mu[e^{-4\beta\psi_\mu}]$ of ψ_μ.

For the quantitative divergence estimate in Proposition 5.10, the constants may depend on β, μ, d, and s, as well as on the density-point construction in Lemma 5.9. More precisely, that construction selects a positive-density level, a density point, a local dyadic cube, and a sufficiently fine scale; the constants c_0, C_0, L_0 and the large-N threshold inherit these choices. The present proof therefore does not assert uniformity over a class of background laws specified only through a common L^∞ bound.

Remark A.1 (Explicit constants)**.** If $\|\nabla v\|_{L^\infty} = 0$, then $\mathsf{A}_n(v, X_N, \mu) = 0$ and Theorem 2.8 holds trivially with $C_{\mathrm{em}} = 1$ and any $\varepsilon_{\beta,n} > 0$. If $\|\nabla v\|_{L^\infty} > 0$, the threshold β_{loc} is determined by the constants in the local numerator estimate of Lemma 7.2, and the proof yields

$$\varepsilon_{\beta,n} = \begin{cases} \dfrac{\beta}{4C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n}, & 0 < \beta \le \beta_{\mathrm{loc}}, \\[2ex] \dfrac{\beta_{\mathrm{loc}}}{8C_{\mathrm{sr},n}\|\nabla v\|_{L^\infty}^n}, & \beta > \beta_{\mathrm{loc}}, \end{cases} \tag{A.8}$$

where $C_{\mathrm{sr},n}$ is the short-range constant in (7.7), depending only on $\mathsf{d}, \mathsf{s}, n$, and

$$C_{\mathrm{em}}(\beta,\mu,v,n) = \begin{cases} e + L, & 0 < \beta \le \beta_{\mathrm{loc}},\ 0 < \mathsf{s} < \mathsf{d}/2, \\ e^{-\beta I_\mu}(e+L), & 0 < \beta \le \beta_{\mathrm{loc}},\ \mathsf{s} = 0, \\ e^{-\beta I_\mu}(e+L)^{1/2} C_{\mathrm{ub}}(2\beta - \beta_{\mathrm{loc}}, \mu)^{1/2}, & \beta > \beta_{\mathrm{loc}}, \end{cases} \tag{A.9}$$

where C_{ub} is the uniform partition-function bound from (5.3); see Appendix A for the associated dependency bookkeeping.

The dependence is locally bounded in explicit scalar controls. More precisely, fix $M, \Lambda < \infty$ and, when $\mathsf{s} = 0$, $J < \infty$. For fixed $\mathsf{d}, \mathsf{s}, n$, the proof of Lemma 7.2 may be run uniformly over all backgrounds and transports satisfying

$$\|\mu\|_{L^\infty} \le M, \qquad \|\nabla v\|_{L^\infty} \le \Lambda, \qquad I_\mu \ge -J \quad \text{if } \mathsf{s} = 0. \tag{A.10}$$

After the trivial case $\|\nabla v\|_{L^\infty} = 0$ is separated, every occurrence of the transport field in the centered commutator kernel and its one-body remainder is controlled by $|v(x) - v(y)| \le \Lambda|x - y|$; the

local numerator proof may therefore be run with Λ in place of $\|\nabla v\|_{L^\infty}$ in (A.8). In the logarithmic case, the remaining background dependence is controlled by

$$\int_{(\mathbb{R}^{\mathsf{d}})^2} |\log|x-y|| \, \mathrm{d}\mu^{\otimes 2}(x,y) \le 2M \int_{|z|\le 1} -\log|z| \, \mathrm{d}z + 2J, \tag{A.11}$$

which follows by decomposing $-\log|x-y|$ into its positive and negative parts. Consequently, β_{loc} has a positive lower bound, and the local numerator constants have finite upper bounds, depending only on $\mathsf{d}, \mathsf{s}, n, M, \Lambda$ and, in the logarithmic case, J.

For a family satisfying (A.10), the exponential-moment constants under $\mathbb{Q}_{N,\beta}(\mu)$ may therefore be chosen uniformly whenever $\mathsf{K}_{N,\beta}(\mu)$ has a common positive lower bound and $\mathsf{K}_{N,2\beta}(\mu)$ has a common upper bound. Indeed, for $0 \le \gamma \le \beta$, log-convexity of the partition function and $\mathsf{K}_{N,0}(\mu) = 1$ give

$$\mathsf{K}_{N,2\beta-\gamma}(\mu) \le \mathsf{K}_{N,2\beta}(\mu)^{1-\gamma/(2\beta)}, \tag{A.12}$$

which uniformly controls the intermediate partition-function factor in the Hölder bootstrap.

For the commutator estimate, once a local threshold $\beta_{\mathrm{loc}}(\mu, v, n)$ has been fixed in Lemma 7.2, the constants in Theorem 2.8 are explicitly

$$C_{\beta,n,v} = \frac{\beta}{\varepsilon_{\beta,n}}, \qquad C_{\beta,\mu,v,n} = \frac{1}{\varepsilon_{\beta,n}}(\log C_{\mathrm{pf}}(\beta,\mu) + \log C_{\mathrm{em}}(\beta,\mu,v,n)), \tag{A.13}$$

with $\varepsilon_{\beta,n}$ and C_{em} given in (A.8) and (A.9). The dependence on the transport field enters through $\|\nabla v\|_{L^\infty}$ and through the local numerator constants $C_{\mu,v,n}$, $M_{\mu,v,n}$, and $C_{\mathsf{K},\mu,v,n}$ appearing in Lemma 7.2. The local-boundedness statement in Remark A.1 shows that common scalar controls yield common choices of these constants and of β_{loc}. For time-dependent applications, $\mathcal{A}^t$ and $\mathcal{B}^t$ in (2.50) are obtained by integrating precisely the resulting $C_{\beta,1,u^t}(\mu^t)$ and $C_{\beta,\mu^t,u^t,1}$.

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